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Bhasha Matters

State of the Education
Report for India 2025

**Mother Tongue and
Multilingual Education**

UNESCO Education Sector

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BHASHA MATTERS MOTHER TONGUE AND MULTILINGUAL EDUCATION

State of the Education Report for India 2025

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Front cover photo

Students hold alphabet cards in their mother tongue and learn to recognize and pronounce the letters. Kalinga Institute of Social Sciences, Bhubaneswar.

Back cover photo

Primary school students learning Odia and English as part of the multilingual curriculum. Kalinga Institute of Social Sciences, Bhubaneswar.

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State of the Education
Report for India 2025

Mother Tongue and Multilingual Education

SHORT SUMMARY

BHASHA MATTERS

State of the Education Report for India 2025: Mother Tongue and Multilingual Education

Language is not merely a medium – it is a bridge between thought, culture, identity and learning. The comprehensive State of the Education Report for India 2025 situates mother-tongue-based multilingual education (MTB-MLE) within an equity-driven social justice framework, emphasizing linguistic rights and educational inclusion in India's diverse cultural and linguistic context. It critically explores language's social, cultural and pedagogical roles in educational justice, blending established MLE practices and community innovations with evolving policy and implementation strategies. Organized into eight chapters, the report covers foundational concepts, India's linguistic diversity, pedagogical approaches, tribal education, teacher preparation, technology use and policy recommendations.

The ten recommendations translate the report's analysis into concrete, actionable steps for government, institutions and partners. They are designed to be adaptable across states while remaining anchored in the National Education Policy 2020 and UNESCO's principles on multilingual education.

The report urges states to adopt clear MTB-MLE language-in-education policies aligned with the National Education Policy 2020, prioritize recruitment and professional development of multilingual teachers, and embed multilingual pedagogy in teacher education. It advocates institutionalizing community participation, developing culturally relevant multilingual materials, integrating gender-responsive frameworks, leveraging digital infrastructure, promoting inclusive language technologies, and ensuring sustainable financing. The report calls for establishing a national mission to coordinate MTB-MLE efforts across government bodies, educational institutions and communities.

By providing actionable, context-sensitive pathways, the report seeks to move beyond rhetoric towards practical solutions that respect linguistic diversity, enhance learning outcomes and foster inclusive education for all learners across India's multilingual landscape.

Readers will find an in-depth analysis of why MTB-MLE is central to educational justice, especially for learners from tribal, minority-language and rural communities in India. They will see how the report links linguistic rights with broader questions of equity, inclusion and social justice in schooling.

The ten recommendations of the report are designed to be adaptable across states while remaining anchored in the National Education Policy 2020 and UNESCO's principles on multilingual education



"Since wars begin in the minds of men and women it is in the minds of men and women that the defences of peace must be constructed"



Message

Secretary

Department of School Education & Literacy
Ministry of Education, Government of India

The National Education Policy (NEP) 2020 recognizes the centrality of language in education and strongly recommends multilingualism and initial learning for children in their home language, local language, or language familiar to the child as the primary medium of instruction. The policy affirms that learning in a language familiar to the child during the foundational years is essential for ensuring cognitive development conceptual clarity, emotional security and equitable learning. This vision has been operationalized through the National Curriculum Framework for the Foundational Stage (NCF-FS) 2022 and the National Curriculum Framework for School Education (NCF-SE) 2023, both of which place education in the child's mother tongue at the centre of early learning and classroom practice.

While India's linguistic diversity appears as a challenge, it is one of its greatest educational strengths. In terms of recommendations of the NEP 2020, India is continuously striving to create learning environments where this diversity is valued, where children's mother tongues are seen as assets, and where multilingualism becomes a bridge to deeper comprehension and wider opportunities. The NEP 2020 underscores this vision through its emphasis on using the mother tongue as the primary medium of instruction till at least Grade 5 and preferably till Grade 8.

The NCF-FS 2022 builds on this foundation by promoting play-based, language-rich learning environments in which children engage and learn in the language most familiar to them. The NCF-SE 2023 provides a clear pedagogical

pathway for additive bilingual and multilingual learning in school education. Together, these frameworks seek to build an inclusive and equitable education system that values India's linguistic diversity and supports the national commitment to ensuring that every child can learn with understanding, confidence and dignity.

Across India, significant initiatives have been undertaken to advance multilingual education – ranging from the development of learning materials in tribal and regional languages to teacher training in multilingual pedagogy, and community involvement in creating culturally rooted resources. These initiatives collectively reflect a shared commitment to enriching multilingual education across the country.

In this context, I am pleased to note that the UNESCO Regional Office for South Asia has chosen mother-tongue-based multilingual education as the theme for its State of the Education Report for India 2025. The report not only highlights innovations and good practices from across the country but also offer insights that can guide future strategies to strengthen multilingualism in our schools. I am confident that this report will serve as a valuable resource for policymakers, educators, researchers and all stakeholders committed to improving the quality of education in India.

Sanjay Kumar

Secretary

Department of School Education & Literacy



Message

Secretary
School and Mass Education Department
Government of Odisha

The emphasis placed by the UNESCO Regional Office for South Asia on the mother tongue and multilingual education in the 2025 State of the Education Report for India resonates strongly with Odisha's long-standing efforts in this area.

Odisha has consistently recognized the centrality of the child's mother tongue in ensuring meaningful and equitable learning, particularly for tribal children who form a significant proportion of the state's population. In keeping with this understanding, Odisha launched its Mother-Tongue-Based Multilingual Education (MTB-MLE) programme in 2007-08, guided by the principle that a child learns best when instruction begins in the language that is most familiar. This approach is firmly aligned with the spirit of the Right to Education Act, 2009, and continues to inform the state's commitment to inclusive and culturally responsive education.

Odisha's MTB-MLE programme today stands as one of the most extensive in the country, covering 21 tribal languages across 17 districts. Through 1,317 MLE schools, nearly 90,000 tribal children receive primary education in their mother tongues. More than 3,000 trained teachers and language instructors support this effort, using contextualized textbooks, primers and supplementary materials developed in all 21 languages. Tribal Resource Centres established

in each MLE district strengthen home-school linkages and help preserve the linguistic and cultural heritage that enriches classroom learning.

Odisha remains steadfast in its commitment to improving foundational learning for tribal children through ongoing investments in teacher development, language resources, and community-centred pedagogies. Odisha will continue advancing multilingual practices to ensure that every child learns with confidence, cultural dignity and a strong sense of identity.

I extend my sincere congratulations to the UNESCO Regional Office for South Asia for the development and production of the State of the Education Report for India 2025. This important report will not only highlight the national significance of mother-tongue-based education but also reaffirm and support the work undertaken by states like Odisha to build an inclusive and linguistically responsive education system.

Dr N. Thirumala Naik

Secretary
School and Mass Education Department
Government of Odisha



Message

Secretary
Department of School Education
Government of Telangana

The Government of Telangana welcomes UNESCO's State of the Education Report for India 2025 on mother-tongue-based multilingual education, a theme deeply aligned with the state's commitment to inclusive, equitable and child-centred learning.

Telangana recognizes that children learn best in the language spoken at home. To strengthen early learning, the state has developed 274 textbooks across 8 Indian languages for Grades 1 to 10, reflecting learners' linguistic environments, cultural contexts and lived experiences. Multilingual and culturally rich materials, including bilingual textbooks in Telugu-English and Urdu-English, QR-enabled content through DIKSHA, and lessons broadcast on T-SAT in Telugu, Urdu and English, ensure smoother transitions from home languages to school languages while affirming children's identities and sense of belonging. Telangana's SCERT has also developed a comprehensive Multilingual Digital Dictionary covering nine languages (Telugu, Hindi, English, Urdu, Kannada, Marathi, Tamil, Gujarati and Bengali), with physical copies available in seven languages.

Teacher capacity building remains central to these efforts. Statewide training in multilingual pedagogy – delivered in Telugu, Urdu and English – equips teachers to effectively support diverse learners. Monitoring through the Telangana Education app strengthens accountability, attendance tracking and digital certification.

Classroom practices such as language corners, daily reading sessions, Bhasha Sanghams, and cultural exchange programmes under Ek Bharat Shreshtha Bharat enrich multilingual exposure and vocabulary development.

The state proudly celebrates innovations emerging from the field. Storybooks have been translated into tribal languages such as Lambadi, Gondi and Kolami to support early literacy and cultural continuity. An AI-based system for translating news into tribal languages has also been developed, an innovation that received national recognition during the Hon'ble Prime Minister Shri Narendra Modi's Mann Ki Baat address. Bharatiya Bhasha Summer Camps, conducted across 1,212 centres and benefiting more than 1,21,000 students, have further deepened foundational literacy, cultural learning and holistic development.

Telangana remains committed to expanding mother-tongue-based multilingual education and ensuring that every child accesses meaningful, culturally responsive and high-quality learning opportunities. We appreciate UNESCO's focus on this vital theme and are confident that the report will offer valuable insights to advance multilingual education across the nation.

Dr Yogita Rana

Secretary
Department of School Education
Government of Telangana



Above: Students learn through Indigenous, regional, English and Hindi languages in class. Government Upper Primary School, Gajapati.



Foreword

Language is more than a means of communication - it is the foundation of thought, identity and learning. When children begin their education in a language they understand, classrooms transform from spaces of struggle into environments of discovery and empowerment.

In India, home to over 1,300 mother tongues, mother-tongue-based multilingual education (MTB-MLE) is not merely a pedagogical choice: it is profoundly connected to educational equity, cultural justice the Right to Education, and to achieving Sustainable Development Goal 4 of ensuring inclusive and equitable quality education for all.

This seventh edition of the UNESCO Regional Office for South Asia's annual State of the Education Report for India reflects UNESCO's commitment to multilingual education as articulated in UNESCO's 2003 position paper, *Education in a Multilingual World*, and reinforced by UNESCO's global guidance on multilingual education. The **International Decade of Indigenous Languages (2022-2032)** represents a global initiative to protect, revitalize, and promote languages that embody centuries of knowledge, culture, and identity. This commitment is further reinforced by **UNESCO's 2003 Convention for the Safeguarding of the Intangible Cultural Heritage**, which recognises language as the vehicle for sustaining cultural identity and knowledge systems across generations. UNESCO's engagement with India's MTB-MLE efforts draws on these interconnected frameworks.

This Report examines the implementation of MTB-MLE across India's diverse educational landscape, presenting evidence from classrooms, teacher-education initiatives, digital innovations, and community-led programmes. Case studies from tribal and rural contexts demonstrate

that when schools embrace learners' home languages, measurable improvements emerge in reading comprehension, classroom participation, conceptual understanding, and in building stronger foundations for lifelong learning.

What distinguishes this report is its comprehensive, evidence-based and policy relevant approach in presenting an integrated analysis of what works, where gaps persist, and why systemic change remains challenging. By examining successful models alongside persistent implementation barriers, this report offers a grounded assessment of both progress and the road ahead. It highlights the need for coordinated, system-wide investments to ensure that every child can access education in a language that supports meaningful learning.

India stands at pivotal moment as it translates the National Education Policy 2020's recognition of home-language-based instruction as central to early learning and UNESCO remains committed to partnering with the Government of India and its partners to advance the insights and recommendations of this Report. I extend UNESCO's appreciation to the researchers, educators, community leaders, and officials whose contributions have made this Report possible and whose help ensures that India's multilingual heritage remain a source of educational strength and national pride.



Tim Curtis
Director
UNESCO Regional Office for South Asia



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Abbreviations and acronyms

AAC	Augmentative and Alternative Communication	GER	Gross Enrolment Ratio
AI	Artificial Intelligence	HEI	Higher Education Institution
AICTE	All India Council for Technical Education	HHSD	Happy Hands School for the Deaf
API	Application Programming Interface	ICT	Information and Communication Technology
ARMS	Action Research Mentoring Scheme	IDIL	International Decade of Indigenous Languages
ASER	Annual Status of Education Report	IGNCA	Indira Gandhi National Centre for the Arts
ASMITA	Anuvad, Shikshan, Margdarshan, Inclusive, Takniki, Anusandhan	IGNOU	Indira Gandhi National Open University
ASR	Automated Speech Recognition	IIIT	International Institute of Information Technology
B.Ed.	Bachelor of Education	IIITDM	Indian Institute of Information Technology, Design and Manufacturing
BBS	Bharatiya Bhasha Samiti	IIT	Indian Institute of Technology
BHASHINI	BHASHa Interface for India	IKS	Indian Knowledge System
CBSE	Central Board of Secondary Education	IME	Input Method Editor
C-DAC	Centre for Development of Advanced Computing	INMT	Interactive Neural Machine Translation
CFEL	Centre for Endangered Languages	ISCI	Indian Standard Code for Information Interchange
CIIL	Central Institute of Indian Languages	ISL	Indian Sign Language
CMW	Community Mobilization Worker	ITEP	Integrated Teacher Education Programme
CSTT	Commission for Scientific and Technical Terminology	ITM	Indigenous Tribal Minority
CUET-UG	Common University Entrance Test for Undergraduate Programmes	JNU	Jawaharlal Nehru University
DIET	District Institute of Education and Training	JSS	Jan Shikshan Sansthan
DIKSHA	Digital Infrastructure for Knowledge Sharing	KGBV	Kasturba Gandhi Balika Vidyalaya
DoEPwD	Department of Empowerment of Persons with Disabilities	KISS	Kalinga Institute of Social Sciences
DRG	District Resource Group	KITE	Kerala Infrastructure and Technology for Education
DTH	Direct-To-Home	LDI	Linguistic Diversity Index
EBSB	Ek Bharat Shreshtha Bharat	LEC	Linguistic Empowerment Cell
ECCE	Early Childhood Care and Education	LLF	Language and Learning Foundation
EFLU	English and Foreign Languages University	LLM	Large Language Model
EGMA	Early Grade Mathematics Assessment	MBA	Master of Business Administration
EGRA	Early Grade Reading Assessment	MIL	Minor Indian Language
ELT	English Language Training	MLE	Multilingual Education
EMRS	Eklavya Model Residential School	MOOC	Massive Open Online Course
EVS	Environmental Studies	MoTA	Ministry of Tribal Affairs
FLN	Foundational Literacy and Numeracy	MPUPS	Mandal Parishad Upper Primary School
		MT	Machine Translation

MTB-MLE	Mother-Tongue-Based Multilingual Education	RISE	Rapidly Improving Standards in Elementary
MTSI	Mother Tongue Survey of India	RTE	Right to Education
NAS	National Achievement Survey	RTR	Room to Read
NCERT	National Council of Educational Research and Training	SALT	Supporting Andhra's Learning Transformation
NCF-FS	National Curriculum Framework – Foundational Stage	SARTHAQ	Students' and Teachers' Holistic Advancement through Quality Education
NCF-SE	National Curriculum Framework – School Education	SCERT	State Council of Educational Research and Training
NCL	National Centre for Literacy	SCF	State Curriculum Framework
NCTE	National Council for Teacher Education	SCL	State Centre for Literacy
NDEAR	National Digital Education Architecture	SCRM	Supporting Community Radio Movement
NEP	National Education Policy	SDG	Sustainable Development Goal
NETF	National Educational Technology Forum	SEO	search engine optimization
NGO	Non-Governmental Organization	SoER	State of the Education Report
NIEPVD	National Institute for Empowerment of Persons with Visual Disabilities	SPPEL	Scheme for Protection and Preservation of Endangered Languages
NIPUN	National Initiative for Proficiency in Reading with Understanding and Numeracy	SPV	Sardar Patel Vidyalaya
NISHTHA	National Initiative for School Heads' and Teachers' Holistic Advancement	SRG	State Resource Group
NIT	National Institute of Technology	STEM	Science, Technology, Engineering and Mathematics
NLP	Natural Language Processing	STT	Speech-To-Text
NLTM	National Language Translation Mission	SWAYAM	Study Webs of Active-learning for Young Aspiring Minds
NMRC	National Multilingual Education Resource Consortium	TDIL	Technology Development for Indian Languages
NPST	National Professional Standards for Teachers	TEI	Teacher Education Institution
OBC	Other Backward Class	TLM	Teaching-Learning Material
OCR	Optical Character Recognition	TRI	Tribal Research Institute
ODL	Open and Distance Learning	T-SAT	Telangana Skills, Academic and Training
OER	Open Educational Resource	TTS	Text-To-Speech
OSEPA	Odisha School Education Programme Authority	UDISE+	Unified District Information System for Education Plus
PARAKH	Performance Assessment, Review, and Analysis of Knowledge for Holistic Development	UDL	Universal Design for Learning
PISA	Programme for International Student Assessment	UGC	University Grants Commission
PLC	Professional Learning Community	UGCF	Undergraduate Curriculum Framework
PLSI	People's Linguistic Survey of India	UNESCO	United Nations Educational, Scientific and Cultural Organization
PM eVIDYA	Prime Minister e-Education for Digital Learning	UNICEF	United Nations Children's Fund
RAA	Regional Academic Authority	UP	Uttar Pradesh
		UT	Union Territory
		VOGA	Vanishing Voices of the Great Andamanese
		WCAG	Web Content Accessibility Guidelines



Executive summary

Around the world, there is growing recognition that language plays a decisive role in shaping educational equity, cognitive development, cultural continuity and social inclusion. Global education frameworks increasingly affirm that children learn best when schools value and use the languages they speak at home. The renewed global focus on mother-tongue-based multilingual education (MTB-MLE), reflected across UNESCO guidance on Languages Matter (2025), the International Decade of Indigenous Languages (2022–2032) and the position paper on “Education in a Multilingual World” (2003), underscores the transformative power of linguistic inclusion in education.

Below: Primary school children engage in language-based activity. Model High School, Osmania University, Hyderabad.

India, home to 1,369 languages, stands at the forefront of this global conversation. Its cultural and linguistic diversity is unparalleled, yet many children still encounter schooling in languages

they scarcely understand. Recognizing this, the National Education Policy (NEP) 2020, the National Curriculum Frameworks (NCFs) for foundational stage (FS) 2022 and secondary education (SE) 2023, and state-level reforms increasingly position mother-tongue-based education at the centre of foundational learning, holistic development and inclusive pedagogy. Across India, educators, communities, governments and civil society partners are exploring new ways to integrate local languages, cultural knowledge systems and multilingual practices into classrooms.

At the same time, India faces a profound implementation challenge: ensuring consistent, high-quality and equitable MTB-MLE across diverse regions, languages, cultural contexts and socio-economic backgrounds. Realizing this vision requires sustained investment, coordination across institutions, strengthened teacher capacity, and pedagogical alignment across school stages.



1,369
mother tongues,
including
121 constitutionally
recognized languages
constitute India's
linguistic landscape.

Census, 2011



Above: Reading course books in Saora language. Government Upper Primary School, Gajapati.

About the report

The UNESCO Regional Office for South Asia State of the Education Report (SoER) 2025 for India examines how mother-tongue-based multilingual education is unfolding across the country's school and teacher-education systems. Framed within the NEP 2020, the NCFs and global commitments such as UNESCO's Languages Matter guidance and the International Decade of Indigenous Languages (2022–2032), the report explores how India's linguistic diversity intersects with classroom practice, curriculum design, teacher preparation, technology ecosystems and community participation. Particular attention is given to the experiences of learners from tribal, minority and lesser-known language communities, whose languages and knowledge systems remain insufficiently represented in formal education.

To build this analysis, the report employs a rigorous mixed-methods approach that integrates quantitative, qualitative and policy evidence. National datasets, including the Census 2011, the Unified District Information System for Education Plus (UDISE+) 2023–24 and the Annual Status of Education Report (ASER) 2024, provide insights into access, participation and early learning in multilingual settings. A detailed review of the NEP 2020, the NCF-FS 2022, the NCF-SE 2023 and state-specific guidelines enables an assessment of how language-in-education commitments are reflected in curriculum, pedagogy and institutional planning. This is complemented by a synthesis of academic research and programme evaluations from multilingual and tribal regions, highlighting

both promising approaches and persistent bottlenecks.

The analysis is further informed by insights from diverse educational settings and multilingual communities across India, offering grounded perspectives on classroom language use, teacher preparedness, community expectations and the realities experienced by learners navigating multilingual environments. Consultations with researchers, educators, State Councils of Educational Research and Training (SCERTs), District Institutes of Education and Training (DIETs), tribal research institutes (TRIs), non-governmental organizations (NGOs), technology partners and cultural institutions enrich the report with practice-based and contextual insights.

To demonstrate how multilingual education is being enacted across the country, the report presents case studies that showcase community-driven initiatives, state-led innovations, Indigenous language revitalization efforts, and digital solutions developed for marginalized languages. These examples illustrate how multilingual materials, local knowledge, oral traditions and accessible technologies can strengthen teaching and learning.

Drawing on this multilayered evidence base, the report identifies key strengths, systemic challenges and emerging opportunities, and concludes with ten actionable recommendations to support policymakers, institutions and practitioners in advancing a more equitable and linguistically responsive education system in India.



UNESCO's framework for multilingual education

Inspired by UNESCO's global guidance on language in education and the International Decade of Indigenous Languages (2022–2032), the report aligns its analysis with **five broad principles**:

GOAL 1

Access, inclusion and equity in and through multilingual education.

Ensuring that all learners, especially tribal children, girls, and learners with disabilities, can meaningfully participate in school through languages they understand.

GOAL 2

Contextual, quality, lifelong and lifewide multilingual learning.

Embedding local languages and cultural knowledge systems across school stages, higher education, teacher training and community learning.

GOAL 3

Appreciation of linguistic diversity and capacity for critical engagement.

Recognizing children's linguistic repertoires as assets and promoting understanding of India's linguistic diversity as part of democratic citizenship.

GOAL 4

Skills for resilient, just and sustainable futures.

Using multilingual learning to build cognitive flexibility, problem-solving and communication skills, and stronger foundations for learning additional languages, including English.

GOAL 5

Institutionalization and valorization of multilingual education ecosystems.

Strengthening policies, institutions, teacher education systems and digital ecosystems to support long-term, sustainable MTB-MLE.

Evolution of global perspectives on MTB-MLE

Global research shows that sustained mother-tongue-based instruction strengthens foundational learning, reduces dropout rates, supports well-being and preserves cultural knowledge. The discourse on multilingual education has shifted substantially over the past three decades. Earlier debates focused narrowly on medium-of-instruction reforms; contemporary frameworks stress holistic multilingual development, the use of learners'

full repertoires, and pedagogy that reflects how children move naturally across languages.

UNESCO's evolving guidance situates MTB-MLE within broader goals of cognitive development, equity, gender inclusion, linguistic rights, digital participation and community empowerment. It resonates strongly with India's linguistic realities and the need to adopt models that reflect everyday multilingual practices, especially in tribal and rural regions.

Indian perspectives and policy analysis in light of UNESCO's framework

India's education policies have long acknowledged linguistic diversity and recent frameworks mark a significant shift towards grounding early education in home and community languages. The NEP 2020 endorses mother-tongue-based instruction, multilingual pedagogy and culturally responsive learning. The NCF-FS 2022 and the NCF-SE 2023 further integrate multilingual pathways into curriculum organization, foundational literacy goals, subject teaching, and assessment in higher classes.

A comparison with UNESCO's principles shows strong conceptual alignment:

- 1 emphasis on inclusive and equitable learning

- 2 recognition of cultural and linguistic diversity
- 3 focus on foundational learning and development
- 4 integration of local knowledge systems
- 5 promotion of multilingual competencies across all levels of education

However, the challenge lies in systemic implementation. While policies consistently affirm the value of children's languages, schools frequently operate in languages unfamiliar to learners. Teacher education, materials, assessment systems, and institutional coordination have not yet fully aligned with multilingual goals, creating gaps between policy intent and classroom reality.

Key findings

MTB-MLE gaining prominence in India's education landscape

The SoER 2025 shows that mother-tongue-based multilingual education is steadily moving from the margins to the centre of India's education reforms. It is increasingly recognized as critical to achieving the goals of the NEP 2020, the NCFs and global commitments to UNESCO's Languages Matter guidance and the International Decade of Indigenous Languages (2022–2032). As the system moves towards this vision, many classrooms, especially in tribal and rural regions, continue to operate in languages unfamiliar to children. Such environments can affect participation and confidence, particularly for girls in remote areas and learners with communication or learning disabilities. Strengthening school-level implementation of MTB-MLE, therefore, remains an important opportunity.

Below: Tactile learning techniques are used to support children with special need. Kalinga Industrial Institute of Technology International School, Bhubaneswar.

Towards more responsive language-in-education planning

Across states, efforts to develop operational frameworks that reflect the principles of MTB-MLE are at varying stages. Systematic

language mapping is essential for decisions on the medium of instruction, the development of teaching-learning materials (TLMs), and teacher deployment, which is expanding but not yet universal. Ensuring that planning processes reflect children's home languages meaningfully will enable more equitable and context-responsive provisioning, particularly in linguistically complex and tribal regions.

Teacher education as the anchor for multilingual and inclusive classrooms

Teachers shape how multilingualism is experienced in everyday classroom interactions. While progress is emerging, multilingual pedagogy, translanguaging and transknowledging are still not consistently embedded across pre-service and in-service programmes. Strengthening teacher preparation to include language-responsive instructional strategies, early-grade assessments in familiar languages, and inclusive approaches that recognize gendered participation patterns and the needs of learners with disabilities will be central to realizing the goals of MTB-MLE.



Integrating curriculum, pedagogy and community knowledge systems

The report highlights compelling examples of schools and communities co-creating learning materials rooted in local knowledge, oral traditions and cultural practices, in languages such as Saora, Kui, Gondi, Santali, Bhili, Khasi, Mizo, Toda and others. These practices demonstrate the educational value of Indigenous epistemologies and multilingual cultural resources. At present, however, such collaborations remain localized. More systematic integration of this knowledge into curricula and pedagogy across school stages can strengthen engagement, comprehension and intergenerational language transmission.

Technology as an expanding space for linguistic inclusion

India's digital public infrastructure, including the Digital Infrastructure for Knowledge Sharing (DIKSHA), PM eVIDYA, ePathshala, BHASHa INterface for India (BHASHINI), Bharatavani, and state platforms, has broadened access to learning resources. Continued efforts to include tribal and lesser-taught languages, improve tool performance in mixed-language environments and enhance accessibility features will enable wider participation. Addressing connectivity gaps and ensuring device access, particularly in remote regions and for girls and learners with disabilities, will be essential for translating digital potential into meaningful learning gains.

Below: Students attend a Gujarati language class. Sardar Patel Vidyalaya, Delhi.

Advancing digital and linguistic inclusion together

Learners in rural, tribal and low-income settings often face limited connectivity, device shortages and a scarcity of digital materials in their home languages. Expanding availability of content in accessible formats, including Indian Sign Language (ISL) and Braille, with multimodal supports, can enable more diverse learners to engage with digital platforms. Viewing digital inclusion and linguistic inclusion as interconnected priorities will help ensure that innovations reach children who stand to benefit the most.

Stronger institutional collaboration for sustainable implementation

Effective delivery of MTB-MLE requires close coordination between multiple actors: the ministries of education, tribal affairs, culture, and technology, the National Council for Teacher Education (NCTE), National and State Councils of Educational Research and Training (NCERT and SCERTs), the Central Institute of Indian Languages (CIIL), DIETs, TRIs, universities, NGOs, community organizations and technology partners. While several promising initiatives demonstrate the value of collaboration, sustained progress will require more formal, long-term mechanisms with clear roles across teacher education, curriculum development, technology integration, research, and community engagement. Strengthening institutional capacities, especially within SCERTs and DIETs, will be key to supporting implementation at scale.



Strengthening research, monitoring and financing for long-term progress

Evidence from various state initiatives consistently shows that children learn more effectively when early instruction is grounded in their home languages. Incorporating such insights more systematically into national monitoring systems, such as the UDISE+ and

the National Achievement Survey (NAS), can enhance planning, programme design and the tracking of learning outcomes. Targeted financing for multilingual teacher preparation, community-developed resources, digital inclusion and accessible formats will further support steady progress. Sustained and predictable investment will be essential to embed MTB-MLE as a durable foundation for equitable, high-quality education across regions.

Challenges in implementing MTB-MLE in India

India's linguistic diversity offers a strong foundation for inclusive education, yet this report notes several ongoing challenges in translating MTB-MLE into classroom practice. In many tribal and minority language regions, young learners still begin school in languages they do not speak at home, affecting early participation and comprehension, an issue felt more acutely by girls in remote settings and by children with communication-related or learning disabilities. Limited availability of textbooks and teaching-learning materials in children's own languages further restricts engagement, especially where scripts or documentation for tribal and lesser-known languages are still developing.

The report also highlights that monolingual teaching practices persist despite multilingual

classroom realities, and many teachers feel underprepared for multilingual lesson design or language mapping. Early transitions to regional or global languages can disrupt learning before strong foundations in the mother tongue are established. Technology offers new possibilities, but tribal and minoritized languages remain under-represented on digital platforms, accessibility features are inconsistent and connectivity gaps continue to affect remote regions, particularly impacting girls and learners with disabilities. Fragmented institutional responsibilities and limited language-disaggregated data further constrain coordinated planning and community engagement, while parental aspirations for dominant languages continue to influence early schooling choices.

Opportunities for advancing MTB-MLE in India

The report highlights several emerging opportunities that position India to strengthen MTB-MLE. A strong policy environment, shaped by the NEP 2020, the NCF-FS 2022, the NCF-SE 2023, and global commitments such as to the International Decade of Indigenous Languages (2022-2032), provides clear momentum for expanding familiar-language learning. Recognition among teachers, parents and communities of its positive impact on learning outcomes is steadily increasing, alongside state-led innovations in Odisha, Andhra Pradesh, Chhattisgarh, Jharkhand, Maharashtra, Kerala, Tamil Nadu and the North-East. The growth of digital public infrastructure, multilingual libraries, translation tools and community-

generated open resources offers new pathways for reaching diverse learners, including girls and children with disabilities, when accessible formats are prioritized. Strengthening partnerships across SCERTs, TRIs, universities, NGOs, cultural institutions and technology teams is creating space for more contextually grounded pedagogy and materials. At the same time, growing interest in documenting Indigenous languages and oral traditions, supported by interdisciplinary approaches linking language, culture, ecology and experiential learning, provides a strong foundation for more inclusive and relevant classroom practices. Together, these developments signal a robust and expanding landscape for advancing MTB-MLE across education systems.



The way forward

The following ten strategic actions address the challenges and risks identified:

RECOMMENDATION 1

Adopt and operationalise clear state-level language-in-education policies grounded in MTB-MLE.

RECOMMENDATION 2

Strengthen teacher recruitment, deployment and professional standards for multilingual competence.

RECOMMENDATION 3

Reform pre-service and in-service teacher education to embed multilingual pedagogy at all stages.

RECOMMENDATION 4

Institutionalise community participation and Indigenous knowledge in school practice.



Above: Children of Government Upper Primary School, Gajapati.

RECOMMENDATION 5

Develop and distribute high-quality multilingual learning materials and assessments across grades.

RECOMMENDATION 6

Integrate a gender-responsive MTB-MLE framework across middle, secondary, and alternative schooling.

RECOMMENDATION 7

Leverage digital public infrastructure for inclusive multilingual teacher support and learning resources.

RECOMMENDATION 8

Invest in Inclusive Language Technologies and Bridge the Digital Divide.

RECOMMENDATION 9

Ensure Sustainable, Equitable and Inclusive Financing for MTB-MLE and Language-Responsive Technologies.

RECOMMENDATION 10

Establish a National Mission for Mother-Tongue-Based Multilingual Education (MTB-MLE) and Strengthen Institutional Coordination.



Language-based learning through puppetry.
Oxford Grammar School, Hyderabad.



CHAPTER

1

About the report

This chapter offers a brief introduction to the context of this report, as well as its structure, purpose and aim, methodology, and scope and limitations.



1

About the report

Introducing mother-tongue-based multilingual education for inclusive and equitable learning

India's linguistic landscape is diverse and fluid, encompassing 1,369 mother tongues and 121 constitutionally recognized languages (Census, 2011). This complexity presents both significant pedagogical possibilities and structural challenges. Most Indians practise functional multilingualism, making linguistic plurality not an exception but the everyday condition of social and economic life. However, this raises a central concern for education: how can educators and policymakers design schooling that affirms children's multilingual repertoires and linguistic identities while ensuring equitable access to high-quality learning?

Language is not merely a medium – it is a bridge between thought, culture, identity and learning. UNESCO's *Languages Matter: Global Guidance on Multilingual Education* (2025)

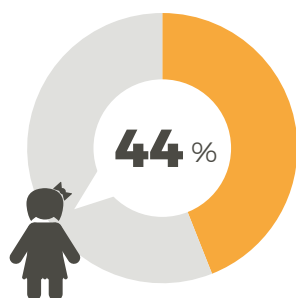
“

over a quarter of a billion learners lack education in the language they understand best.

”

underscores that leaving them at a structural disadvantage in comprehension, engagement and progression. This insight resonates powerfully in India, where nearly 44 per cent of children enter school speaking a language different from the medium of instruction (NCERT, 2022). National surveys (NCERT, 2024; ASER, 2024) reveal persistent learning deficits in rural, tribal and minority-language regions. The Annual Status of Education Report (ASER) 2024 reports that only 23.4 per cent of Grade

Below: Primary school children sing in different languages while learning about animals. Model High School, Osmania University, Hyderabad.



In India, nearly 44 per cent of children enter school speaking a language different from the medium of instruction.

NCERT, 2022





Above: Tamil class for children from different grades, Sardar Patel Vidyalaya, Delhi.

3 students in government schools could read a Grade 2 text, a pattern strongly associated with home-school language mismatch. Similarly, PARAKH (Performance Assessment, Review, and Analysis of Knowledge for Holistic Development) (NCERT, 2024) finds steep declines in mathematics proficiency from Grade 3 to Grade 9, attributing poorer performance among tribal and minority-language learners to weak foundational literacy caused by instruction in unfamiliar languages. These patterns reflect the inadequate preparation of schools and teachers for multilingual pedagogy, the scarcity of locally grounded learning materials, and limited community confidence in the educational value of mother tongues and home languages. The National Education Policy (NEP) 2020 recognizes this as a systemic fault-line, mandating the use of the mother tongue/home language/ local language as the medium of instruction until at least Grade 5 and preferably until Grade 8. Building on UNESCO's call for multilingual and culturally responsive systems, this State of the Education Report (SoER) for 2025 situates mother-tongue-based multilingual education (MTB-MLE) as central to achieving equity, learning continuity and identity affirmation – particularly for tribal, Indigenous and marginalized learners whose languages embody India's living heritage.

Since 1953, UNESCO has played a pivotal role in advancing multilingual education as a framework for achieving inclusive, equitable and quality learning across the globe. In its Education 2030 Framework for Action, adopted in 2015, UNESCO emphasizes that education systems must accommodate linguistic diversity to promote learning outcomes, cultural identity and social cohesion (UNESCO, 2015). Building on this foundation, UNESCO declared 2022–2032 the International Decade of Indigenous Languages, underscoring the urgent need to preserve endangered languages and ensure that educational systems support mother-tongue-based instruction as a means of empowerment (UNESCO, 2022). UNESCO's recent report, *Languages Matter: Global Guidance on Multilingual Education*, further consolidates global expertise by offering evidence-based recommendations for curriculum design and teacher training, and for policy frameworks that address the challenges of implementing mother-tongue-based education while promoting language rights (UNESCO, 2025). These efforts align with the United Nations Sustainable Development Goal (SDG) 4, which calls for inclusive and equitable education for all (United Nations, 2015), and affirm that linguistic diversity is not a barrier but a resource to be leveraged for learning, participation and cognitive development. UNESCO situates multilingual education at the intersection of human rights, sustainable development and educational equity, advocating for transformative policies that reflect the sociolinguistic realities of diverse learners.



Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

United Nations Sustainable Development Goal 4

This report centres mother-tongue-based multilingual education within a broader equity-driven social justice framework, thereby advancing the discourse on linguistic rights and educational inclusion.



Above: Preliminary exams are offered in both Urdu and Telugu. Model High School, Osmania University, Hyderabad.

Purpose and aim

Since 2019, the UNESCO Regional Office for South Asia State of the Education Report has endeavoured to address critical gaps in educational access and quality in India by translating empirical research and field-based insights into pragmatic policy recommendations. This edition of the report deepens this trajectory by centring mother-tongue-based multilingual education within a broader equity-driven social justice framework, thereby advancing the discourse on linguistic rights and educational inclusion. It critically examines the social, cultural and pedagogical dimensions of language as a vehicle for

educational justice, integrates existing MLE practices and community-led innovations with evolving policy and implementation frameworks, and charts actionable pathways for state and non-state actors, aiming to move beyond declarative commitments towards robust, context-sensitive and sustained interventions. It highlights the imperative of embedding linguistic diversity within educational planning, ensuring that curriculum design, teacher preparation and institutional structures are responsive to the lived realities of learners across India's heterogeneous cultural and linguistic landscape.

OBJECTIVES

- Assess the current status, models and quality of mother-tongue-based multilingual education across Indian states, with systematic attention to equity indicators such as gender, ethnicity, socio-economic status, disability, location (rural/urban/tribal) and school type.
- Leverage linguistic fluidity and everyday multilingual practices for child-centred critical pedagogy.
- Identify system bottlenecks and enabling conditions across curricula, teacher education, assessment, language technology and community participation.
- Translate proven ground-level innovations into policy and programme options for scale, including financing and governance arrangements.
- Support ongoing NEP 2020 reforms (e.g. NIPUN Bharat, FLN and STEM frameworks)¹ with clear implementation pathways and capacity-building needs.
- Propose a monitoring and indicator framework to track progress and accountability at national and state levels.
- Advance linguistic, cultural and epistemic justice by recommending mechanisms for sustained community participation in curricular design, resource development, language revitalization, and the long-term use of children's mother tongues/home languages across stages of schooling.

¹ NIPUN Bharat = National Initiative for Proficiency in Reading with Understanding and Numeracy; FLN = foundational literacy and numeracy; STEM = science, technology, engineering and mathematics.

Methodology

This State of the Education Report for 2025 employs a mixed-methods research design that integrates large-scale quantitative datasets with qualitative, field-based insights. Quantitative analysis draws on major national datasets, including the Unified District Information System for Education Plus (UDISE+) 2022-23; ASER; learning assessments conducted by the National Council for Educational Research and Training (NCERT); the database of the National Council for Teacher Education (NCTE); and international sources such as UNESCO. These datasets are used to map patterns of linguistic diversity, participation and learning outcomes; track correlations between language use, gender, ethnicity, disability and socio-economic status; and conduct comparative analyses of states that are already implementing some form of mother-tongue-based multilingual education. The aim of this analysis is to highlight how constitutional provisions and policy environments shape the implementation of MTB-MLE.

The qualitative component deepens this analysis by examining the contextual and lived realities of multilingual education across India. A systematic analysis is carried out of national and state policy documents, curriculum frameworks, teacher-training materials, and new initiatives in different media and digital platforms. Case studies are invited both from government and non-governmental organization (NGO) partners involved in implementing stronger forms of multilingual education in linguistically diverse districts with emphasis on gender, class and ethnicity. A few case studies are developed through document analysis and interviews with teachers, students and community members. These qualitative data shed light on actual classroom practices, pedagogical innovations, systemic challenges and community-led initiatives, and help in triangulating evidence across sources. Together, these methods enable a nuanced understanding of both macro-level trends and micro-level experiences, producing a comprehensive and contextually grounded assessment of multilingual education and its future in India.

Scope and limitations

This report presents a triangulated, evidence-rich account of India's multilingual education landscape.

The scope of this report encompasses both the formal education system and community-based practices that shape multilingual education in India. It focuses on government and government-aided schools (Grades 1 to 12), with an emphasis on the foundational and primary stages, where language plays a critical role in determining learning trajectories.

This report also analyses teacher education systems, both pre-service and in-service, and the role of curriculum agencies like the NCERT and State Councils of Educational Research and Training (SCERTs) in developing multilingual learning materials and frameworks. It includes documentation from Eklavya Model Residential Schools (EMRSs), Kasturba Gandhi Balika Vidyalayas (KGBVs), Ashramshalas and other community-focused institutions.

LIMITATIONS

- Language disaggregation in major datasets [e.g. the UDISE+, National Achievement Survey (NAS) and ASER] is currently inadequate. Many local languages, including tribal and minority languages, are grouped under larger linguistic categories, making micro-level analysis challenging.
- Comparative assessments of learning outcomes by language of instruction are limited and often anecdotal, especially in Grades 1 to 3.
- There is a lack of gender-disaggregated data on how the language of instruction affects girls and gender-diverse learners, especially in tribal schools. Non-formal education (e.g. adult literacy programmes) and alternative schooling systems are under-represented due to limited documentation.

Despite these constraints, this report presents a triangulated, evidence-rich account of India's multilingual education landscape.



Structure of the report



This report is organized into nine interlinked chapters, each examining a critical aspect of multilingual education, from its conceptual foundations to programmatic innovations and policy recommendations.

CHAPTER 1

About the report

This provides a contextual, methodological and structural overview of the report.

CHAPTER 2

Multilingual education in India: concepts, policies and practices

This chapter introduces theoretical concepts relevant to MTB-MLE and discusses global and Indian policy developments.

CHAPTER 3

Linguistic landscape of India

This chapter maps the linguistic landscape of India, drawing on the Census of India, the People's Linguistic Survey of India (PLSI), UNESCO's *World Atlas of Languages*, and select sociolinguistic surveys undertaken by various agencies in India.

CHAPTER 4

Mother-tongue-based multilingual education: reimagining language in curriculum and pedagogy in India

This chapter examines how languages function across educational stages and assesses curricular and pedagogical practices from a language-inclusive perspective.

CHAPTER 5

Education of tribal children

This chapter focuses on the education of tribal students, including language mapping, curricular and pedagogic innovations, community-led models, and teacher development in tribal languages.

CHAPTER 6

Teacher education for mother-tongue-based multilingual education

This chapter analyses teacher preparation for MTB-MLE, with attention to curriculum reform, multilingual pedagogy, resource development and community partnerships.

CHAPTER 7

Use of technology in mother-tongue-based multilingual education

This chapter explores how technology platforms and digital innovations are advancing or inhibiting language inclusion, with special attention to the digital divide.

CHAPTER 8

Way forward: teaching and learning in every tongue

This chapter presents a synthesis of findings and forward-looking recommendations for a comprehensive, system-wide framework that embeds mother-tongue-based multilingual education across all stages of education.

CHAPTER 9

Recommendations

This chapter presents recommendations on multiple pathways that India may adopt to develop a systemic, adequately funded MTB-MLE strategy that links policy, teacher education, community knowledge, gender equity measures, multilingual learning resources, technology integration and governance mechanisms to ensure equitable, high-quality education for all children, particularly those from tribal and minority language communities.

Facing page: Students are given the choice to write an exam in Urdu or Telugu. Model High School, Osmania University, Hyderabad.



Children learn the alphabet through activities that teach proper pronunciation and writing skills.
Kalinga Institute of Social Sciences, Bhubaneswar.



CHAPTER

2

Multilingual education in India: concepts, polices and practices

India's 1,369 mother tongues are not barriers to learning – they are its most powerful classroom tools.

What if the secret to solving India's learning crisis lies not in English- or regional-medium instruction, but in the mother tongues children already think with and think through to create new imaginations and new worlds?

What is multilingual education and why is it important for India?



2

Multilingual education in India: concepts, policies and practices

What is multilingual education?

Multilingual education (MLE) refers to the systematic use of more than one language in schooling to facilitate learning, literacy and cognitive development. It begins with the learner's mother tongue or first language as the medium of instruction and progressively introduces regional, national or international languages in pedagogically sequenced ways (UNESCO, 2003a and 2021; Heugh, 2021; Benson, 2017; Mohanty et al., 2009). This approach rests on the principle that children learn best in a language they understand, enabling more effective conceptual development and

participation in school. By integrating local languages into classroom instruction, MLE bridges home and school cultures, and enhances comprehension, confidence and identity formation among children from linguistic minority communities. Empirical studies across the Global South demonstrate that multilingual education founded on children's mother tongues leads to improved literacy and numeracy outcomes, reduced dropout and repetition rates, and increased community engagement in education (Panda, 2025; Mohanty, 2019).

Below: Multilingual reading session in library. Sardar Patel Vidyalaya, Delhi.

Children learn best in a language they understand, enabling more effective conceptual development and participation in school.





Above: Teachers are trained to use the mother tongue for teaching. Government Upper Primary School, Gajapati.

Why does India need multilingual education?

India is home to 1,369 languages (Census of India, 2011), yet only a few of the officially recognized languages included in the Eighth Schedule of the Constitution are used in formal schooling. This has historically privileged a few dominant regional and global languages, leading to alienation and poor learning outcomes among children from linguistically less privileged homes, including tribal and other minority language communities (Mohanty, 2019; Benson, 2002; Panda, 2021). Studies reveal that when learners are introduced to literacy in an unfamiliar medium, they experience cognitive overload, diminished classroom participation and higher dropout rates (NCERT, 2022; NCERT, 2023; NEP, 2020). Conversely, instruction in the mother tongue during early schooling enhances comprehension, foundational literacy and socio-emotional well-being (UNICEF, 2019a; 2019b; Panda, 2022 and 2025; Mohanty, 2019; NCERT, 2022; Cummins, 2018). It also enhances students' intersubjective capacities (Panda, 2022) and strengthens intergenerational cultural transmission (Abbi 2006 and 2008, Skutnabb-Kangas, 2000).

As UNESCO (2021) states,



“education through the language of the learner's home is a foundation for equity and quality learning.”

The Annual Status of Education Report (ASER) 2024 underscores that many children in India continue to struggle with basic reading and numeracy, and that the language of instruction contributes to these learning gaps. In rural government schools, in 2024, only 23.4 per cent of Grade 3 students, 44.8 per cent of Grade 5 students and 67.5 per cent of Grade 8 students were able to read a Grade 2 text. Similarly, arithmetic performance remains weak, with 33.7 per cent of Grade 3 students, 30.7 per cent of Grade 5 students, and 45.8 per cent of Grade 8 students able to perform Grade 2 subtraction, division and division-and-subtraction (arithmetic composite) tasks in 2024.



Language outcomes remain relatively strong across all grades, suggesting the greater familiarity and contextual accessibility of linguistic tasks.

TABLE 2.1
Reading skills (ASER tool: Grade 2 story) (in percentage)

Grade	Indicator	2018	2022	2024	Notes
3 (government schools)	Can read Grade 2 text	20.9	16.3	33.4	Highest in ASER history for government schools
5 (government schools)	Can read Grade 2 text	44.2	38.5	44.8	Clear recovery in government schools
5 (private schools)	Can read Grade 2 text	65.1	56.8	59.3	Small improvement from 2022
8 (government schools)	Can read Grade 2 text	69	66.2	67.5	Slight improvement from 2022; state variation wide

Source: ASER, 2024

TABLE 2.2
Arithmetic performance [ASER tool: number recognition, subtraction (borrowing) and division (3 digits ÷ 1 digit)] (in percentage)

Grade	Indicator	2018	2022	2024	Notes
3 (all)	At least subtraction	28.2	25.9	33.7	Highest in 10 years
3 (government schools)	At least subtraction	20.9	20.2	27.6	Big gains in Tamil Nadu and Himachal Pradesh (>15 per cent)
5 (all)	At least division	27.9	25.6	30.7	Driven mainly by government schools
5 (government schools)	At least division	—	—	—	
8 (all)	Basic arithmetic composite	44.1	44.7	45.8	Essentially flat vs. 2018 and 2022

Source: ASER, 2018, 2022 and 2024



The findings indicate a gradual weakening of conceptual understanding as learners progress through school.

The National Achievement Survey Report, PARAKH 2024, reveals a clear and consistent decline in average proficiency levels from Grade 3 to Grade 9, with the sharpest deterioration in mathematics (NCERT, 2024). Nationally, average scores in mathematics drop from approximately 60 per cent in Grade 3 to 46 per cent in Grade 6 and about 37 per cent in Grade 9, indicating a gradual weakening of conceptual understanding as learners progress through school. By contrast, language outcomes remain relatively strong across all grades, suggesting the greater familiarity and contextual accessibility of linguistic tasks compared to numeracy skills. However, within mathematics, specific competencies such as understanding fractions in Grade 6 and advanced algebraic or percentage operations in Grade 9 show particularly low proficiency levels, often falling below 30 per

cent to 40 per cent. These findings underscore the urgent need to strengthen foundational learning, especially in mathematics, through early-grade interventions in children's mother tongues to build conceptual clarity, fluency and problem-solving capacity, thereby preventing the cumulative learning deficits that manifest in later stages of schooling. The Performance Assessment, Review, and Analysis of Knowledge for Holistic Development (PARAKH) attributes the relatively poor performance of tribal students in language and arithmetic tasks to language gaps and weak foundational literacy. It substantiates the position of the National Education Policy (NEP) 2020 and argues in favour of leveraging the home language as the medium of instruction in early years of schooling to help consolidate a stable base in literacy, numeracy, comprehension and intellectual engagement (NCERT, 2024).

In multilingual settings, people constantly switch and mix languages, which requires them to manage complex cognitive processes.



Above: Children participate in multilingual narration as an integral component of their puppet show script. Oxford Grammar School, Hyderabad.

Cognitive and intersubjective benefits of multilingualism

Research shows that using multiple languages strengthens core executive functions such as attention, memory and self-control (Tsimpli et al., 2020; Panda, 2022). In multilingual settings, people constantly switch and mix languages, which requires them to manage complex cognitive processes and enhances their metalinguistic awareness (Mohanty, 2019). This mental flexibility, being able to shift between tasks, perspectives and rules, is closely linked to higher intelligence (Panda, 2022; Bialystok, 1999, Bialystok et.al. 2005; Bialystok et al., 2012; Green,

1998). Studies with bilingual and multilingual learners show that they often perform better than monolingual peers on tasks tapping working memory, inhibition and attentional control (Vogelzang et al., 2025; Bialystok, 2011; Cummins, 2018), a finding that has important implications for low-income multilingual communities in India. They also showed stronger metacognitive and theory-of-mind abilities. Overall, “multilinguaging” supports both language and cognition, fostering creativity, adaptability and learning success (Panda, 2022).

Educational advantage of multilingualism and mother-tongue-based instruction

The educational advantage of multilingualism is well established in contemporary research provided it is founded on children’s mother tongues (MHRD, 2020; UNESCO, 2003a and 2025; UNICEF, 2019). Learning in the mother tongue enables children to access prior knowledge, comprehend abstract concepts more effectively and participate meaningfully in classroom discourse; it fosters deeper cognitive engagement and literacy development (Cummins, 2000; Mohanty, 2019; UNESCO, 2025). Studies consistently show that early education in the mother tongue/home language enhances reading comprehension (NCERT, 2024; ASER, 2024; Room to Read, 2018; Vogelzang et al., 2022), academic achievement and self-esteem, while reducing dropout and repetition rates (NCERT, 2024 and 2011; Panda et al., 2011; Benson, 2009;

Heugh, 2011). Moreover, multilingual education cultivates **additive bilingualism** – the expansion rather than replacement of linguistic repertoires – that facilitates transfer of skills across languages and strengthens metalinguistic awareness (Bialystok, 2011; Cummins, 2001, Mohanty, 2019). In diverse societies like India, mother-tongue-based multilingual education also advances social inclusion and equity by validating learners’ linguistic identities and bridging discontinuities between the home and the school (UNESCO, 2025; Pattanayak, 1981; Jhingran, 2024; Panda, 2012a and 2021). As affirmed by the NEP 2020 and UNESCO’s *Languages Matter* (2025) report, the pedagogical use of children’s home languages is both an instrument of quality learning and a right that sustains cultural continuity and linguistic justice.



Above: Teacher helps a student read a Bengali book. Sardar Patel Vidyalyaya, Delhi.

Language-in-education policy in India

India's language-in-education policy reflects a long struggle between linguistic pluralism and homogenizing state practices (Annamalai, 2001; Panda and Mohanty, 2015). The Constitution of India (Articles 343–351) and the Three-Language Formula (1956) recognized the pedagogical value of learning in the mother tongue. Even though the national education policies of 1968 and 1986 emphasized teaching in children's mother tongues/home languages, classroom practices overwhelmingly shifted towards dominant regional and global languages, leading to alienation and high dropout rates among speakers of non-dominant languages.

From the 1990s onward, multilingual education began to re-emerge as a corrective framework within both research and practice (Skutnabb-Kangas, 2000; Benson, 2002; UNESCO, 2003a;

Mohanty, 2008 and 2019; Mohanty et al., 2009). Multilingualism is no longer seen as a deficit but rather a cognitive and cultural resource, and mother-tongue-based multilingual education (MTB-MLE) is accepted as the most equitable pathway for inclusion. Experimental programmes in Odisha, Andhra Pradesh and Chhattisgarh, supported by institutions such as the National Council of Educational Research and Training (NCERT), State Councils of Educational Research and Training (SCERTs), the Central Institute of Indian Languages (CIIL), and state MLE missions, demonstrated improved comprehension and retention when children were taught through their home languages before transitioning to regional and national languages (NCERT, 2011; Panda and Mohanty, 2013). The NEP 2020 states, "As... multilingualism has great cognitive benefits to young students, children will be exposed to different languages early on (but with a particular emphasis on the mother tongue), starting from the Foundational Stage onwards..." (NEP 2020, Section 4.12). The National Curriculum Frameworks (NCFs) 2022 and 2023 institutionalize these insights, envisioning multilingual pedagogy as central to learning and cognitive development.

Multilingualism is no longer seen as a challenge, but as India's greatest pedagogical resource. The historical arc has moved from linguistic assimilation toward a rights-based, pedagogically grounded recognition of India's multilingual reality (Nag, 2023; Mohanty, 2023; Panda, 2021; Skutnabb-kanga and Phillipson, 2023).

FIGURE 2.1
Benefits of mother-tongue-based and multilingual pedagogy

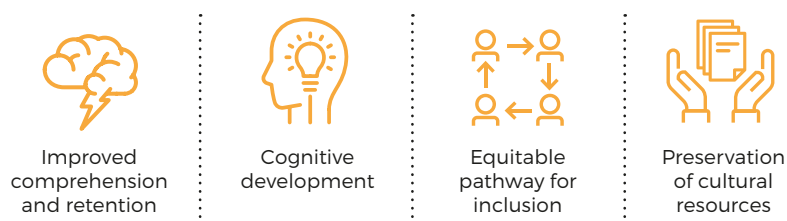


FIGURE 2.2

India's framework for mother-tongue-based multilingual education**1****CONSTITUTIONAL FOUNDATIONS****Articles 29(1) and 30(1)**

Safeguard the rights of linguistic and cultural minorities to preserve and promote their languages

Eighth Schedule

Recognizes 22 scheduled languages; encourages inclusion in education and administration

Articles 345, 347, 350, 350A, 350B

- » Empower states to designate official languages
- » Ensure citizens' rights to communicate in their mother tongues
- » Mandate mother-tongue-based instruction at the primary level

2**POLICY MILESTONES**

Year	Policy / commission	Key recommendations for language-in-education
1964–66	Kothari Commission	Advocated the Three-Language Formula for linguistic equity and national integration
1968, 1986	National Policy on Education (NPE)	Endorsed mother-tongue-based primary education; promoted regional languages
2009	Right of Children to Free and Compulsory Education (RTE) Act, Section 29(2)(f)	Recommended the mother tongue as the medium of instruction wherever feasible
2020	National Education Policy (NEP)	Advocates for the mother tongue/home language/local language as the medium of instruction up to Grade 5, preferably till Grade 8 and beyond; supports bilingual resources and bridge strategies

3**INSTITUTIONAL AND PROGRAMMATIC SUPPORT**

• **National Curriculum Framework for Foundational Stage (2022) and NCF for School Education (2023)**

Embeds multilingual pedagogy and local-language learning resources

• **National Initiative for Proficiency in Reading with Understanding and Numeracy (NIPUN Bharat (2021)**

Strengthens foundational literacy and numeracy through multilingual approaches

• **Samagra Shiksha**

Integrates teacher training, resource creation and information and communication technology (ICT) for multilingual classrooms.

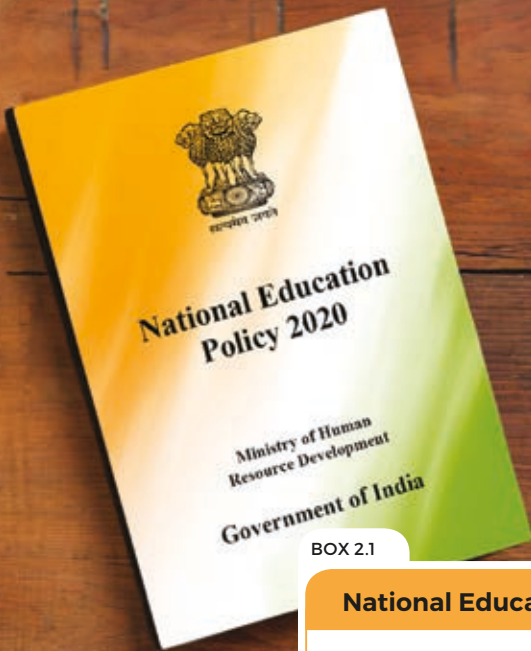
• **Digital ecosystem**

- » **Digital Infrastructure for Knowledge Sharing (DIKSHA)** – open platform for multilingual open educational resources (OERs)
- » **PM eVIDYA** platform – for inclusive learning
- » **BHASHa Interface for India (BHASHINI)** – AI-driven language translation and digital access in Indian languages

4**CORE PRINCIPLES OF INDIA'S MLE FRAMEWORK**

- **Equity.** Ensure inclusion of linguistic minorities
- **Cultural continuity.** Preserve Indigenous knowledge systems
- **Cognitive development.** Build learning through the language a child understands best
- **Bridging strategies.** Smooth transition from mother tongue/home language/local language to regional/national languages

Source: NMRC, JNU



BOX 2.1

National Education Policy 2020: multilingualism and the power of language

4.11. Children learn best in their home or mother tongue; schools should use it as the medium of instruction till at least Grade 5 (preferably Grade 8) and encourage bilingual methods where needed.

4.12. Early exposure to multiple languages enhances cognition; teaching should begin at the foundational stage through play-based, interactive and bilingual approaches.

4.13. The Three-Language Formula continues with flexibility; at least two of the three languages must be Indian, and no language will be imposed on any state.

4.14. Bilingual textbooks for science and mathematics will help students think about and express concepts in both their mother tongues and in English.

4.15. Mastery of one's own language and culture strengthens intellectual and national development; India's linguistic heritage is a source of pride and identity.

4.16. Students (Grades 6 to 8) will undertake a joyful project on "Languages of India" to learn about linguistic unity, diversity and cultural richness.

4.21. Language teaching will be enriched through experiential and cultural methods like storytelling, theatre, films and music, supported by digital tools.

4.22. Indian Sign Language will be standardized nationwide, with curriculum materials developed, while respecting and including local sign languages.

Source: NEP 2020, pp. 13-15



National Curriculum Framework for Foundational Stage 2022

The National Curriculum Framework for Foundational Stage (NCF-FS) 2022 highlights that



the use of the home language or mother tongue of the child is the most effective medium of teaching and learning at the foundational stage (NCERT, 2022, p. 12).



It recognizes that learning in the mother tongue



builds a strong foundation for conceptual understanding, self-expression, and cognitive growth (p. 13).



The framework further notes that multilingualism is a natural feature of Indian classrooms and that



children's languages should be viewed as assets and used as resources for learning new languages and concepts (p. 15).



By encouraging multilingual and translingual practices and teacher preparedness for multilingual settings, the NCF-FS positions mother-tongue-based multilingual education as central to achieving joyful, inclusive and equitable foundational learning for all children.



National Curriculum Framework for School Education, 2023

The National Curriculum Framework for School Education (NCF-SE) 2023 asserts that

“

children learn and comprehend best in their home language or mother tongue,

”

positioning multilingualism as a “rich resource for learning” rather than a barrier (NCERT, 2023, pp. 14, 18).

In line with the NEP 2020, it recommends that the

“

mother tongue, home language, or regional language be used as the medium of instruction up to at least Grade 5, but preferably till Grade 8 and beyond (p. 17).

”

FIGURE 2.3

Learning Standards

1

LEARNING THREE LANGUAGES

The NCF-SE aims to enable all Indian students to learn at least three languages, fully leveraging India's sociocultural context and resources. India's languages are among the

country's greatest heritages. This provides Indian students with a remarkable opportunity to learn multiple languages when complemented by the education system.

Language	Description	Purpose
R1 The mother tongue / most familiar language	This is the language in which literacy is first acquired in school and the language that the student already knows. This enables the full use of the linguistic and cultural knowledge (and resources) from one's home and community, which the student already has	• Deeper engagement and relevance • Effectiveness and efficiency in attaining literacy
R2 Second language	R2 is the state language, which would be the second-most-familiar language	• Enabling literacy in R2 • Understanding cultural richness
R3 Third language	Any language other than R1 and R2	• Expanding communication • Building multiple cognitive capacities

2

CORE ASPECTS

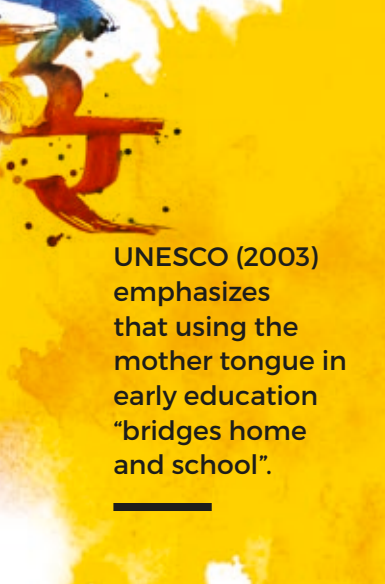
- At least two of R1, R2 and R3 must be native to India.
- Choice of R1, R2 or R3 to be determined by the state or other relevant bodies.
- An Indian language must be available to students as medium of instruction throughout school education (till Grade 12).

3

PRACTICAL CONSIDERATIONS

- Diversity of mother tongues in a class/school/ community/region.
- The complexity and difficulty of developing written resources for languages spoken by relatively small populations.
- There may be regions in a state where the linguistic diversity is such that the most familiar language for students (after their mother tongues) is another widely spoken local language, which may also be different from the state language. In such circumstances, if the mother tongue cannot be R1, then the most familiar local language may be used as R1.

Source: NCF-SE, 2023, Section 2.4 Box 2.2



UNESCO (2003) emphasizes that using the mother tongue in early education “bridges home and school”.



Above: Classroom bulletin board showcases paintings by students and their descriptions in their mother tongues.. Sardar Patel Vidyalaya, Delhi.

International perspectives on mother-tongue-based and multilingual education

UNESCO's role in promoting multilingual education

UNESCO has consistently championed mother-tongue-based multilingual education as a cornerstone of inclusive, equitable and quality learning. Beginning with its 1953 landmark report *The Use of Vernacular Languages in Education* (UNESCO, 1953), the Organization established the foundational principle that learners understand and learn best when taught in their first language. This stance was reaffirmed in subsequent global frameworks such as Education for All (EFA), Sustainable Development Goal (SDG) 4, and the Education 2030 Framework for Action, which collectively position language as a vehicle for inclusion and lifelong learning. UNESCO (2003a) emphasizes that using the

mother tongue in early education “bridges home and school”, enhances cognitive development and preserves cultural identity. More recently, UNESCO's policy guide, *Mother Tongue Matters: Local Language as a Key to Effective Learning* (2008), articulates that language choice in education is not merely a pedagogical issue but also a matter of social justice and human rights, underscoring that linguistic inclusion is essential for achieving SDG 4 and its vision of education that “leaves no one behind”. In 2003, UNESCO adopted the Recommendation concerning the Promotion and Use of Multilingualism and Universal Access to Cyberspace, providing a normative framework to ensure that all languages are represented in digital spaces and that linguistic diversity is fully integrated into knowledge and communication technologies.

BOX 2.2

The new UNESCO guidance (2025)

RECOMMENDATIONS FOR PRACTICAL ACTIONS TO IMPLEMENT MOTHER-TONGUE-BASED MULTILINGUAL EDUCATION

- Collect sociolinguistic and educational data to inform policy design and resource allocation.
- Formalize political commitments to multilingual education through supportive policies and legislation.
- Integrate multilingual education into curricula from early grades, prioritizing mother-tongue-based instruction.
- Develop learning materials in learners' languages and align assessment strategies with multilingual education principles.
- Recruit teachers fluent in both the students' mother tongue and the official language of instruction and provide training to deliver engaging multilingual education adapted to specific contexts.
- Ensure the meaningful engagement of parents, caregivers and local communities, including Indigenous peoples, to design and deliver multilingual education programmes that build on the knowledge, skills and opportunities to succeed in today's multilingual and interconnected world.

Source: UNESCO, 2025

Non-state partners play a vital role in advancing mother-tongue-based multilingual education.

Migration, poverty and multilingual education

UNESCO and UNICEF emphasize that migration-induced linguistic diversity is one of the defining educational challenges of the twenty-first century, requiring systems to move beyond assimilationist models toward inclusive multilingual pedagogies (UNESCO, 2021b; UNICEF, 2019). In many high-income countries, the initial policy responses to migrant learners – such as “language support” or “integration” programmes – often focused on rapid acquisition of the dominant national language, assuming linguistic homogeneity as the norm. However, contemporary research in sociolinguistics and education (Cummins, 2018; García and Wei, 2014; Busch, 2017) argues that such approaches disregard the dynamic repertoires migrant children bring with them, which include hybrid and transnational language practices. Recognizing these repertoires as cognitive and cultural assets, rather than deficits, has become central to multilingual education frameworks adopted in Europe, North America and Australasia, where “translanguaging” and

“plurilingual” pedagogies now inform curricula and teacher training (Cenoz and Gorter, 2017).

Non-state partners for mother-tongue and multilingual education

Non-state partners such as non-governmental organizations (NGOs), community-based organizations and social enterprises play a vital role in advancing mother-tongue-based multilingual education by reaching learners often excluded from public systems. The *Global Education Monitoring Report* (UNESCO, 2021a) reiterates “If cost-efficient practices exist, they should be diffused throughout the education system and practised by all schools, state and non-state alike” (p. 3). It emphasizes that the efforts of non-state partners must align with state policies; share innovative multilingual practices; and strengthen public accountability for equitable learning. It states that “Efficiency and innovation, rather than being commercial secrets, should be diffused and practised by all” (p. xi), underscoring the collaborative spirit essential for inclusive education.



Right: Child reading in her mother tongue, Marathi. The book is published by Pratham Books.

Key concepts of multilingual education

Multilingual education has evolved globally from a narrow focus on language subjects to a comprehensive, learner-centred framework that treats language as both epistemic resource and cultural capital. Across regions, from Africa and Southeast Asia to Latin America and Europe, successful multilingual education systems map local linguistic ecologies, identify learners' strongest languages at entry, and sequence additional languages in ways that deepen comprehension rather than fragment it. International research emphasizes that multilingual education is not a label but a pedagogical design: it requires careful alignment of curricula, materials, teacher preparation and assessments so that each new language builds on existing linguistic knowledge (UNESCO, 2003; Heugh, 2019; Hornberger, 2003). In these contexts, models such as mother tongue education (MTE), mother-tongue-based multilingual education (MTB-MLE), bilingual/biliteracy programmes, translanguaging pedagogy, and the continua of biliteracy provide flexible pathways for learners to mobilize their full repertoires and transfer knowledge across languages.

India's multilingual education landscape mirrors this global evolution while responding to its own exceptional linguistic diversity. With over 1,000 mother tongues and highly localized language ecologies, Indian models of MLE emphasize planned progression, beginning with the learner's mother tongue/first language (L1) as the

medium of instruction, and gradually introducing state/regional languages (L2) and English or other Indian languages (L3). Frameworks such as MTB-MLE (Mohanty et al., 2009), multilingual and translingual pedagogies (Panda 2022 and 2025; García and Wei, 2014), additive multilingualism (Cummins, 2000; Mohanty, 2023), and sequential progression supported by bi/multiliteracy practices have shaped both classroom experimentation and state-level initiatives. Theoretical models, such as Hornberger's continua of biliteracy and Heugh's (2021) concept of transknowledging, are increasingly used in India to understand how learners draw on home, community and school languages to construct meaning.

Mother tongue education (MTE)

Mother tongue education refers to any form of schooling that uses children's first or native language as the primary medium of instruction.

Mother-tongue-based multilingual education

Mother-tongue-based multilingual education begins in the learner's first language, using it as the primary medium of instruction to build strong conceptual and linguistic foundations before introducing additional languages in an additive, multilingual framework.

Below: Student writes in Tamil in a language class, Sardar Patel Vidyalaya, Delhi.

With over 1,000 mother tongues, Indian models of MLE emphasize planned progression.





Above: Student learns Odia alphabets. Kalinga Institute of Social Sciences, Bhubaneswar.

Code-switching and code-mixing

Code-switching is alternating between languages or dialects across sentences/clauses, depending on the context, audience or topic (Poplack, 1980): “मैंने homework कर लिया, now I just need to review it before submission.”

Code-mixing blends elements (words/phrases/morphemes) from different languages within a single sentence (Muysken, 2000; Shridhar and Shridhar, 1980): “मुझे assignment submit करना है urgently.”

Translingual pedagogy

Translingual pedagogy enables learners to mobilize their full repertoire within tasks to secure meaning, then consolidate it in the target language (e.g. brief peer explanations in L1, bilingual anchors for key terms, and structured movement back to the target language) (Canagarajah, 2011 and 2022; Panda and Bhagat, 2017).

Translanguaging

Translanguaging refers to the dynamic and fluid use of multilingual resources to create meaning and learn, without rigid separation of languages (García and Wei, 2014).

Bi/multiliteracy

Bi/multiliteracy refers to proficiency in reading and writing in two or more languages, encompassing the ability to navigate diverse textual and cultural conventions (Cummins, 2007).

Continua of biliteracy

Continua of biliteracy refers to literacy development across multiple continua rather than as a static binary (Hornberger, 2003). The continua comprises contexts (formal schooling and informal home/community learning), varieties/registers (dialects and styles), skills/strategies (decoding, comprehension, writing, oral expression, symbol navigation), cultural-

social practices (identities, traditions, narratives) and modalities/media (print, digital, visual, oral).

Multi/ translingual pedagogy

Multilingual/translingual pedagogy recognizes the fluid, interconnected use of multiple languages in learning, both spontaneous and deliberate (Panda, 2025).

Transknowledging

Transknowledging is an integrative process whereby learners weave linguistic, cultural and experiential resources across languages to make meaning and engage critically (Heugh, 2021).

Additive versus subtractive approaches

Additive approaches build upon L1 alongside additional languages, enhancing cognition, identity and achievement (Cummins, 2000). Subtractive approaches replace or devalue L1 with a dominant L2, thereby risking language loss, erosion of identity and reduced outcomes (Skutnabb-Kangas, 2000; Mohanty, 2019; Annamali, 2001).

Simultaneous and sequential MLE

SIMULTANEOUS MLE

This refers to a multilingual education approach whereby two or more languages are introduced and used as mediums of instruction from the very beginning of schooling, allowing learners to develop proficiency in multiple languages concurrently.

SEQUENTIAL MLE

Sequential MLE introduces languages in planned progression, beginning with the learner's mother tongue as the primary medium and gradually adding regional, national or international languages at later stages, to ensure conceptual clarity before linguistic transition.

Recent intervention research in India underscores that the strongest forms of multilingual education, though drawing on different models, are systematic and additive; they use the mother tongue as a foundation, strengthen higher-order thinking and support learners' transition to additional languages without erasing their linguistic identity (Mohanty et. al.).

Facing page: Student translates Hindi and English to Urdu in class. Sardar Patel Vidyalaya, Delhi.

Below: A book composed in the Saora language. Government Upper Primary School, Gajapati.





Summary

MTB-MLE involves the planned use of multiple languages in education, starting with the learner's mother tongue and gradually introducing regional, national and international languages.

In India's highly diverse linguistic context, mother-tongue-based education improves comprehension, literacy and classroom participation, especially for tribal and marginalized learners.

National policies, from the Kothari Commission to NEP 2020, along with the NCF-FS 2022, NCF-SE 2023 and NIPUN Bharat, emphasize the mother tongue/home/local language as the medium of instruction up to at least Grade 5 and promote bilingual materials and multilingual teacher preparation.

MTB-MLE is globally recognized as a rights-based, inclusive and equitable educational approach.

Pedagogical strategies such as multilanguaging/translanguaging, biliteracy and additive multilingualism enhance cognitive flexibility, affirm identity and improve learning outcomes, framing multilingualism as a basis for educational equity and social justice.



Students stand in front of the district map of Telangana.
Model High School, Osmania University, Hyderabad.



CHAPTER

3

Linguistic landscape of India

What does it mean to live in a country where nearly every village, city and street corner speaks differently?

To read its linguistic map is to read the pulse of its diversity and its inequalities.

India's linguistic diversity is the backbone of its ecological wisdom.



3

Linguistic landscape of India

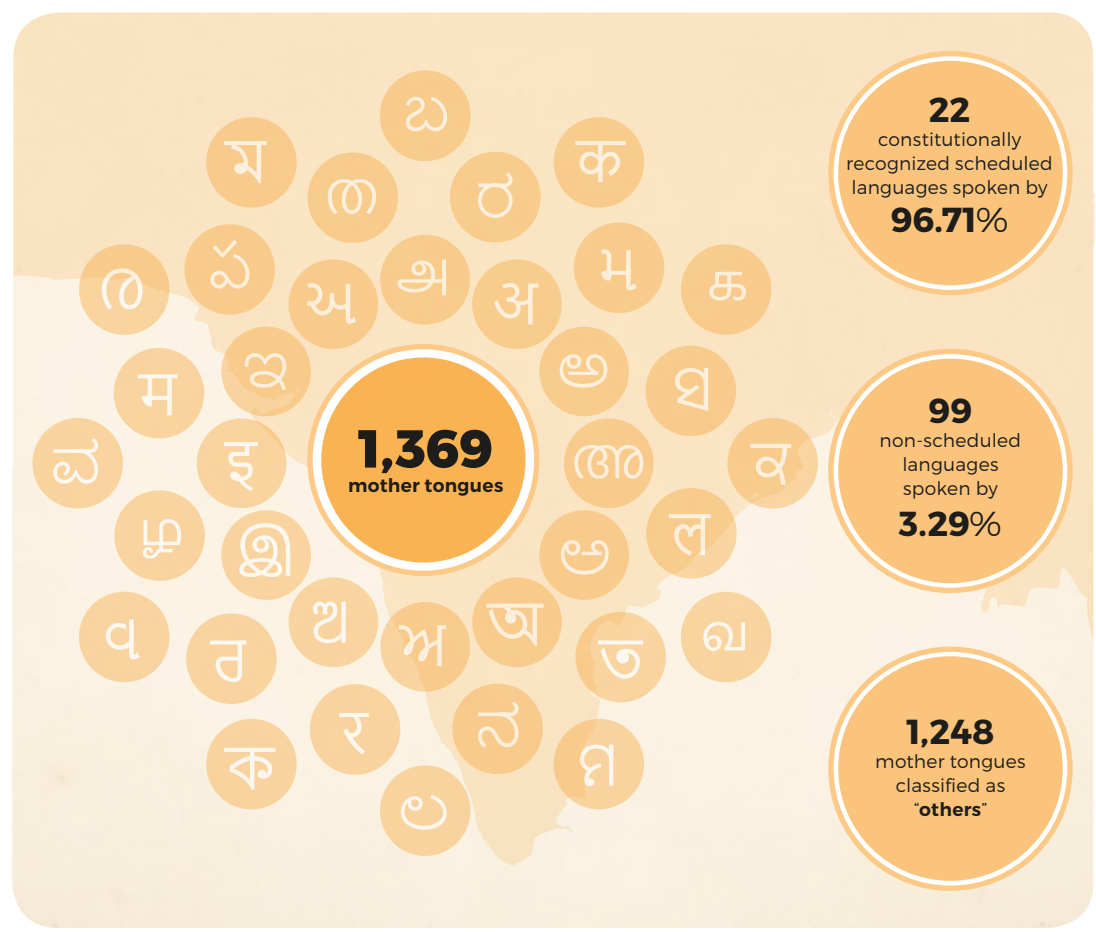
Linguistic demographic profile

India is home to 1,369 mother tongues, of which 121 are spoken by over 10,000 people. Among these, the 22 constitutionally recognized languages in the Eighth Schedule of the Indian Constitution are spoken by 96.71 per cent of the population, while the remaining 3.29 per cent – around 37.8 million people – speak one of 99 non-scheduled languages and 1,248 mother tongues classified as “others”. These smaller, often marginalized communities display high multilingualism and fluid identity affiliations, illustrating dynamic,

adaptive patterns of language use across diverse social domains (Census of India, 2011; Office of the Registrar General and Census Commissioner, 2011).

India is the most populous country in the world and also one of the most linguistically and culturally diverse. The Linguistic Diversity Index (LDI) 2025, which measures the probability that two randomly selected individuals speak different mother tongues, found the probability at 0.914 in India, one of the highest in the world.

FIGURE 3.1
Linguistic profile of India





Above: Children learn the alphabet through activities. Kalinga Institute of Social Sciences, Bhubaneswar.



India is one of the most linguistically and culturally diverse country

0.914

is the Linguistic Diversity Index value of India, which is one of the highest in the world.

TABLE 3.1
Linguistic Diversity Index by country

Country	Linguistic Diversity Index 2017
Papua New Guinea	0.988
Cameroon	0.974
Vanuatu	0.973
Solomon Islands	0.968
Central African Republic	0.959
Democratic Republic of the Congo	0.948
Chad	0.933
Benin	0.933
Uganda	0.929
South Sudan	0.929
Kenya	0.927
Mozambique	0.926
Liberia	0.917
India	0.914
Togo	0.905

Source: World Population Review (<https://worldpopulationreview.com/country-rankings/linguistic-diversity-index-by-country>)

TABLE 3.2
Language families in India

Language family	Languages	Total speakers (approximately)
Indo-Aryan	Hindi, Odia, Bengali, Marathi, Gujarati, Punjabi	78%
Dravidian	Telegu, Tamil, Kannada, Malayalam	20%
Austro-Asiatic	Santali, Mundari, Birhor, Ho, Korku	1.2%
Tibeto-Burman	Bodo, Mizo, Khasi, Nyishi, Ao, Garo, Manipuri, etc.	0.8%

Source: Census, 2011

India's languages belong to four major genealogical families. Indo-Aryan languages account for about three-quarters of all speakers; Dravidian languages around 20 percent; Austro-Asiatic approximately 1.2 percent; and Tibeto-Burman about 0.8 percent, with a few isolates such as Nihali and Great Andamanese. Hindi is the first language of 43.6 percent of the population, followed by Bengali (8.3 percent), Marathi (6.9 percent), Telugu (6.7 percent) and Tamil (5.7 percent). Thirty languages each have over 1 million first-language speakers, while hundreds of languages survive only in small, often remote communities.

India's linguistic diversity often mirrors inequality. Scheduled languages enjoy constitutional recognition and institutional visibility in education, administration and media. By contrast, many non-scheduled languages – spoken by smaller or geographically isolated communities – receive little official support. Their exclusion from education and governance reduces their symbolic and material value, hastening language shift and loss of intergenerational transmission. Ensuring that linguistic diversity translates into educational inclusion remains a central policy challenge.

Below: Students outside their classroom. Government Upper Primary School, Gajapati.





Above: A student reads a Hindi-Urdu dictionary. Model High School, Osmania University, Hyderabad.

Language endangerment and the “endangerment-support paradox”

UNESCO's *Atlas of the World's Languages in Danger* (2010) identifies close to 200 Indian languages as vulnerable or endangered, with several dozen classified as critically endangered. Among the critically endangered languages are Toto (a Tibeto-Burman language of West Bengal) and the Great Andamanese languages of the Andaman Islands, each with only a few elderly speakers left.

TABLE 3.3
Endangerment-support paradox

Language family	Endangered (%)
Indo-Aryan	14%
Dravidian	22%
Austro-Asiatic	52%
Tibeto-Burman	73%

Source: UNESCO 2010

Analyses of UNESCO's *Atlas* and Indian language surveys show a clear inverse relationship between a language's speaker population and its risk of endangerment. India's smallest language families, especially Tibeto-Burman, Austro-Asiatic and the Andamanese group, contain a disproportionately high number of endangered and critically endangered languages. In contrast, the larger Indo-Aryan and Dravidian families, spoken by the majority of the population, have comparatively fewer endangered languages. This pattern reflects deeper structural vulnerabilities: smaller Indigenous and tribal languages often lack institutional support, digital presence, official recognition, and access to education or media in the mother tongue, making them more susceptible to decline (UNESCO, 2010; Abbi, 2006 and 2008; PLSI, 2013–2022).



Above: A student reads a textbook developed in the Saora language. Government Upper Primary School, Gajapati.

Language documentation and revitalization efforts

The Scheme for Protection and Preservation of Endangered Languages (SPPEL), launched in 2013 and implemented by the Central Institute of Indian Languages (CIIL) is a flagship programme of the Government of India. The SPPEL targets non-scheduled and endangered languages, particularly those with fewer than 10,000 speakers. Currently, the SPPEL has identified 117 endangered languages and is working towards documenting approximately 500 lesser-known languages. The programme includes the development of bilingual and trilingual dictionaries, primers, pictorial glossaries, grammatical sketches and ethnolinguistic profiles, accompanied by high-quality audio and video recordings.

The CIIL collaborated with the National Council of Educational Research and Training (NCERT) and produced primers in 117 endangered languages, many of which are indigenous to tribal communities. These are archived in the CIIL's digital repository, Sanchika, which was launched in July 2025. Sanchika now hosts hundreds of samples and audio files of endangered languages, drawing on fieldwork that is community-inclusive, with native speakers participating in both data collection and material review.

The Bharatiya Bhasha Samiti (BBS), established by the Ministry of Education in November 2021, is a high-powered committee tasked with advancing Indian languages. Its mandate is to chart pathways aligned with the National Education Policy (NEP) 2020 by advising on revitalizing

language teaching, developing terminology, and ensuring linguistic compatibility with modern technologies. The committee focuses on strengthening the Three-Language Formula, creating instructional materials in Indian languages, and enhancing their visibility in education and public life. Its flagship initiative ASMITA (Anuvad, Shikshan, Margdarshan, Inclusive, Takniki, Anusandhan), a major national effort to develop high-quality textbooks and learning materials in all twenty-two scheduled languages, seeks to dismantle long-standing linguistic hierarchies in higher education by ensuring that students across disciplines, engineering, medicine, sciences, social sciences, law and humanities, can access core academic content in their mother tongues. Aligning with the NEP 2020's vision of multilingual and equitable higher education, this initiative mobilizes a wide network of scholars, translators, subject experts and digital production teams to create linguistically accurate, pedagogically sound and culturally relevant materials.

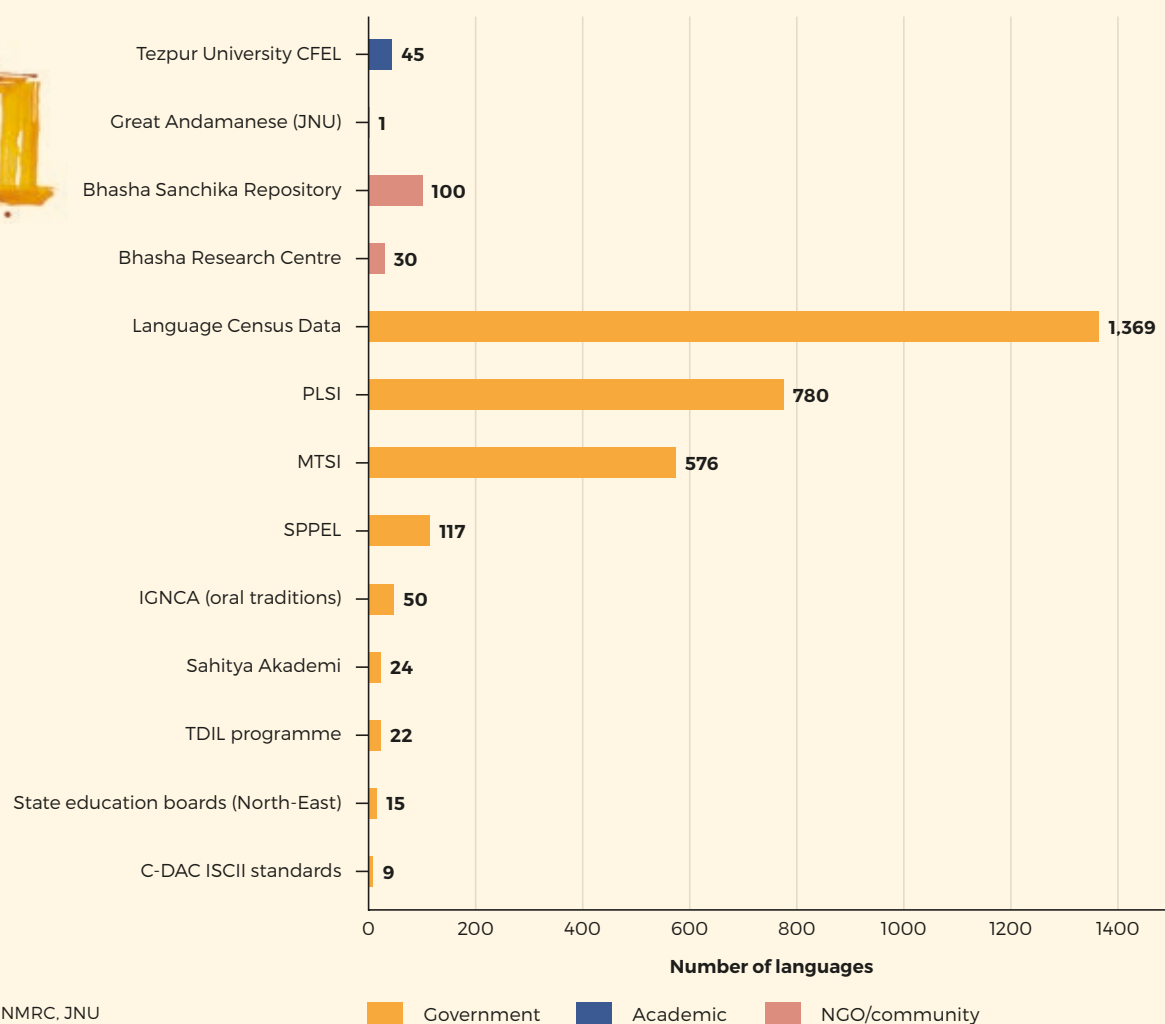
The Mother Tongue Survey of India (MTSI), completed in 2022 by the Office of the Registrar General of India, represents more than a linguistic census – it is a visual linguistic archive aiming to preserve language features and oral testimonies, forming the foundation of India's first video atlas of languages. The 576 languages videographed through this initiative reflect centuries of mobility, contact and adaptation, revealing how linguistic forms travel, transform and carry embodied knowledge systems across regions.

Academic and community-led efforts

Outside government channels, many universities and non-governmental organizations (NGOs), such as the People's Linguistic Survey of India (PLSI), Jawaharlal Nehru University (JNU) and

the English and Foreign Languages University (EFLU), Hyderabad, have contributed to documenting and nuancing the linguistic landscape of India.

FIGURE 3.2
Major language documentation initiatives in India
Number of languages documented or covered (1950s-2025)



Source: Compiled by NMRC, JNU

Community and grassroots efforts, such as the Native Tongue Title movement in Australia,² have enabled tribal youth to record oral histories and create YouTube content. Similarly in India, organizations like the Bhasha Centre in Vadodara have developed trilingual dictionaries and textbooks in collaboration with communities. State education boards in the North-East have co-developed orthographies for languages such as Mizo and Kokborok.

There are initiatives supported by the Sahitya Akademi, which organizes meetings of tribal writers and publishes dictionaries, grammars and folklore. The Indira Gandhi National Centre for the Arts (IGNCA) documents oral traditions and rituals. The Anthropological Survey of India's "People of India" project (1985-92) remains a foundational ethnographic and linguistic resource.

² <https://researchportalplus.anu.edu.au/en/publications/native-tongue-title-compensation-for-the-loss-of-aboriginal-langu/>

1

Reviving Gondi and Kolami through artificial intelligence

The work of Thodasam Kailas



Location: Adilabad district, Telangana.



Duration: 2001–present.



Objective: To preserve and promote endangered tribal languages such as Gondi and Kolami by using digital tools and artificial intelligence (AI) for education, music and communication, thereby inspiring tribal youth to value and learn in their mother tongues.



Focus areas: Language revitalization, educational technology, and community engagement.



As a government school teacher in a multilingual region, Thodasam Kailas observed that Gondi and Kolami-speaking students struggled when instruction occurred only in Telugu or English. To address this gap, he began integrating mother-tongue-based multilingual education (MTB-MLE) principles into classroom practice, using the child's home language as the foundation for concept formation and gradually bridging to state and national languages.

Kailas designed lessons where students could first discuss and understand new topics in Gondi or Kolami, then relate them to terms in Telugu and English. He created bilingual and trilingual learning aids, local story-based materials, and audio-visual resources that validated students' languages in the formal learning space. His approach mirrors the core MLE sequence of L1 → L2 → L3, promoting comprehension and confidence.

To make mother-tongue learning sustainable, Kailas harnessed digital and AI-based tools. He developed Sungal Thorso, an AI news anchor that delivers news in Gondi, proving that tribal languages can exist in digital domains. He also produced hundreds of songs, short videos and educational clips in Gondi, Kolami, Hindi and English, which he shares through his YouTube channel and school sessions. These multimedia resources extend the reach of MTB-MLE

beyond the classroom, showing children that their languages have technological and cultural value.

Kailas's work demonstrates how teacher-led innovation can translate the philosophy of MTB-MLE – learning that begins in the mother tongue and connects to broader linguistic worlds – into day-to-day practice.

PROCESS AND IMPLEMENTATION

- Classroom innovation.** Introduced bilingual explanations, local-language visuals, and audio aids for social-science and science lessons.
- Digital resource creation.** Purchased and used a projector and Tutor Plus device to display PowerPoint slides, AI-generated songs and Gondi-language materials.
- AI-based content.** Developed Gondi AI news anchor "Sungal Thorso" and produced multilingual educational songs.
- Broadcast and outreach.** Provided Gondi-language public-awareness messages through All India Radio Adilabad; uploaded educational videos for rural students.
- Community mentoring.** Conducted spoken-Gondi programmes for non-Gondi speakers.

ACHIEVEMENTS AND IMPACT

- Strengthened MTB-MLE practice.** Enabled tribal children to learn through Gondi and Kolami while building bridges to Telugu and English.
- Digital inclusion of tribal languages.** Created the first AI-generated Gondi news anchor and digital learning content.
- National recognition.** Praised by the Prime Minister in the Mann ki Baat radio programme and honoured by former Vice President M. Venkaiah Naidu.
- Cultural preservation.** Authored *Pandokna Mahabharat* (2024) and *Kandirang Vesuding* (2019), documenting tribal narratives for classroom and radio use.
- Educational innovation.** Improved student participation and curiosity by connecting local culture with technology.

KEY PEOPLE AND ORGANIZATIONS INVOLVED

Thodasam Kailas, teacher-innovator, MPUPS, Gourapur, with support from **District Collector**, Adilabad; **All India Radio**, Adilabad, broadcast partner for Gondi programmes; **Government of India** – public acknowledgment through *Mann Ki Baat*.

Indigenous and tribal language, biodiversity and human rights

Linguistic diversity and biodiversity are closely linked (Skutnabb-Kangas, 2000). The tribal and Indigenous languages of India embody ecological knowledge accumulated through generations, preserving practices of sustainable resource use and environmental management. The loss of such languages represents not only cultural erosion but also the disappearance of localized environmental understanding (Abbi, 2006).

UNESCO has long recognized linguistic rights as integral to human rights (the Universal Declaration of Linguistic Rights, 1996; UNESCO, 2003). Skutnabb-Kangas and Phillipson (2023) identify the denial of mother-tongue-based education as a form of linguistic discrimination, with social and ecological consequences.

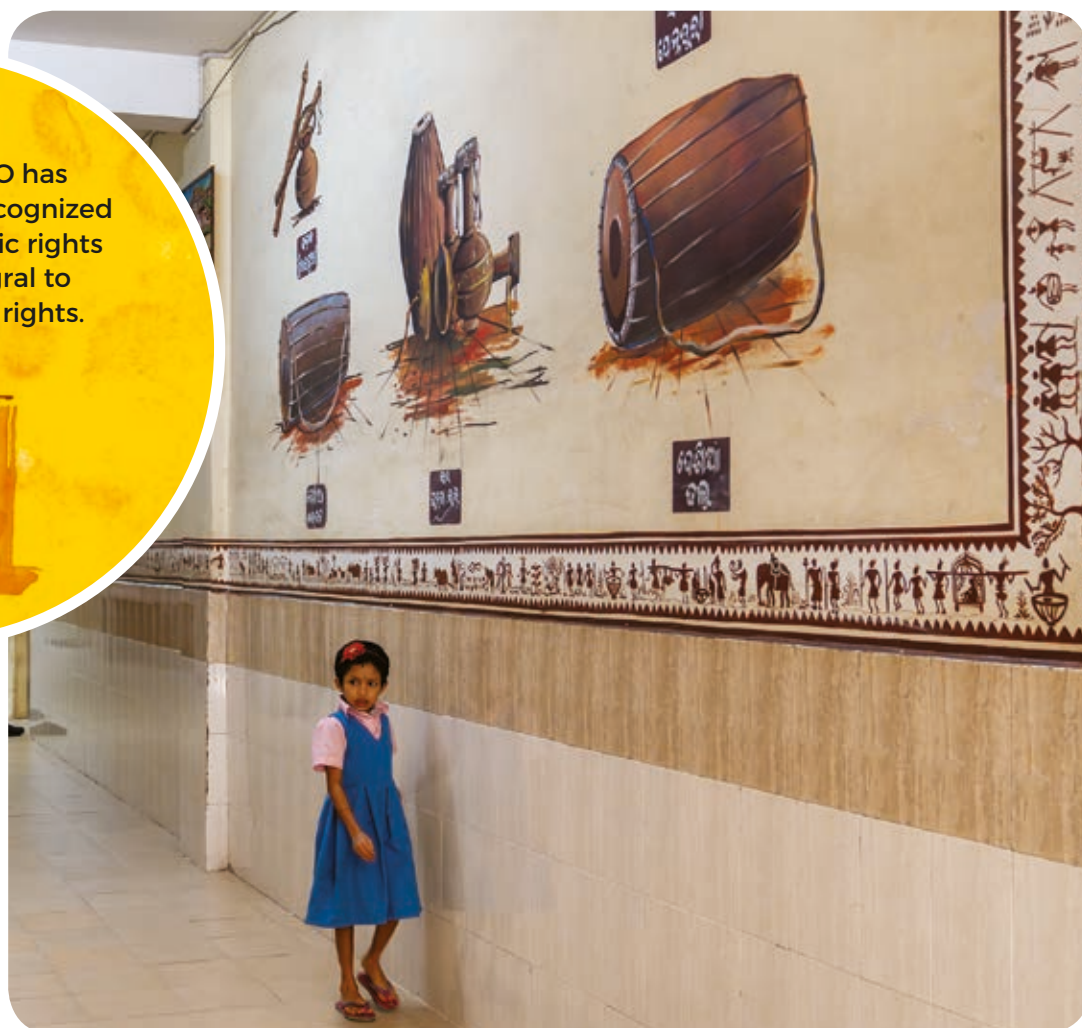
The Mother Tongue Survey of India, completed in 2022, goes beyond a linguistic census to

document how India's 576 recorded languages embody migration, knowledge and cultural exchange. It reveals how linguistic forms evolve through movement and contact, challenging the notion of languages as static and instead portraying them as dynamic systems of lived knowledge. Migration and economic mobility reshape languages, sometimes causing loss, as local terms fade in new contexts, but also fostering **transknowledging**, where people blend vocabularies and practices to create shared meaning. Acts of revival, such as Dhaniram Toto's development of the Toto script in West Bengal, demonstrate how communities transform linguistic decline into renewal, aligning with the International Decade of Indigenous Languages (2022–2032) in sustaining India's linguistic diversity as a living, adaptive archive of human experience.



UNESCO has long recognized linguistic rights as integral to human rights.

Right: A student stands in the corridor, surrounded by paintings of musical instruments labelled in the mother tongue. Kalinga Institute of Social Sciences, Bhubaneswar.





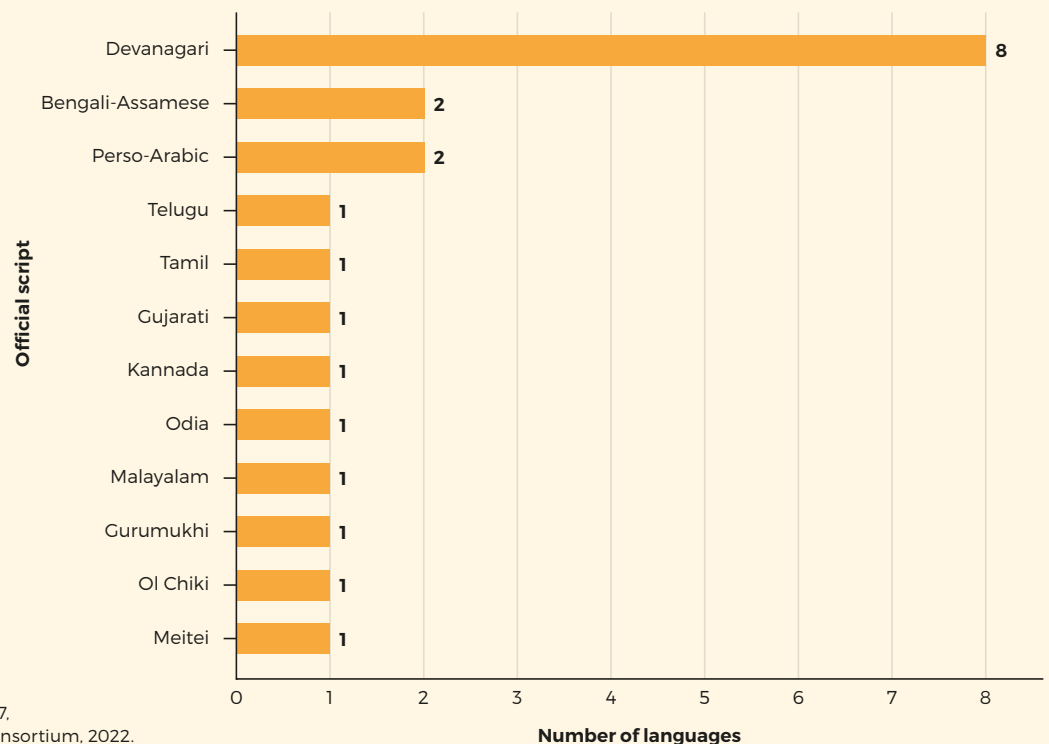
Right: Students learn to write Devnagari script. Model High School, Osmania University, Hyderabad.

India's complex script ecology

India's remarkable linguistic diversity extends into its writing systems. Its relationship with scripts and identities is complex and political. Census 2011 identifies 14 major scripts with significant contemporary usage: Devanagari (used by over 120 languages) Bengali-Assamese (38 languages) Tamil, Malayalam, Telugu, Kannada, Gujarati, Odia, Gurmukhi, Perso-Arabic (Urdu), Roman (growing usage, over 34 languages), Ol Chiki (Santali) and Meitei Mayek. Additionally, several smaller

scripts, including Siddham, Modi, Grantha and Sharada, continue to have limited usage in specific communities. Many major Indian scripts evolved from the ancient script Brahmi that dates to the third century BCE. Major scripts have full Unicode support, keyboard input methods and publishing infrastructures, while smaller scripts face technical implementation challenges despite their cultural significance (Unicode Consortium, 2022).

FIGURE 3.3
Indian scripts and writing systems



Source: Ministry of Home Affairs, 2017, in collaboration with the Unicode Consortium, 2022.

BOX 3.1

Status of script in India and its regional distribution**SCRIPT DOMINANCE**

- Devanagari serves over 120 languages, making it the most widely used script in India.
- Bengali-Assamese script supports 38 languages across Eastern India.
- Roman script is used for over 34 languages.
- Nine scripts serve only single languages, showing script-language specificity.

REGIONAL DISTRIBUTION

- Northern and central India are dominated by the Devanagari script.
- Each southern and eastern Indian state maintains its distinct script traditions.
- North-eastern India shows mixed script usage (Bengali, Roman, Indigenous).

Source: NMRC, JNU

Marginality and the oral-written language continuum

The oral-written language continuum feeds into the major-minor language divide in India. While major scripts benefit from standardized fonts, keyboard layouts and Unicode integration, several minor or community-specific scripts

still face technological marginalization. Such technical limitations restrict speakers' access to digital learning resources and government services, inadvertently excluding speakers of smaller languages.

While major scripts benefit from standardized fonts, keyboard layouts and Unicode integration, several minor scripts still face technological marginalization.

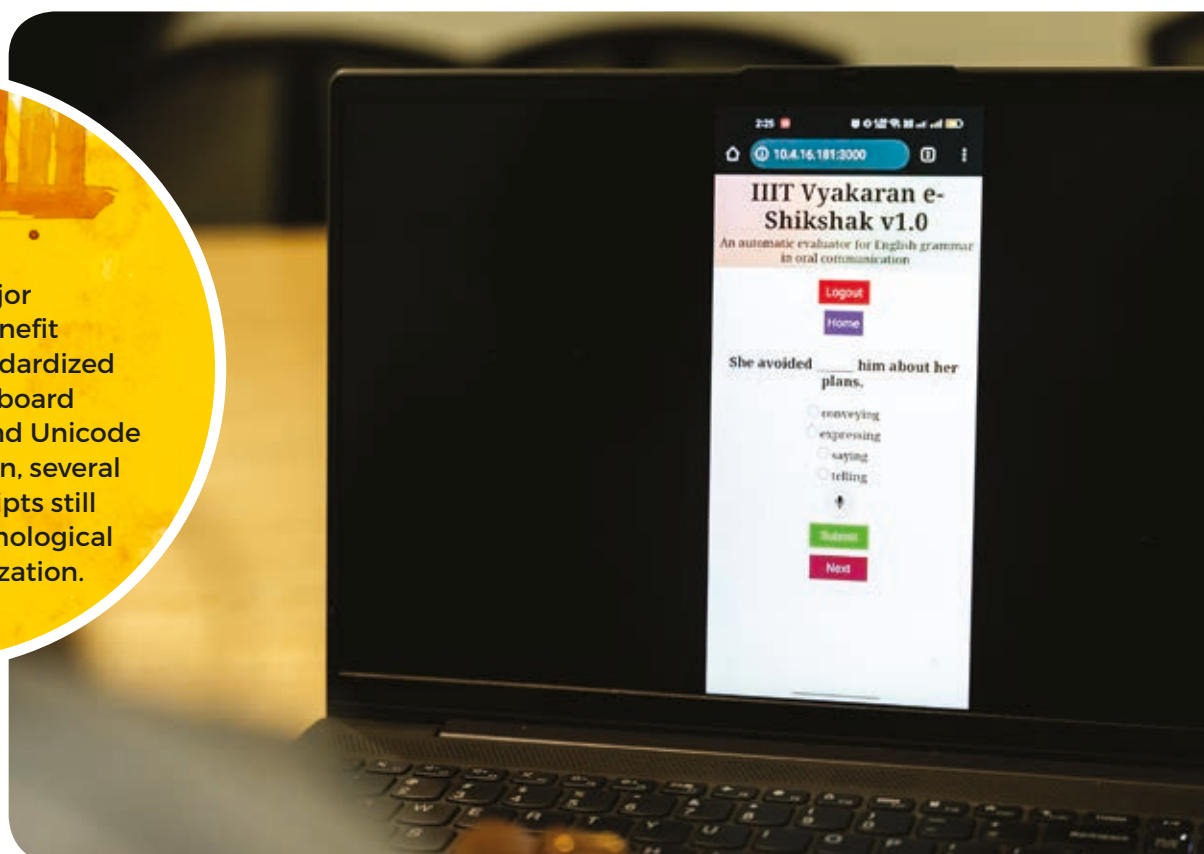


TABLE 3.4

Script-language relationships in India (contemporary use and digital readiness)

Script (Unicode block)	Major languages using the script (non-exhaustive)	Primary geographies (administration/education/media)	Indicative reach	Digital readiness (summary)
Devanagari	Hindi, Marathi, Nepali, Sanskrit, Konkani, Dogri, Bodo, Maithili	North, central and west India; also Nepal	Very high	High Mature fonts/IMEs, OCR widely available, strong NLP toolchains (tokenizers, ASR, MT)
Bengali-Assamese	Bengali, Assamese	West Bengal, Assam, diaspora	Very high	High Robust fonts/IMEs, OCR good, growing NLP/ASR resources
Gurmukhi	Punjabi	Punjab, diaspora media	High	High Stable fonts/IMEs, good OCR, improving ASR/MT
Gujarati	Gujarati	Gujarat, business networks	High	High Fonts/IMEs mature, OCR solid, decent NLP coverage
Odia	Odia	Odisha and adjacent districts	High	Medium-high Good fonts/IMEs, OCR improving, growing NLP assets
Tamil	Tamil	Tamil Nadu, Puducherry, global diaspora	Very high	High Extensive fonts/IMEs, strong OCR/ASR, rich web content
Telugu	Telugu	Andhra Pradesh, Telangana	Very high	High Mature fonts/IMEs, good OCR, expanding NLP/ASR
Kannada	Kannada	Karnataka	High	High Stable fonts/IMEs, OCR reliable, ASR/MT improving
Malayalam	Malayalam	Kerala, Gulf diaspora media	High	High Excellent fonts/IMEs, strong OCR, active NLP/ASR
Urdu (Perso-Arabic, Nastaliq/Naskh in India)	Urdu, Kashmiri (also written in Devanagari), Sindhi (also in Devanagari)	Jammu and Kashmir, Uttar Pradesh, Telangana, Delhi, press and literature	High	Medium-high fonts/IMEs solid, OCR in Nastaliq harder but improving, decent NLP
Meitei Mayek	Manipuri (Meitei)	Manipur (revived in schools/media)	Medium	Medium Unicode standardized, fonts/IMEs available, OCR/NLP emerging
OI Chiki (OI Cemet')	Santali	Jharkhand, Odisha, West Bengal, Assam (education/media growth)	Medium	Medium Unicode support stable, fonts/IMEs available, OCR/NLP early-stage
Latin (Roman)	English; also used informally for Konkani, Nagamese, tribal/online writing	Pan-India (higher education, information technology, signage, private media)	Very high	Very high Complete ecosystem (fonts, OCR, NLP, ASR/MT, search/SEO)

Source: Census of India, 2011: Language Tables, Language Atlas of India (CIIL, Mysuru); MEIT, 2019; UNESCO, 2022; and Unicode Consortium, 2022.

Note: IME = input method editor; OCR = optical character recognition; NLP = natural language processing; ASR = automatic speech recognition; MT = machine translation; SEO = search engine optimization



Above: Students studying in Saora language. Government Upper Primary School, Gajapati.

Language prestige, social hierarchies and the double divide

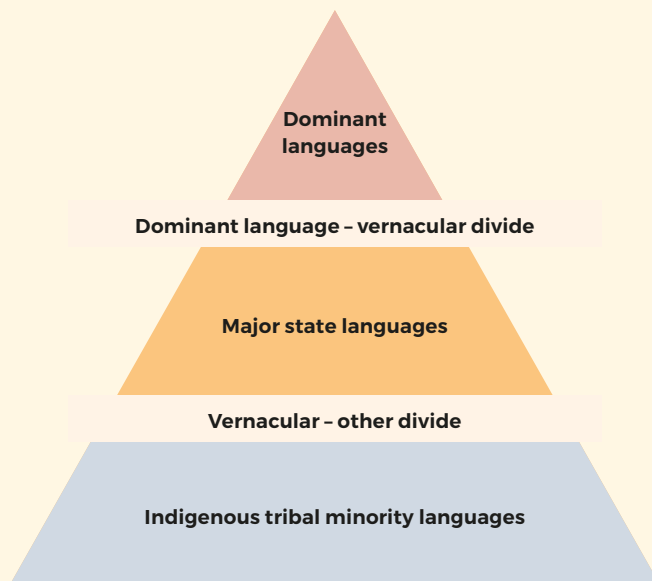
India's linguistic hierarchy places English at the apex, followed by Hindi and other dominant regional languages, with tribal and minority languages at the bottom (Mohanty, 2010; Panda and Mohanty, 2015). Meganathan (2018) describes this as a "prestige gradient", where language choices are shaped by aspirations for mobility

and social advancement. Research shows strong correlations between English proficiency, income, academic achievement and social mobility, reflecting the material power associated with high-status languages.

Panda and Mohanty (2015) conceptualize this stratification as a double divide. The first divide – English versus regional languages – is marked by English dominating education, employment and high-value domains, pushing major Indian languages into subordinate positions. The second divide – regional languages versus Indigenous tribal minority (ITM) languages – captures how dominant regional languages marginalize smaller mother tongues, excluding them from schooling, governance, media and digital spaces. This hierarchy is reflected clearly in India's school system: private schools privilege English, government schools use regional languages, and ITM languages rarely appear in textbooks, assessments or teacher education. Even where mother-tongue-based multilingual education programmes have been introduced, they face early-exit pressures that shift children quickly into dominant mediums unless backed by sustained policy and financial support.

These dynamics drive systematic language shifts, reducing the vitality of smaller languages and limiting the capabilities of the communities that speak them (Mohanty and Panda). The double divide explains persistent educational and social inequities and underscores the need for structural reform rather than short-term, project-based approaches to linguistic inclusion.

FIGURE 3.4
The double divide in multilingualism in India



Source: Mohanty, 2019

Urban multilingualism and the challenges of mediums of instruction

Facing page: Medicinal plants garden with both scientific names and names in the mother tongue. Kalinga Institute of Social Sciences, Bhubaneswar.

Below: Telugu, Urdu and English alphabets turned into a hopscotch game for interactive learning. Model High School, Osmania University, Hyderabad.

Urban India presents a complex paradox for the implementation of mother-tongue-based multilingual education. Cities are vibrant linguistic contact zones where migrants, regional-language speakers, English-medium educated youth, and speakers of smaller Indigenous languages interact daily, producing fluid repertoires marked by code-switching, translanguaging and mixed urban varieties. Yet, schooling in these spaces remains largely monolingual in practice – dominated by English in private schools and by state languages in government schools – leaving little institutional room for the diverse home languages of urban learners. For millions of first-generation migrants and speakers of non-scheduled or tribal

languages living in cities, the school language seldom matches the language of the home or neighbourhood. This creates a pedagogic contradiction: children who navigate multilingual realities outside the classroom encounter rigid language boundaries inside it. Urban MLE efforts struggle because classrooms comprise heterogeneous linguistic mixes, teachers are rarely trained to work with such diversity, and policy frameworks assume stable, territorially bound language communities rather than mobile urban populations. Consequently, urban multilingualism exposes both the necessity of MLE and the structural barriers that prevent its effective implementation in India's rapidly expanding urban classrooms.





Summary

India's linguistic landscape is one of the world's richest, with 1,369 mother tongues, yet this diversity is unevenly distributed, with scheduled languages enjoying institutional visibility while 99 non-scheduled and over 1,200 smaller mother tongues remain marginalized.

Language endangerment is severe and uneven, disproportionately affecting Tibeto-Burman, Austro-Asiatic and Andamanese languages, where lack of recognition, digital presence and educational support accelerates decline and the erosion of ecological and cultural knowledge systems.

Government-led documentation and revitalization efforts such as SPPEL, CIIL's Sanchika archive, the Mother Tongue Survey of India, and the Bharatiya Bhasha Samiti's ASMITA initiative are expanding linguistic resources, orthographies and digital materials, especially for endangered and Indigenous languages.

Academic and grassroots initiatives, including by PLSI, JNU, EFLU the Bhasha Centre and community-driven digital innovations, demonstrate how collaborative documentation and technology can preserve and strengthen vulnerable languages.

India's complex script ecology reflects major-minor hierarchies, where widely used scripts have full digital readiness, while smaller community scripts face technological exclusion, limiting access to digital learning and government services.

Linguistic hierarchies and the double divide between English and regional languages and between regional and Indigenous languages shape access to education, literacy and opportunity; while everyday multilingualism and urban linguistic fluidity highlight both the necessity and the challenge of implementing equitable MLE across India.





Children learn Odia using maps and descriptive visuals.
Government Upper Primary School, Gajapati.



CHAPTER

4

Mother-tongue-based-multilingual education:

reimagining language in curricula and pedagogy in India

When instruction and assessment are predominantly in a non-native medium, many children confront the double burden of decoding an unfamiliar language and grasping new concepts. In such contexts, mother-tongue-based education is not an optional support but the starting point of curricular reform: without it, concepts remain foreign, participation remains fragile and learning remains shallow.



4

Mother-tongue-based multilingual education: reimagining language in curricula and pedagogy in India

Language(s) in school education

The Annual Status of Education Report (ASER) 2024 underscores that many children in India continue to struggle with basic reading and numeracy tasks, and that the language of instruction contributes to these learning gaps. In rural government schools, in 2024, only 23.4 per cent of Grade 3 students were able to read a Grade 2 text, while 33.7 per cent of Grade 3 children could perform a basic subtraction task (ASER, 2024). One plausible explanation for such weak performance is a mismatch between the language of instruction and students' mother tongues/home languages. When instruction and assessment are predominantly in a non-native medium, many children confront the double

burden of decoding unfamiliar language and grasping new concepts. This linguistic barrier reduces comprehension, slows response time and leads to errors, not because students lack conceptual understanding but because they struggle with language. In effect, they may understand the content in their mother tongue but underperform when required to process it in an unfamiliar language.

Recent curriculum frameworks in India mark a decisive recognition of children's first or strongest language as central to learning, particularly in the formative years. The National Curriculum Frameworks (NCFs) for the foundational stage

Below: Assembly of students, where prayer songs are sung in multiple regional languages. Kalinga Institute of Social Sciences, Bhubaneswar.





Above: Ascending and descending order are taught through activities. Model High School, Osmania University, Hyderabad.

Teachers are not merely conveyors of knowledge but mediators who validate children's lived experiences.

(FS) and for school education (SE) (NCERT, 2022; NCERT, 2023) underscore this principle unambiguously: "Since children learn concepts most rapidly and deeply in their mother tongue/home language, the primary medium of instruction would optimally be the child's mother tongue/home language/local language (also referred to below as L1) in the Foundational Stage. This should be the approach in both public and private schools" (NCERT, 2022, Section 3.2). This statement situates the child's first language as the natural epistemic base for cognition, concept-building and early learning. It recognizes that scientific and mathematical abstractions, as well as literacy, emerge most securely when rooted in familiar linguistic repertoires.

Closely linked is the insistence on teachers' linguistic and cultural competence. The NCF-FS 2022 states: "To ensure that each child has continued proper use of their L1 when they begin at the Foundational Stage, it is essential to have teachers (e.g., from the local community) who not only understand the language but also the local culture and traditions. More than at any other Stage, teachers of the Foundational Stage should be proficient in the child's L1" (NCERT, 2022, Section 3.2). This emphasis acknowledges that effective pedagogy is inseparable from cultural familiarity and linguistic proximity. Teachers are not merely conveyors of prescribed knowledge but mediators who validate children's lived experiences, Indigenous categories and repertoires of meaning-making.

At the systemic level, the NCF for School Education (NCERT, 2023) extends this vision by situating curriculum and pedagogy within India's sociocultural fabric: "It is deeply rooted in India. This is in content and learning of languages, in the pedagogical approaches including tools and resources, and most importantly in the cultural, philosophical, and contextual underpinnings" (NCERT, 2023, p. 4). This articulation calls for rejecting homogenized templates and privileging curricular forms that are responsive to Indian languages, traditions and contexts.

The NCF-SE 2023 highlights the critical role of teaching-learning materials (TLMs): textbooks and supplementary resources must reflect local contexts, providing flexibility for states and schools to adapt them to regional languages, cultural references and community practices (NCERT, 2023). Such flexibility ensures that materials not only support multilingual pedagogy but also affirm children's sociocultural identities in their learning journeys.

Together, the NCF-FS 2022 and the NCF-SE 2023 provide a coherent vision: early childhood education must rest on mother-tongue-based instruction, supported by culturally grounded teachers, while the broader school curriculum must be indigenized and linguistically responsive. This represents a paradigmatic shift from seeing multilingualism as a barrier to recognizing it as India's greatest educational resource.

Multilingual framework for different stages of education

A multilingual framework across the four stages of schooling must recognize learners' evolving linguistic, cognitive and disciplinary needs. In the foundational stage, children learn best through their mother tongue, using oral language, play and multilingual exposure to build early literacy and numeracy. The preparatory stage expands this base by strengthening reading comprehension, introducing additional languages through integrated, meaning-based approaches. In the middle stage, multilingual pedagogy enables subject learning, encouraging multi/translingual scaffolding, critical thinking and gradual entry into academic registers. By the secondary stage, students engage with complex concepts across languages, developing advanced

bilingual or multilingual proficiency essential for higher education, vocational pathways and positive social identity. Organizations such as the National and State Councils of Educational Research and Training (NCERT and SCERTs), the National Council for Teacher Education (NCTE), the Language and Learning Foundation (LLF), the National Multilingual Education Resource Consortium (NMRC), Room to Read, Pratham and the British Council strengthen this progression through language mapping, curriculum design, material development, teacher education and community involvement. Together, they operationalize policy vision into concrete pedagogic tools that support equitable language learning across stages.

Mother-tongue-based multilingual education at foundational and preparatory stages

Since children learn concepts most rapidly and deeply in their mother tongue, the primary medium of instruction should optimally be the child's mother tongue/home language/familiar language.

UNESCO's *Languages Matter: Global Guidance on Multilingual Education* states that "learners must first acquire a solid foundation in a language they understand before they transition to additional languages of instruction" (UNESCO, 2025, p. 23). Children should not be burdened by linguistic decoding while acquiring literacy, numeracy and conceptual foundations. The guidance further emphasizes that the layering of additional languages should be gradual, supported by scaffolded pedagogies that respect learners' cognitive loads and linguistic trajectories (p. 24). It provides concrete country-level examples to illustrate how foundational multilingual curricula have been operationalized, including an example from the Philippines' mother-tongue-based multilingual education (MTB-MLE) programme, where children are instructed in their mother tongues from kindergarten through Grade 3, while Filipino and English are introduced thereafter (p. 63). Another example from Latin America uses Indigenous languages for initial literacy and concept introduction before transitioning to national or dominant languages (p. 67).

The NCF-FS (NCERT, 2022) situates the foundational literacy and numeracy (FLN) agenda firmly within the early years of school education: three years of early childhood care and education (ECCE) plus Grades 1 and 2. It emphasizes that "early learning must include early literacy,

numeracy, and awareness of environment, and that these must be mediated through mother tongue/familiar languages of children. Since children learn concepts most rapidly and deeply in their mother tongue, the primary medium of instruction should optimally be the child's mother tongue/home language/familiar language (also referred to below as L1) in the Foundational Stage. This should be the approach in both public and private schools" (NCERT, 2022, Section 3.2). It frames the mother tongue not only as the medium of instruction but as a cognitive and cultural scaffold and mandates that teachers be proficient in the child's first language and understand local culture and traditions.

A linear "curriculum" in the form of multilingual textbooks is less helpful. Initial phonemic awareness and decoding can be mapped in children's mother tongues/familiar languages before multi/translingual transfer. The FLN curriculum encourages the use of oral repertoires and play-based methods, engaging children in exploration, dialogue, multimodal experiences and learning through the environment and local contexts. Children engage with numerical and everyday science literacy via their linguistic and embodied repertoires. Teachers are envisaged as contextualizing materials, choosing familiar examples from children's life environments, and relating to community knowledge.

The framework underscores that materials, illustrations and stories should be regionally and culturally relevant; curriculum designs include bilingual or multilingual glossaries, side-boxes in regional languages, bi/multilingual notes, and parallel texts.

The NCERT and FLN initiatives

The NCERT leads in conceptualizing, assessing, training and supporting FLN-implementation across the linguistic and regional diversity of India. Recently, it has developed 121 primers for early readers in 117 languages. It assessed about 86,000 Grade 3 students across 10,000 schools in 20 languages, generating baseline data on reading and numeracy competencies in preparation for the national mission (PARAKH/ NCERT, 2024) and offered a National Initiative for School Heads' and Teachers' Holistic Advancement (NISHTHA 3.0) FLN professional development course, which trains teachers and head-teachers in multilingual pedagogy and competency-based assessment (NAERT,

2021). The NCERT collaborates with the National Initiative for Proficiency in Reading with Understanding and Numeracy (NIPUN) Bharat mission of the Ministry of Education to align FLN benchmarks with policy targets.

Strategies adopted by SCERTs for teaching languages under the NCF-FS 2022

USE OF THE CHILD'S MOTHER TONGUE (L1) FOR INSTRUCTION AND INTERACTION

In accordance with the NCF-FS, which emphasizes that "the primary medium of instruction would optimally be the child's home language/mother tongue/familiar language"(NCERT, 2022, Section 3.2), several state governments have introduced classroom practices that prioritize the use of mother tongues for teaching and interaction. Teachers are encouraged to conduct early literacy activities, storytelling and oral comprehension tasks in children's mother tongues to strengthen cognitive and intersubjective foundations.



FLN TARGETS : AY 2025 - 2026 (Class - 2)

MATHEMATICS					TELUGU				
Status for AY 2024 - 2025					Status for AY 2024 - 2025				
Targets for AY 2025 - 2026					Targets for AY 2025 - 2026				
Competency	At Class level	Non Performers	At Class level	Non Performers	Competency	At Class level	Non Performers	At Class level	Non Performers
Identification of numbers	64%	4%	80%	0%	Word Reading	40%	12%	60%	0%
Comparison of numbers	61%	2%	80%	0%	Oral Reading Fluency (ORF)	9%	31%	20%	0%
Addition (+)	35%	25%	60%	0%	Word Writing	41%	19%	60%	0%
Subtraction (-)	24%	35%	50%	0%					

ENGLISH					URDU				
Status for AY 2024 - 2025					Status for AY 2024 - 2025				
Targets for AY 2025 - 2026					Targets for AY 2025 - 2026				
Competency	At Class level	Non Performers	At Class level	Non Performers	Competency	At Class level	Non Performers	At Class level	Non Performers
Word Reading	25%	23%	45%	0%	Word Reading	42%	12%	60%	0%
Oral Reading Fluency (ORF)	9%	23%	20%	0%	Oral Reading Fluency (ORF)	18%	20%	30%	0%
Word Writing	44%	23%	60%	0%	Word Writing	56%	11%	65%	0%

At Class level means :

1. Telugu, English & Urdu : For competencies word reading & word writing, percentage of students answering 80% of the questions correct & for Oral Reading Fluency, percentage of students reading 25 correct words per minute.
2. For Maths, Percentage of students answering 80% of the questions correct in the given competency.

Non Performers means :

1. Percentage of students not answering even a single question in the given competency (applies for all subjects).





State curricula integrate children's cultural practices, local stories, songs and folk expressions into learning materials.



Above: Multilingual puppet show. Oxford Grammar School, Hyderabad.

MULTILINGUAL AND TRANSLINGUAL SCAFFOLDING

Many states now promote “mixing of languages” in classroom communication to aid comprehension and engagement. Teachers use bi/multilingual blackboard writing, explain difficult concepts through the mother tongue/home language, and maintain the linguistic fluidity of the region to bridge the gap between home and school languages.

GRADUAL INTRODUCTION OF ADDITIONAL LANGUAGES (L2, L3)

States have started adopting a phased approach to multilingual education – introducing additional languages only after children achieve literacy and conceptual fluency in L1. Classroom activities such as bilingual story reading, narrative retelling, picture labelling and transknowledging in mathematics and environmental studies (EVS) using multilingual repertoires help children transition smoothly to regional or national languages.

CONTEXTUALIZED LANGUAGE EXPERIENCES

Increasingly, state curricula integrate children's cultural practices, local stories, songs and folk expressions into learning materials. Language textbooks and teaching-learning materials are being localized to reflect community life, ensuring that children encounter familiar words, images and contexts in the classroom. The language pedagogy respects the multilingual fluidity of a region and encourages creativity and freedom among students to discover linguistic principles and literary expressions.

TEACHER CAPACITY BUILDING AND COMMUNITY ENGAGEMENT

State education departments are organizing teacher training programmes to strengthen multilingual pedagogies. Teachers are trained to recognize linguistic diversity as an asset, use local cultural references, and engage parents through community reading sessions. States such as Odisha, Chhattisgarh and Andhra Pradesh have developed multilingual primers and digital resources to support this vision.

Neev Multilingual Education Programme for foundational learning in Bastar, Chhattisgarh

The Government of Chhattisgarh supported by the Language and Learning Foundation

Location: 1,524 government primary schools across all seven blocks of Bastar district, Chhattisgarh: Bakawand, Bastar, Bastanar, Darbha, Lohandiguda, Tokapal and Jagdalpur.

Duration: May 2022 – present

Objective: To improve foundational literacy and numeracy outcomes of Adivasi children by integrating their home languages, Halbi, Gond and Bhatia, into early grade teaching and learning.

Focus areas: Foundational learning, multilingual education, teacher capacity building.

The Neev Multilingual Education Programme, Bastar, is an initiative of the Government of Chhattisgarh, supported by the Language and Learning Foundation, aimed at the integration of children's home languages in early grade classrooms. In Bastar, nearly 70 per cent of the population belongs to Adivasi communities and more than 90 per cent of Grade 1 children enter school with no exposure to Hindi, the school language. This linguistic gap has contributed to low early learning achievement, low engagement and high learning poverty.

To address this, the programme introduces children's home languages in classroom processes from the foundational stage. Teachers use Gond, Halbi and Bhatia in a bilingual education model along with Hindi to build conceptual understanding, encourage participation and build competencies in both the child's home language and the school's language of instruction.

The programme was piloted in 200 schools and later expanded to 1,524 schools across Bastar district under Samagra Shiksha, establishing MLE as the district's approach to early learning. The initiative integrates local culture, oral traditions, stories and community knowledge into pedagogy through bilingual TLMs, teacher training, and community-led cultural activities.

PROCESS

1 Baseline assessment and language mapping. A district-wide mapping of Grade 1 students and teachers was conducted in 2022 as part of language mapping across the state. It highlighted significant linguistic diversity and a mismatch between home and school languages. This informed the design of bilingual materials and pedagogical processes.

2 Teacher training and academic support. State and District Resource Groups (SRGs and DRGs), teachers and headmasters were trained on multilingual pedagogy, the

use of home languages for scaffolding, and structured lesson planning. Regular school visits, classroom observations and block review meetings helped maintain quality.

3 Material development: The LLF supported the state in developing bilingual and culturally rooted TLMs, including story-based materials, local songs, oral narratives and classroom resources that reflect the lived experiences of Adivasi children.

4 Community engagement: Over 900 storytelling sessions and 25 Shala Sangrahalayas (School Museums) were created, bringing community culture into classrooms. Parents and elders contributed oral stories, artefacts and local knowledge, strengthening home-school connections.

5 Assessment: Baseline and midline assessments were conducted to measure changes in literacy, numeracy, listening comprehension, oral language and writing skills.

ACHIEVEMENTS/RESULTS

- 68,000+ students reached across Bastar.
- 2,000+ teachers trained in multilingual pedagogy.
- 1,500+ schools implementing MLE strategies.
- Substantial improvements in mean score percentage of key early grade reading assessment (EGRA) and early grade mathematics assessment (EGMA) indicators. For instance, the mean scores for
 - listening comprehension increased from 42 per cent to 63 per cent;
 - accurate oral reading fluency improved from 28 per cent to 68 per cent;
 - word-level dictation scores rose from 31 per cent to 64 per cent;
 - basic addition/subtraction scores rose from 38 per cent to 69 per cent; and,
 - complex addition/subtraction scores rose from 26 per cent to 50 per cent.
- Teachers reported increased student participation, confidence and engagement.
- Government support led to district-wide scaling and embedding of MLE within the state's FLN mission.

PARTNER ORGANIZATIONS

Language and Learning Foundation, Government of Chhattisgarh (SCERT, Samagra Shiksha), HCL Foundation, Deloitte.



Role of NGOs and universities in implementing FLN goals

Many non-governmental organizations (NGOs) across India have taken proactive roles in helping states realize FLN goals. The Language and Learning Foundation partners with the state governments of Chhattisgarh and Jharkhand to reimagine classroom approaches based on the sociolinguistic profiles of children and community knowledge, and to build systemic capacity to scale FLN outcomes across millions of children.

The National Multilingual Education Resource Consortium, Jawaharlal Nehru University, has supported states in conducting sociolinguistic surveys to inform their language-in-education policies for early grade reading and numeracy; organized critical pedagogy workshops for teaching literacy and numeracy to children from linguistic minority communities; helped states and NGOs develop bi/multilingual readers and storybooks; developed the first language-in-education policy for the Government of Odisha; and advocated mother-tongue-based multilingual education.

The NMRC developed a series of activities for teaching Indigenous minority children numeracy in early years of schooling using their number systems, games, arts, crafts and mother tongues. It trained teachers in Saora and Kondh areas of Gajapati and Phulbani district of Odisha to begin teaching with Saora and Kui number systems and oral algorithmic practices and to integrate these with school mathematics. The MLE approach adopted by the NMRC helped to interweave children's spontaneous concepts with academic concepts within a multilingual-multiliteracy paradigm. Once conceptual clarity was achieved, a gradual replacement of the Saora number system and oral algorithms ensured transition into formal mathematical discourse. The dynamic interplay between Indigenous cognition and academic mathematics resulted in greater conceptual retention, engagement and confidence among learners (Panda, 2004; 2006).

Above: Promoting multilingual education within higher learning. The School of Social Sciences at Jawaharlal Nehru University, Delhi.

3

Teaching mathematics through the Saora number system and Indigenous heuristics

The NMRC (with support from UNICEF) and the Government of Odisha

Location: Gajapati district, Odisha, where 90 per cent children speak Saora at home.

Duration: 2009-2015.

Objective: The specific objective was to teach mathematics through the Saora number system and Indigenous heuristics. The broader objective was to not teach mathematics by erasing children's cultural and linguistic worlds but by interweaving them into the very structure of conceptual reasoning.

Focus area: Numeracy in Grades 1 to 4 (foundational and preparatory stages).

This case study is based on an MTB-MLE programme in the Saora-speaking areas of Gajapati district, Odisha, where the NMRC designed a series of mathematics activities for teaching Saora children using the Saora number system, local heuristics, games, arts, crafts, etc. and integrating these with school mathematics, including the universal number system.

In the MLE classrooms, Saora children began to learn to count orally from 1 to 20 using their own number system – *abay* (1), *bagu* (2), *yagi* (3), *unji* (4), *manlai* (5), *туру* (6), *gulji* (7), *tanji* (8), *tinji* (9), *galji* (10), *galmuai* (11), *migal* (12), etc. – before extending the count to 30 in Grade 1. The Saora system operates on two base values, 12 and 20, unlike the decimal system used in Odia and English. Numbers between 13 and 19 are derived by addition to 12: thus, *migalbay* (12 + 1), *migalbagu* (12 + 2), *migalgulji* (12 + 7). Similarly, *kudi* (20) functions as the next base for generating numbers up to 39 (Panda, 2004).

Earlier teaching materials introduced the base 10 system prematurely, using pictorial representations of “bundles of ten sticks” with loose sticks for ones. However, this conflicted with the Saora child's linguistic logic and Indigenous counting patterns. The MLE intervention began by strengthening oral numeracy through meaningful everyday contexts before moving towards written representation.

- Children counted seeds, pebbles and bamboo strips in Saora, exploring patterns of grouping based on their natural base values (12 and 20).
- Teachers guided learners in deconstructing numbers from 13 to 19 as 12 *plus* 1, 12 *plus* 2, etc. thereby facilitating conceptual understanding of number composition.
- Only after developing the concept of “base” through their own system were children introduced to the Odia number system, where comparative reflection helped them grasp the abstract notion of place value.

Since Saora lacks a writing system, all initial numeracy activities were oral and embodied. Children learned through regrouping strategies and reasoning rather than memorization, using fingers, bundles and natural objects.

In higher classes, concepts such as odd and even numbers, number sequences, patterns and infinity were explored through dialogue about Saora art, crafts and daily life, like counting the bamboo strips in baskets and hats, observing their symmetrical designs and identifying regular progressions.

The mother tongue intervention distinguished between replacement (substituting everyday understandings with abstract academic concepts) and interweaving (allowing two systems to coexist and inform each other).

ACHIEVEMENTS

- A longitudinal study conducted by the NMRC (Panda and Mohanty, 2013) showed improvement in mathematics learning with understanding among Saora students.
- The “difficulty” of introducing the concept of a base 10 to Saora children did not arise from cognitive limitations but from pedagogical gaps.
- Understanding of base values, patterns and operations improved.
- This model was incorporated into mathematics pedagogy in non-MLE schools in neighbouring villages.



Right: Open-licensing allows users to translate and print picture books from Pratham Books' StoryWeaver platform across languages and formats.

The NGO Pratham has complemented FLN initiatives by expanding rich resources and innovation within multilingual contexts. Through its Read India and StoryWeaver platforms, Pratham has developed large-scale, evidence-based models for improving early literacy and numeracy in under-resourced communities. The StoryWeaver initiative has created the world's largest repository of multilingual children's stories, hosting over 64,000 books in more than 370 languages. StoryWeaver's open-source model allows free access, translation and adaptation of stories, empowering communities to create culturally relevant, age-appropriate content for children. Educators can translate popular stories into local dialects, bridging literacy gaps in underserved communities. The platform facilitates code-switching by offering the same stories in multiple languages, helping students transition between languages.

Similarly, Room to Read (RTR), with its emphasis on children's literature, reading

pedagogy and libraries in multiple Indian languages, strengthens FLN initiatives. It has developed high-quality bilingual and local-language storybooks, teacher capacity-building strategies to promote reading, and community engagement strategies that normalize the use of mother tongues/home languages in foundational classrooms. Its reading pedagogy comprises three core elements:

- ① oral language development in the child's own language
- ② systematic *akshara* ("alphabetic") instruction
- ③ use of multilingual storybooks and discussion tools (e.g. conversation charts) that begin in the local language and gradually incorporate other instructional languages.

RTR's reading curriculum begins with literacy instruction in a familiar language to build phonemic, vocabulary and narrative competence, deploying levelled texts and conversation supports simultaneously, including support for other languages.

Multilingual assessment at the FLN stage

At the FLN stage, multilingual and multiliteracy assessment values children's full linguistic repertoires. It allows them to demonstrate understanding in their mother tongue or familiar language, using oral, pictorial and written responses. Such assessment captures meaning-making across languages, ensuring fairness, reducing linguistic bias, and supporting the NCF-FS 2022's vision of inclusive learning. The states that are implementing FLN use a combination of the following multilingual methods for assessment:

ORAL AND PICTURE-BASED ASSESSMENT

Use storytelling, picture descriptions and oral retellings in the child's mother tongue/home language to gauge comprehension and vocabulary.

BI/MULTILINGUAL QUESTIONING

Ask questions in both the familiar language (L1) and school language (L2) to check conceptual understanding rather than language decoding.

TRANSLATION AND PARAPHRASING TASKS

Encourage children to explain a text or concept

in their own words or another known language to assess depth of understanding.

COMMUNITY WORD LISTS AND LOCAL CONTEXTS

Incorporate local words, counting systems and cultural references in assessment items to validate mother/home language literacy.

MIXED-LANGUAGE WORKSHEETS

Combine L1 and L2 terms or instructions in one task, allowing multi/translanguaging while testing numeracy or literacy skills.

PEER AND GROUP ASSESSMENTS

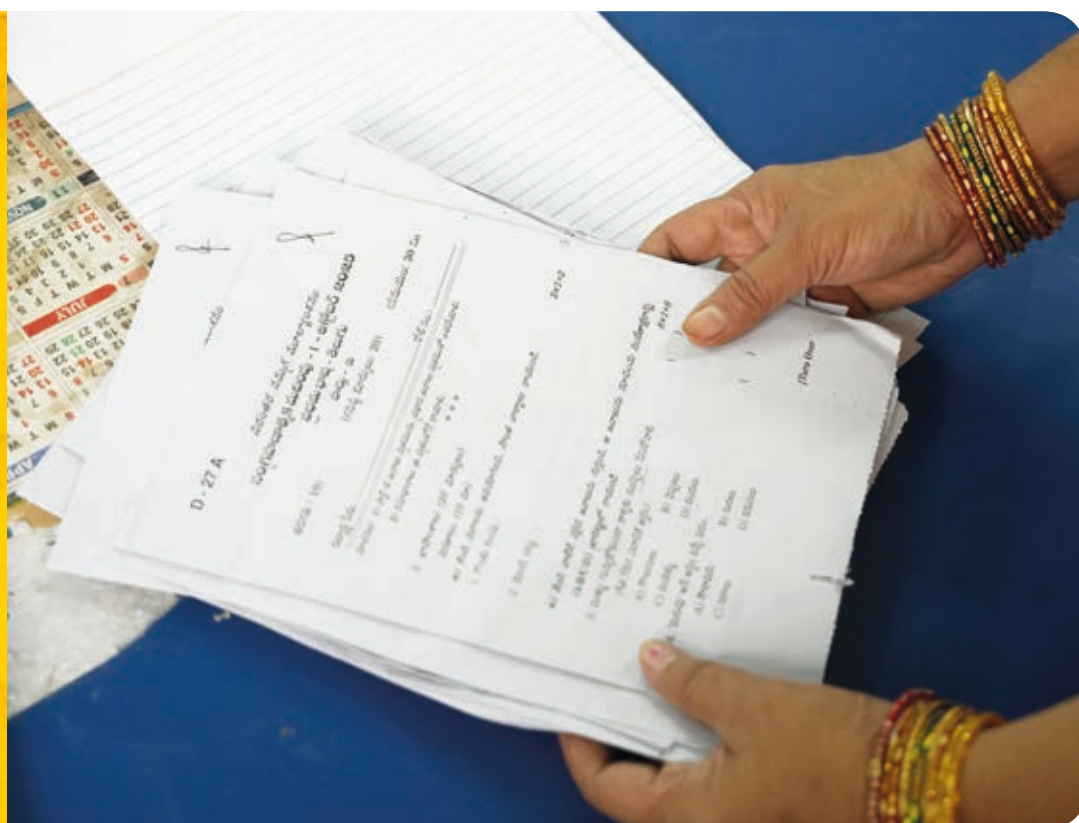
Let children discuss and solve problems collaboratively in L1, observing how meaning is co-constructed across languages.

PORTFOLIO AND ANECDOTAL RECORDS

Maintain continuous observation notes on children's progress in both home and school languages.

Below: Teacher assesses Telugu answer sheets. Model High School, Osmania University, Hyderabad.

Multilingual and multiliteracy assessment values children's full linguistic repertoires. It allows them to demonstrate understanding in their mother tongue or familiar language, using oral, pictorial and written responses.



... availability of an Indian language as a medium of instruction through Grade 12 sustains continuity in conceptual development and strengthens students' capacity for logical argumentation and problem solving.



Above: Books are made available in multiple Indian languages. Kalinga Institute of Social Sciences, Bhubaneswar.

Language in the middle stage

The NCERT formalizes a trilingual framework for preparatory, middle and secondary stage education and calls these three languages R1, R2 and R3. R1 is the language in which literacy is first learned in school (mother tongue or the state language). R2 and R3 are languages other than R1. At least two of these three languages must be native to India (NCERT, 2023). In science/mathematics classrooms, this allows teachers to design instruction materials that move students across the mother tongue (R1), the regional / state language (R2) or English, and English / another Indian language (R3).

Mother tongue, Indigenous knowledge and teaching of STEM

The NEP 2020 states: "It is well understood that young children learn and grasp non-trivial concepts more quickly in their mother tongue/home language/local language" (NEP, 2020, Section 4.11). The NCF-SE 2023 translates this vision into concrete curricular and pedagogic plans. It views the curriculum not as a neutral or universal template but as one "deeply rooted in India... in content and learning of languages, in pedagogical approaches including tools and resources, and most importantly in the cultural, philosophical, and contextual underpinnings" (NCERT, 2023, p. 4). For science, it states that learning goes beyond theories and facts to

include connecting concepts with real life, developing scientific reasoning, and applying knowledge to understand the world (Section 4.6). The framework thus emphasizes epistemic engagement over rote learning and calls for cultural, linguistic and cognitive scaffolds that help children express understanding first in their home languages and then in formal scientific discourse. It encourages pedagogical designs such as multilingual scaffolding, translanguaging and culturally grounded inquiry to foster deeper conceptual thinking.

In mathematics, the use of the mother tongue supports the transition from concrete experiences in the preparatory stage to abstraction in the middle and secondary stages. Learners begin to generalize properties, justify claims and model phenomena through algebraic reasoning. To build mathematical discourse, the framework recommends introducing and standardizing new terminology in Indian languages. These practices promote computational thinking, including decomposition, representation and algorithmic reasoning, without the added cognitive strain of unfamiliar linguistic structures. Ensuring the availability of an Indian language as a medium of instruction through Grade 12 sustains continuity in conceptual development and strengthens students' capacity for logical argumentation and problem solving.

BOX 4.1

Pedagogical design offered by NCF-SE 2023 for the preparatory stage**TEACHING SCIENCE THROUGH THE MOTHER TONGUE AND MULTILINGUAL MEDIATION**

- Begin lessons in the mother tongue (L1) for observation and description of everyday phenomena (e.g. explaining phases of the moon, properties of soil, etc.).
- Encourage translingual and multilingual discussions, where students hypothesize in L1 but gradually articulate concepts in the regional language (L2) and English (L3) to access scientific registers.
- Use local metaphors and idioms in the mother tongue to anchor abstract concepts (e.g. comparing evaporation with local practices of drying grain).
- Make science books available in different languages through AI platforms.
- AI-based multimedia scaffolds can be accessed through specially designed online science activities and translation platforms.

TEACHING MATHEMATICS THROUGH MULTILINGUAL MEDIATION

- Teach basic numeracy and operations using culturally embedded artefacts (millet seeds, shells, sticks, games and local number systems) with L1 explanations, then introduce formal symbols and English terms. Use oral stories and narrations in local languages as much as possible in the early years and gradually move to written practices without abandoning multilingual oral practices completely in higher classes.
- Enable problem-solving discussions in multiple languages. Students first narrate solutions in L1, then co-construct explanations in bilingual mode and switch

gradually to formal mathematical registers in the school language.

- Develop bilingual glossaries of key mathematical terms (e.g. “fraction”, “denominator”) linking them to local language equivalents. Children’s everyday experiences can be used.
- Prepare flipped classrooms in bilingual mode and use mathematical tensions arising from classroom dialogue to advance mathematical understanding.

TEACHING-LEARNING MATERIALS

- Textbooks and worksheets should include parallel explanations in L1 and L2, and side-boxes in English.
- Teachers’ handbooks should provide translanguaging strategies: when to allow free L1 expression, when to scaffold to L2/L3, and how to validate multiple answers expressed in different languages.
- Incorporate local contexts in TLMs (measurement through paddy bundles, geometry through weaving patterns) to link mathematics and science to lived experience.

INTEGRATING INDIGENOUS KNOWLEDGE SYSTEMS

- Use Indigenous community-based mathematical concepts, ideas and practices to scaffold learning of higher concepts in mathematics.
- Use Indian mathematical systems to create a richer transdisciplinary dialogue on mathematical concepts and ideas.

Source: NCF-SE 2023

Sardar Patel Vidyalaya (SPV) in Delhi employs a bilingual medium of instruction: Hindi medium at primary levels (Nursery to Grade 5) and English medium from Grade VI onwards, with Hindi and Sanskrit continuing as important languages. This offers a natural linguistic scaffold: students entering middle school already have conceptual grounding in Hindi and continue learning content in English. In practice, middle school classes prefer code-switching and translanguaging between Hindi and English. Similarly, Kendriya Vidyalayas adopt a bilingual

approach that integrates Hindi and English across subjects to support comprehension, participation and disciplined language progression. At the primary and middle levels, teachers strategically use both languages to explain concepts, build vocabulary and enable smooth transition into academic English without compromising conceptual clarity. This flexible bilingual pedagogy helps learners from diverse linguistic backgrounds adapt to a common curriculum while maintaining cognitive continuity and equitable access to learning.



MODULE 4.1

Teaching science bilingually in middle school, Kendriya Vidyalaya, Delhi

Location: Kendriya Vidyalaya, Delhi.

Duration: 2017-2019.

Objective: Teaching science using multilingual pedagogy.

Focus area: Middle school

TEACHER INTRODUCES GRAVITY WITH EVERYDAY EXPERIENCES IN HINDI AND ENGLISH

- Teacher holds an object (chalk/ball)
- Teacher (in Hindi): अगर मैं ये गेंद छोड़ दूँ तो क्या होगा? ये नीचे गिरेगी या ऊपर जाएगी?
- Students: नीचे गिरेगी।
- Teacher reinforces: हाँ, ये नीचे गिरेगी। क्यों? ज़मीन इसे अपनी तरफ खींचती है। इसी को गुरुत्वाकर्षण कहते हैं।
- Teacher adds in English: This pulling force of the Earth is called gravity.

INTRODUCING KEY VOCABULARY BILINGUALLY

- गुरुत्वाकर्षण शक्ति (gravity force)
- द्रव्यमान (mass)
- भार (weight)
- पृथ्वी का केंद्र (centre of the earth)

Source: NMRC, JNU

Teacher writes both **Hindi terms** and **English terms** on the board side by side to build a glossary.

DEMONSTRATION AND DIALOGUE

- Drop two objects (e.g. chalk and notebook).
- Ask: दोनों वस्तुएँ एक साथ क्यों गिरीं? क्या भारी चीज़ जल्दी गिरती है?
- Then in English: Do heavier objects fall faster than lighter ones?
- Teacher clarifies using both languages: In reality, both fall with the same acceleration due to gravity, which we call $g = 9.8 \text{ m/s}^2$.

CONTEXTUAL EXAMPLE

- Hindi: जब आम पेड़ से गिरता है, वो ज़मीन की तरफ क्यों आता है?
- English: This happens because of the Earth's gravitational pull.
- Link to real life: इसी कारण हमें वज़न (weight) महसूस होता है। Weight is the force of gravity acting on our mass.

SUMMARIZE BILINGUALLY

- Hindi: हर वस्तु को पृथ्वी अपने केंद्र की ओर खींचती है। यही गुरुत्वाकर्षण है।
- English: Gravity is the universal force of attraction that pulls objects towards the centre of the earth.

Teaching English using the mother tongue and translingual practices

In India's multilingual classrooms, teaching English becomes richer when students' home languages are part of the interpretive process. The British Council's MultiEd initiative emphasizes that using familiar local languages supports academic success and views learners' L1 as a resource, not a limitation. Teachers can invite students to discuss imagery, tone and metaphor in Hindi, Tamil or other home languages before expressing their ideas in English, enabling deeper engagement with texts.

Translingual practices are already common in Indian English classrooms (Anderson and Lightfoot, 2018), though few teachers integrate them into literary analysis. Allowing students to connect English lines with local idioms or proverbs expands interpretation and expression. For example, when exploring Shakespeare's imagery, a student might translate a metaphor into Hindi, unpack its layered meanings, and then reformulate the insight in English, showing how multilingual reflection enhances literary understanding.

BOX 4.2

Multilingual pedagogy and assessment

MULTILINGUAL EDUCATION FOR PRIMARY GRADE INDIAN CLASSROOMS

Multilingual education in primary grade Indian classrooms is the result of twin projects led by the University of Cambridge, UK, in collaboration with the English and Foreign Languages University (EFLU) in Hyderabad, India. Both projects were conducted in Hyderabad, the capital city of Telangana, across ten low-cost government schools with multilingual teachers and children who were proficient in Telugu, Dakkhini and English.

TEACHER TRAINING IN MULTILINGUAL PEDAGOGY

A British Council ELTRA-funded project (2021–2023) trained fourteen teachers (twelve female, two male) from ten low-cost government schools to implement multilingual pedagogy through workshops and classroom collaborations. Over nine months, observations revealed teachers' growing confidence in using learners' home languages and translanguaging to explain concepts and expand multilingual vocabularies. Teachers began designing multilingual reading and peer-interaction activities, strengthening comprehension and participation. However, they expressed concern over the sustainability of administrative support for multilingual pedagogy in English-medium schools and highlighted the need for continued capacity-building from MLE practitioners.

DEVELOPMENT IN READING COMPREHENSION THROUGH MULTILINGUAL PEDAGOGY

Building on this, a British Academy-funded project (2021–2023) assessed reading comprehension among 248 Grade 5 students taught by the trained teachers. Reading performance, measured at three stages (pre-, mid- and post-intervention), improved by 30 per cent, particularly in expository texts that contain academic language critical for cognitive growth (CEFR, 2020). The gains reflected teachers' use of multilingual strategies (translanguaging, multilingual discussions, and integrated English and science activities) that bridged conceptual understanding and enhanced linguistic transfer between English and the learners' home languages.

MULTILINGUAL MULTIMODAL ASSESSMENTS

Reading assessments included English texts with bilingual questions in Telugu and English, delivered both orally and in print. Students responded in their preferred language, Telugu, Dakkhini or English, promoting accessibility and reducing linguistic bias. This design replaced monolingual assessments, allowing children to process and express comprehension authentically. Increased engagement and confidence demonstrated the ethical and equitable value of multilingual assessment frameworks, offering a valid model for inclusive classroom evaluation in linguistically diverse contexts.

Source: EFLU, Hyderabad



Facing page: Science class taught bilingually, in Telugu and English, through model-making projects. Model High School, Osmania University, Hyderabad.

5

Teachers leading multilingual and evidence-based practice in India

Action Research Mentoring Scheme



Location: Tamil Nadu, Telangana, Assam.



Duration: 2024–2025



Objective: To strengthen teachers' professional capacities as reflective practitioners and classroom researchers, enabling them to use multilingual and translanguaging pedagogies to improve learning outcomes, especially in subjects like mathematics and science.



Focus areas: Professional development of teachers, classroom-based research, multilingual pedagogy, mathematics education, translanguaging strategies, and reflective teaching.

The Action Research Mentoring Scheme (ARMS), implemented by the British Council in India, supports teachers in systematically studying and improving their classroom practices through action research. The 2024–25 edition of ARMS focused on multilingual education, encouraging teaching in the home language and the use of multilingual approaches in foundational and subject learning.

The scheme was implemented through three academic partners: Gauhati University (Assam), the Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kanchipuram (Tamil Nadu), and the National Institute of Technology (NIT), Warangal (Telangana). These institutions mentored fifty-four teacher-researchers guided by sixteen mentors who supported teachers across disciplines and linguistic contexts.

Each participating teacher developed and conducted an action research project in their classroom. The projects explored how using students' home languages (Tamil, Telugu, Assamese, Bodo, Bengali, Hindi, Urdu and Lambadi) could enhance engagement and comprehension. The scheme provided teachers with structured mentoring, technical guidance on research design, and peer review spaces, culminating in the publication *From Insight to Action: Teachers Leading Change* (2025), which documents seventeen selected research stories.

Several ARMS teachers focused on mathematics classrooms, where language barriers often prevent children from grasping abstract concepts. In many government schools, children whose home language differs from the medium of instruction struggle to understand mathematical terminology and reasoning expressed in unfamiliar linguistic forms.

Through ARMS, teachers in Telangana and Tamil Nadu implemented translanguaging approaches in mathematics to make concepts accessible. They encouraged students to discuss problems in their home language first, translate their reasoning into English, and finally write the mathematical solution using formal terminology. Teachers also co-created bilingual mathematical glossaries and displayed word walls showing key concepts (e.g., "perimeter," "volume," "fraction") in both English and the local language.

In one classroom in Warangal, for instance, a mathematics teacher observed that her Grade 7 students often misinterpreted English word problems because they did not understand relational terms such as "less than", "equal to" or "difference". By allowing discussions in Telugu and inviting students to express a problem situation in their own words before solving it, she found that students' participation increased from 10 per cent to over 70 per cent, and the class average in formative assessments improved significantly.

Similarly, in Tamil Nadu, a teacher used dual-language story problems in which everyday contexts, such as shopping in local markets, were narrated in Tamil first, followed by the school's medium of instruction. Students demonstrated better conceptual understanding when they could reason in their mother tongue and then transfer their logic to the target language. These mathematics-focused action research projects reaffirmed that conceptual reasoning and linguistic comprehension are interlinked. Using multilingual strategies not only built cognitive connections but also reduced students' fears and anxieties about mathematics as a subject taught in a non-native language. Teachers also reported that peer discussions in the home language encouraged collaborative problem-solving, making classrooms more inclusive and dialogic.

IMPLEMENTATION PROCESS

- 1 Mentor-mentee partnerships.** University-based mentors supported teachers in identifying context-specific problems, such as students' difficulty in understanding word problems in mathematics or their conceptual gaps in science.
- 2 Teacher-led inquiry.** Teachers designed classroom-based interventions, encouraging group discussions in both the home language and the school's medium of instruction, for instance, or translating mathematical terms into local equivalents.

- 3 **Data collection and reflection.** Teachers recorded classroom interactions, student reflections, and learning outcomes before and after interventions.
- 4 **Peer dissemination.** Teachers presented findings in state-level sharing sessions and to local education departments, contributing to a community of reflective practitioners.

ACHIEVEMENTS

- 1 **Teacher empowerment.** Over fifty teachers became practitioner-researchers capable of designing, implementing and analysing classroom interventions.
- 2 **Improved learning outcomes.** Evidence from mathematics and science projects showed measurable improvements in student comprehension and engagement.
- 3 **Pedagogical innovation.** Teachers applied translanguaging to introduce bilingual problem-solving, oral reasoning and multilingual maths journals.
- 4 **Multilingual resources created.** Teachers developed bilingual flashcards, trilingual word walls, home-language learning charts, and mathematics dictionaries.
- 5 **Professional network building.** A mentor network spanning Assam, Tamil Nadu and Telangana established a model for ongoing teacher-led research in multilingual contexts.

IMPACT (QUALITATIVE AND QUANTITATIVE)

- 1 **Classroom participation.** In Telangana, mathematics class participation increased from 10 per cent to over 70 per cent following bilingual instruction.
- 2 **Comprehension gains.** Teachers in Assam reported marked improvement in students' ability to solve word problems after discussions in Assamese and Bodo.
- 3 **Confidence and expression.** Students in Tamil Nadu began volunteering to explain mathematical reasoning in both Tamil and English.
- 4 **Shift in teacher mindset.** Teachers moved from viewing home languages as barriers to recognizing them as cognitive resources.
- 5 **Institutional sustainability.** Several mentors and teachers plan to extend the ARMS methodology into district-level professional development programmes.

KEY PEOPLE AND ORGANIZATIONS INVOLVED

- 1 Lead organization and technical support provider: British Council India.
- 2 Partner institutions: Gauhati University (Assam), IIITDM (Tamil Nadu) and NIT Warangal (Telangana).
- 3 Mentors: sixteen academic mentors guiding fifty-four teacher-researchers.
- 4 Teacher-researchers: practising teachers from primary, upper primary and secondary schools working in multilingual settings.





Above: Teacher teaches science bilingually, in Telugu and English. Model High School, Osmania University, Hyderabad.

MTB-MLE in Eklavya Model Residential Schools

Eklavya Model Residential Schools (EMRSs), established under the Ministry of Tribal Affairs, are a flagship intervention designed to provide high-quality residential schooling for Scheduled Tribe (ST) children from Grades 6 to 12. EMRS guidelines state that each school have a capacity of 480 students from Grades 6 to 12 and aim to replicate the standards of the Navodaya Vidyalaya Samiti residential schools (MoTA, 2022). As of 25 March 2025, 788 EMRSs were approved under the scheme, of which 378 were reported as functional across India, serving students from diverse tribal communities, including Santali, Gondi, Saora, Munda, Bhili, Bodo, Khasi, Mizo, Wancho, Kurukh and Nicobarese (MoTA, 2025).

Although EMRSs showed increasing enrolments and retention despite English-medium instruction, Halder and Guha's (2023) study reports concerns about academic struggle due to language gaps between home and school. In 2025, Tamil Nadu and West Bengal decided to teach tribal students in EMRSs in their mother tongues with emphasis on good quality English education and have refused to switch to CBSE (Sharma, 2025). Several states like Odisha, Chhattisgarh, Jharkhand and Rajasthan are apprehending that linguistic discontinuity may contribute to disengagement, delays in conceptual learning, and gender-differentiated vulnerabilities in EMRS schools and are therefore attempting to gradually shift to a mother-tongue-sensitive classroom pedagogy.

Successful EMRSs share enabling conditions:

- local teacher recruitment;
- peer-mentoring systems;

- community knowledge integration (folklore, agricultural cycles, forest ecology);
- trilingual learning aids; and
- supportive school leadership.

Aligning EMRS pedagogy with NEP 2020's emphasis on home-language instruction in early grades and NCF 2023's multilingual pathways can systematically strengthen outcomes. Partnerships with the NCERT, the Central Institute of Indian Languages (CIIL), the NMRC and state tribal departments can support the creation of trilingual textbooks, bridging materials, and teacher-training modules.

Kasturba Gandhi Balika Vidyalayas and equity compliance

Kasturba Gandhi Balika Vidyalayas (KGBVs) are residential schools under the Samagra Shiksha programme, for girls from Grades 6 to 12 belonging largely to SC, ST, OBC, minority and below-poverty-line households in educationally backward blocks. In districts with high ST populations, KGBVs cover 69 per cent of ST girls, underlining their importance for first-generation and tribal learners (GoI, 2017). NITI Aayog's (2019) evaluation of the KGBV scheme notes that "The scheme provides opportunity to reach Hard to reach groups and vulnerable girls (who required long term socio-psychological support to resume their schooling and join KGBVs) only once during the enrolment drive", (p. viii). The NEP 2020 explicitly calls for KGBVs to be "strengthened and expanded to increase the participation... of girls from socio-economically disadvantaged



Integrating mother tongue support emerges, therefore, as a key lever for promoting equity in girls' schooling.

backgrounds" (section 6.9, p.27) and simultaneously emphasizes language-inclusive education in the middle and secondary years. Recent equity guidance from the Ministry of Education links KGBVs, tribal girls' education and the appointment of teachers with knowledge of tribal languages for tribal-area schools as part of inclusive and equitable education reforms (NEP, 2020).

Implementation of MTB-MLE in KGBVs is still emerging and largely takes the form of bilingual and bridging strategies rather than full mother-tongue-medium schooling. The NCERT's 2022-23 national evaluation study of KGBVs notes that, in several states, teachers reported using the local mother tongue as the medium of instruction in early grades and were encouraged to adopt a bilingual approach, combining home language with Hindi or the regional language. Teacher-training efforts also carry an implicit multilingual orientation. The NCERT's customized teacher-training packages for KGBV teachers in language and subject areas emphasize contextualized, activity-based pedagogy and encourage teachers to connect classroom content with learners' linguistic and cultural backgrounds. In some tribal regions, NGOs and language resource centres have organized outreach programmes on "awareness of mother tongues" in KGBVs. For instance, the Research Institute of World's Ancient Traditions Cultures and Heritage (RIWATCH's) mother-language programme in a KGBV in the Lower Dibang Valley, Arunachal Pradesh, signals growing recognition of local languages as educational resources rather than obstacles (RIWATCH, 2023).

LEARNING OUTCOMES, GAPS AND THE NEED FOR STRONGER MTB-MLE

Evidence from learning assessments shows why deeper MTB-MLE implementation in KGBVs is needed. Singh and Singh's (2017) study on the Hindi language competency of KGBV students in Bihar found that only about 40 per cent of Grade 6 girls achieved satisfactory performance in Hindi, and that there was a "loss in transition (about 6–24%) at the upper primary level" despite the residential support provided. The study also reported acute shortages of teachers and wardens, making it difficult to provide sustained language support. Parashar (2018) found that tribal girls in Jharkhand face significant educational barriers when the medium of instruction diverges from their home languages. She writes that "the language of instruction acts as a hidden filter, limiting tribal girls' access to learning even after enrolment" (p. 5). The study highlights that when instruction shifts prematurely to Hindi or English, comprehension

drops and girls' retention and achievement suffer. Integrating mother tongue support emerges, therefore, as a key lever for promoting equity in girls' schooling.

National-level evaluations of the KGBV scheme document important gains in access, attendance and completion for marginalized girls, but they rarely report systematic, large-scale MTB-MLE practices within KGBVs. Some studies report scattered bilingual materials and teacher-training inputs, and localized initiatives on mother language awareness, but no universal, structured MTB-MLE model across KGBVs. Strengthening KGBVs as hubs of MTB-MLE would, therefore, require: (i) explicit language-in-education guidelines for KGBVs; (ii) recruitment and professional development of teachers who speak learners' home languages; (iii) development of graded bilingual materials; and (iv) routine monitoring of language-wise and gender-wise learning outcomes.

GENDER, ALTERNATIVE SCHOOLING AND EDUCATION IN THE MOTHER TONGUE/HOME LANGUAGE

The Unified District Information System for Education Plus (UDISE+) 2023-24 estimates that about 47.44 million children aged 6 to 17 are out of school (MoE, 2024). Evidence indicates that many out-of-school children re-enter school through alternative pathways, and that girls are more likely than boys to be channelled into alternative/non-formal schooling rather than formal schools as these centres are located within habitations, easing parental concerns about distance, safety and domestic responsibilities. For children re-entering school through alternative pathways, the NEP 2020 mandates "bridging courses in the child's home language" (Section 3.5) to support learning continuity and retention. The policy also emphasizes multilingual learning materials, community involvement and flexible pathways that allow children to rejoin formal schooling without linguistic disadvantage.

For children in flexible, residential, community-run, seasonal migrant or alternative schooling settings, the NCF-SE 2023 emphasizes multilingual bridging strategies – bi/multilingual materials, oral narratives and context-sensitive pedagogies – to ensure that "learning gaps arising from discontinuity of language are minimised" (NCERT, 2023, pp. 27–28). Alternative schooling programmes provide home-language-based foundational learning, bridge courses in the child's familiar language and transition materials from home language to state/English mediums, and ensure curriculum flexibility for children entering from diverse linguistic backgrounds.

Linguistic inclusion of children with special needs



Teacher teaches child with disability. Kalinga Institute of Industrial Technology, Hyderabad.

Linguistic inclusion for children with disabilities in India requires curriculum, pedagogy and assessment aligned with both disability-related and language-related needs. The NEP 2020 (MoE, 2020) emphasizes that “children learn best in their home language” (Section 4.12) and calls for “appropriate curriculum and pedagogy for children with special needs” (Section 6.8). Consistent with the NEP 2020 and the NCFs 2022 and 2023, schools must ensure access to learning in the child’s home language or preferred communicative mode, including Indian Sign Language (ISL), Braille, Augmentative and Alternative Communication (AAC), and simplified, multimodal texts. Teacher preparation must include specialized training in multilingual, visual, tactile and activity-based strategies so that learners with hearing, speech, cognitive or other specific learning disabilities can meaningfully engage with classroom processes. Embedding these supports within a Universal Design for Learning (UDL) framework ensures that inclusion is not an add-on but integral to quality education. Government programmes on inclusive education must, therefore, treat linguistic and disability inclusion as co-equal components of educational equity.

CASE STUDY

6

Inclusion of Braille in education

State and central interventions

India’s efforts to include Braille-based education across levels has translated into concrete institutional and state-level initiatives, ensuring that visually impaired students receive textbooks, learning materials and assistive support from primary school through higher education. The National Institute for Empowerment of Persons with Visual Disabilities (NIEPVD) in Dehradun, Uttarakhand, runs the Central Braille Press that produces Braille textbooks, reference books and educational materials in multiple Indian languages, including regional and minority languages.

A central scheme called the Project for Financial Support for Development of Accessible Learning Materials, funded by the Department of Empowerment of Persons with Disabilities (DoEPwD) produces Braille textbooks and assistive materials using large-print tactile pages and digitally accessible formats, including talking (narrated) and e-pub/digital books. These materials are provided free of cost to students with visual impairments up to Grade 12 nationwide. A total of 25 Braille Press/implementing agencies have produced Braille materials across the country and distributed them to 159,628 beneficiaries under this project as of 2023-24 (DoEPwD).

Besides national-level provisions, several state-level initiatives demonstrate localized commitment to producing Braille material. The Government of Uttar Pradesh (UP) established five Braille presses in order to produce UP Board textbooks for visually impaired students up to high school. In southern India, the Kerala Federation of the Blind runs a Braille press in Thiruvananthapuram that prints textbooks and reading materials for all schools for the blind in Kerala, distributing them free of cost. Some institutions provide specialized Braille resources to support science and mathematics education. Efforts are made to adopt a uniform Braille mathematics code (Nemeth/Bharati Braille) and develop tactile kits for teaching science, addressing long-standing barriers in STEM education for visually impaired learners.

India is moving towards developing a multilayered ecosystem comprising central policy, national institutions, state-level presses and community-based organizations working to embed Braille-based learning in mainstream education. Challenges, however, persist in the timely production and distribution of Braille materials across states.

7

Unique eco-school for children with hearing disabilities

The Happy Hands School for the Deaf

- Location:** Subarnapur district, Odisha.
- Duration:** 2016 – present
- Objective:** Academic learning, ecological stewardship and linguistic empowerment among students with hearing disabilities.
- India's only eco-school for children with hearing disabilities, integrating environmental engagement with daily learning.
- A research-led institution, partnering with national and international scholars to document and analyse its innovative pedagogy.

The Happy Hands School for the Deaf (HHSD), founded in 2016 under the Rural Lifeline Trust by Dr Sibaji Panda, stands as a pioneering model of inclusive, bilingual and eco-centric education for children with hearing disabilities in rural Odisha. Located within the Shikha Ecovillage in Subarnapur district, the HHSD provides a fully residential learning environment for 72 students aged 3 years to 16 years. The school's approach combines academic learning, ecological stewardship and linguistic empowerment, positioning it as a transformative initiative in education for children with hearing disabilities in India.

DESCRIPTION

- Managed and staffed almost entirely by educators with hearing disabilities, centring leadership by persons with disabilities in the education of children with hearing disabilities.

Through its rich bilingual environment, the HHSD addresses one of the gravest challenges faced by children with hearing disabilities: language deprivation. In India, nearly 90 per cent of children with hearing disabilities grow up in homes without sign language exposure, resulting in delayed cognitive and linguistic development. The HHSD combats this by establishing a sign-bilingual model in which Indian Sign Language (ISL) serves as the first language, providing the cognitive base for learning English literacy as a second language. This ensures that children acquire linguistic competence, conceptual clarity and self-confidence. By Grade 3, students who once lacked any functional language can sign fluently, comprehend complex ideas and engage in creative expression. By middle school, they demonstrate advanced literacy skills, research independently and express ideas both visually and textually.

The HHSD's success lies in its ecosystemic pedagogy: the seamless blending of sign language immersion, environmental engagement, and thematic, cross-curricular learning. Children participate in eco-campus activities such as gardening and small design projects that are woven into maths, science and art lessons, grounding abstract concepts in lived experience. This holistic design fosters academic achievement, emotional well-being and a strong sense of deaf identity.



Language in secondary education



Bilingual and multilingual scaffolds enhance comprehension of science, mathematics and social sciences. . .

Multilingual curriculum and pedagogy in secondary education promises to move beyond transitional language use and embrace an additive, concept-rich multilingualism that supports higher-order thinking. The NEP 2020 clearly states that students should have “the opportunity to learn at least two languages” and develop strong proficiency in one Indian language (Sections 4.12 and 4.16). Research shows that bilingual and multilingual scaffolds enhance comprehension of science, mathematics and social sciences by enabling students to move fluidly between everyday reasoning and academic registers (Panda and Bhagat, 2017; Heugh, 2021). Recently, circulars issued by the Central Board of Secondary Education (CBSE)

to operationalize the NCF-SE language guidelines reaffirm the primacy of the mother tongue at all stages and promote the use of multilingual resources. They encourage secondary schools to adopt bilingual texts, multilingual explanations and translanguaging practices that strengthen comprehension while ensuring alignment with examination requirements in English or regional languages. These directives also shape feeder practices and guide resource allocation, influencing how multilingual pedagogy is integrated into secondary classrooms. At the secondary level, multilingual pedagogy expands disciplinary access, strengthens identity and prepares students for higher education where multilingual academic practices are increasingly necessary.

Several states, including Kerala and Tamil Nadu, already support mother-tongue-based educational practices infrastructurally. Teachers translanguage and code-switch to bridge local conceptual discourse with pragmatic pathways that align with the NCF-SE 2023’s multilingual vision while meeting secondary-stage exam demands.

MODULE 4.2

Classroom activity: bilingual algebra lesson in Tamil Nadu classrooms (Grade 9)



Objective: To help students understand and apply factoring using both Tamil and English, bridging everyday reasoning with mathematical terminology.

Tamil Nadu’s Samacheer Kalvi system publishes Tamil-medium mathematics and science textbooks (and parallel English-medium versions) through secondary grades. This enables teachers to launch topics, proofs and applications in Tamil, then rehearse key English technical terms for exam readiness (e.g. உயிரியல் → Biology; வட்ட சமன்பாடு → circle equation). Many schools use bilingual exercises, mixed-language examples and parallel glossaries to connect Tamil conceptual talk to English symbolization and graphing conventions.

ACTIVITY

The teacher begins the lesson by writing an algebraic expression on the board, e.g. $x^2 + 5x + 6$. In Tamil, she asks, இந்த சமன்பாட்டை கூறுகளாகப் பிரிக்க முடியுமா? (Can we break this into factors?). Students discuss in Tamil how two numbers multiply to give 6 and add to 5. They identify 2 and 3, saying இரண்டு, மூன்று — இரண்டு கூட்டினால் ஐந்து, பெருக்கினால் ஆறு. The teacher affirms their reasoning, then transitions to English: Good! So, we factorize as $(x + 2)(x + 3)$.

Students rewrite the steps in English using algebraic notation, marking the Tamil explanation alongside. This deliberate Tamil→English shift helps students internalize the concept through reasoning in their home language and formalize it in the academic register of English. The session concludes with a short bilingual worksheet to reinforce comprehension.

Source: NMRC, 2024



Above: Students engage in a reading session of various books in English and regional languages at the library. Sardar Patel Vidyalaya, Delhi.

Teaching English literature

In Indian secondary school English literature classroom, using mother-tongue-based and translanguaging practices can make literature more accessible and culturally grounded, and cognitively richer. Instead of insisting on English-only responses, teachers invite students to explore meanings in their home languages, draw cultural parallels, and then negotiate how those ideas map into English literary language. For example, when analysing a poem's metaphor or imagery, a teacher might ask: कैसे यह छवि आपके भाषा (हिंदी / तमिल / बंगाली) में बोले जाने वाली कहानियों से मिलती है? Students may then articulate the metaphor in their language first, and collaboratively translate or adapt it into English, reflecting on what is gained or lost in translation.

Research backed by the British Council shows that many English teachers in India already use occasional L1 practices to explain difficult words, classroom instructions or clarifying concepts, and some facilitate translanguaging in classroom talk (Anderson and Lightfoot, 2018). A more recent comparative study of expert secondary school English teachers found that all engaged in complex translanguaging practices inclusive of learners' languages, often prioritizing learner participation and outcomes over rigid English use (NMRC, 2024). For instance, a teacher might allow students to discuss character motivations in Hindi first, then guide them to frame their arguments in English for writing, thus scaffolding the bridge between local language and knowledge and academic English literacy.



The NEP 2020 supports expanding the use of Indian languages in higher education and providing language support to students who need it.

Multilingual policy and practices in higher education

As students enter higher education (college and university), the linguistic landscape changes yet again. English becomes the dominant medium, especially in technical and professional courses. There is, however, a growing momentum, supported by the NEP 2020, to expand the use of Indian languages in higher education and to provide language support to students who need it. The multilingual context of higher education spells out specific needs of technical students, the availability of materials, fostering a reading culture, and linking school and higher education for a seamless multilingual learning continuum.

Linguistic profiles of students in higher education

University campuses in India are linguistic melting pots. Students come from all over the country with varied mother tongues and from diverse schooling mediums. A typical state university may have a mix of urban, English-medium-educated students and rural, regional-medium-educated students. For example, in a Bachelor's course in Rajasthan, you may have students who completed their Grade 12 education in Hindi medium and some in English



Many students are effectively bilingual, but not always confident in academic English. Some may prefer writing an exam in Hindi.

Above: Masters students in a linguistics class. Jawaharlal Nehru University, Delhi.

medium; a few may speak Rajasthani dialects at home, while others speak Punjabi or Urdu. Similarly, central universities draw students from multiple states – a place like JNU, Delhi, has historically had students conversing in a mix of Hindi, Bengali, Malayalam and other languages, even if classes were conducted in English. Many students are effectively bilingual, but not always confident in academic English. Some may prefer writing an exam in Hindi if allowed, or discussing concepts in their mother tongues with peers. Recognizing this profile is important so that universities can plan bridge courses and not assume all freshmen have uniform language abilities. There is also a subset of students who pursue higher studies in their regional language (some universities offer certain courses in languages such as Hindi, Tamil, etc.). Their challenge comes when references or competitive exams are in English. In essence, higher education institutions face the task of catering to students who think primarily in Indian languages and those fluent in English, as well as those who are bilingual, and everything in between.

The National Education Policy and follow-up actions

The NEP 2020 positions language as both an instrument of equity and a pathway to knowledge-building across all levels of education. Although its clearest guidance on medium of instruction appears in the school sections, the chapters on higher education and the dedicated language section signal a broader shift: universities are expected to integrate language studies within multidisciplinary frameworks and to take renewed, system-wide responsibility for sustaining India's classical, tribal and endangered languages. The policy also emphasizes strong bilingual and multilingual approaches in teaching and learning, affirming that all languages must be taught well regardless of the medium of instruction and that high-quality bilingual materials, including for mathematics and science, are essential. While articulated in the school context, these principles increasingly inform regulatory decisions shaping multilingual admissions, curricular design and resource development in higher education.

Regulatory moves shaping higher education practice

Admissions and assessment languages (CUET-UG). As of 2024, the Common University Entrance Test for undergraduate programmes (CUET-UG) is offered in thirteen languages (Assamese, Bengali, English, Gujarati, Hindi, Kannada, Malayalam, Marathi, Odia, Punjabi, Tamil, Telugu, Urdu), with thirty-three language tests available, explicitly framed as drawing from the NEP 2020.

Teaching/learning in Indian languages (UGC/AICTE). The University Grants Commission (UGC) has repeatedly urged higher education institutions (HEIs) to permit the writing of examinations and coursework in mother tongues/local languages and to expand Indian-language content. The All India Council for Technical Education (AICTE) runs a large technical book-writing and translation programme to seed materials across twelve scheduled languages.

Indian knowledge systems and language mainstreaming. The UGC has issued guidelines to incorporate Indian knowledge systems (IKSs) into higher education curricula, an adjacent driver that normalizes Indian-language scholarship, teaching materials and faculty development within degree programmes.

Below: Digital Santali dictionary used by students. International Institute of Information Technology, Hyderabad.



Implementation strands now visible on campuses

Medium and pedagogy pilots in STEM. Since 2021-22, AICTE-affiliated colleges have offered Bachelor of Technology seats in regional language cohorts (e.g. Hindi, Tamil, Telugu, Marathi, Bengali), supported by translation and faculty mentoring. Early cohorts were small (typically sixty seats per programme) but established feasibility in computer science/mechanical engineering/electrical engineering tracks.

Curriculum frameworks that require Indian-language study. A prominent example is Delhi University's Undergraduate Curriculum Framework (UGCF) 2022, which requires study of two languages across the first two years, of which at least one must be an Indian language, with offerings across Tamil, Telugu, Bengali, etc. This embeds language competence as a graduate attribute rather than a remedial add-on.

Materials ecosystems (translation, textbooks, and open and distance learning). The AICTE has funded technical textbook writing/translation; official updates note twenty-two books in nine languages in the first year, with expansion to twelve languages and eighty-eight subjects identified for year two. The Indira Gandhi National Open University (IGNOU), India's largest open and distance learning (ODL) university, has launched Master of Business Administration (MBA) programmes in Hindi and Odia and is expanding to more Indian languages, often using AI-assisted translation toolchains (e.g. Anuvadini). This signals that professional education can be delivered at scale in Indian languages.

Campus-wide language inclusion policies. Institutes are experimenting with trilingual signage and communication, including targeted Hindi/Indian-language instructional support in select courses (e.g. IIT Kharagpur's three-language policy for campus communications). While symbolic, these actions lower everyday language barriers and support first-generation learners.

English language support programmes at universities. The Linguistic Empowerment Cell (LEC) at JNU is a university-level initiative designed to support students who face challenges with academic English. Similar units exist in other Indian universities (e.g. the Language Cell at Ambedkar University; the Centre for Writing and Communication at Ashoka University). Although not formally mandated by the UGC, these centres draw legitimacy from the Thorat Committee Report (2007), which highlighted the need for remedial English-language support for marginalized students, particularly those from SC/ST backgrounds.

8

Linguistic empowerment through a multilingual education framework in higher education

Linguistic Empowerment Cell

Location: Linguistic Empowerment Cell, Jawaharlal Nehru University.

Duration: 2024–2025.

The Linguistic Empowerment Cell (LEC) in JNU addresses the English proficiency of marginalized students, particularly those from SC/ST backgrounds. The LEC classrooms are highly diverse, bringing together students from different academic programmes, linguistic backgrounds and levels of English proficiency. It offers two non-credit courses, leading to fluctuating attendance and variability in classroom continuity. Despite these constraints, faculty members employ a multilingual, student-responsive pedagogy that combines vocabulary-building, grammar support and oral communication with training in academic skills such as summarizing, paraphrasing and avoiding plagiarism. The teaching approach foregrounds interaction, confidence-building and learner agency rather than rigid coverage of a prescribed syllabus.

BEST PRACTICES AND INSIGHTS

- **Building self-confidence.** Instruction centred not only on language skills but also on enhancing students' confidence, addressing the social power of English, and preparing learners for professional communication demands.

- **Participation-driven pedagogy.** The classroom functioned as a space where student participation actively shaped curricular pacing and content, enabling faculty to align instruction with learners' needs in real time.
- **Collaborative pedagogic ecosystems.** The case study underscored the need for cross-university communities of practice, as remedial programmes often operate in isolation without shared resources or collective problem-solving structures.
- **Interactive and multimodal learning.** Given the multilingual mix of the class, teachers used visual prompts, storytelling, image descriptions, crosswords and role-play to support comprehension and expression in ways not dependent on any single language.
- **Resource and funding needs.** Effective implementation requires sustained material investment for diagnostic assessments, monitoring learning trajectories and developing context-specific teaching tools.

OUTCOMES

The LEC model demonstrates the value of remedial and support programmes in addressing the inequities of India's multilingual higher education landscape. By creating inclusive spaces that recognize linguistic diversity and provide structured academic support, such initiatives strengthen learners' confidence, improve language proficiency and contribute to more equitable participation in university life.

Source: Vrinda Bhatia, OP Jindal Global University





Above: Students looking through a telescope at the annual science festival. Kalinga Institute of Social Sciences, Bhubaneswar.

Facing page: Masters students receive linguistics training. Jawaharlal Nehru University, Delhi.

Language needs of technical students

Technical and scientific fields pose special challenges because the terminology and literature are overwhelmingly in English. A student may possess the intellectual capability to grasp complex engineering concepts but struggle due to language barriers. Recognizing this, support can take forms like:

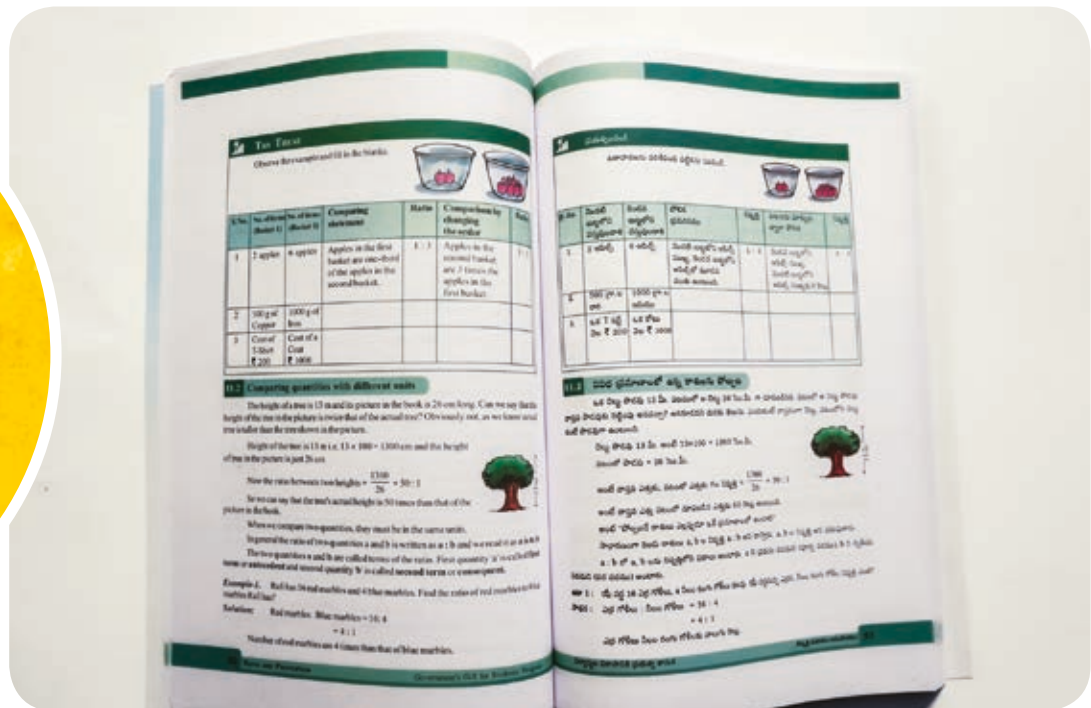
- **Glossaries and reference guides.** India's Commission for Scientific and Technical Terminology (CSTT) has developed glossaries of technical terms in many Indian languages. For example, there is a Hindi engineering glossary that provides Hindi equivalents for thousands of terms. Distributing and using these materials in classes can be helpful. A professor can mention both terms: e.g. "entropy (*randomika* in Hindi)". Knowing the vernacular term can make the concept less intimidating, even if the student ultimately uses the English term in exams.
- **Bilingual study material.** Increasingly, supplementary notes or lab manuals are being made bilingual. A college in Maharashtra may distribute a physics lab manual with instructions in both Marathi and English. This ensures that

safety and procedure understanding are not compromised by language confusion.

- **Peer study groups.** Technical students often form study groups. If they share a mother tongue, they naturally discuss problems in that language. This is actually a valuable resource – some colleges encourage grouping students who are comfortable with each other so they can help each other overcome language hurdles. It is informal, but effective.
- **Faculty sensitization.** Many professors are used to English-medium teaching. Now, orientation programmes for new faculty increasingly include sensitizing them to multilingual classrooms – e.g. encouraging them to accept answers in Hindi in exams if the university allows, or not to penalize minor English grammar issues if the core content is correct (so students are not scared to attempt answering).
- Ultimately, the need is to ensure that language does not prevent talented students from excelling in technical fields. Language support may be a transitional need – after some years of bilingual exposure, most students become comfortable with English terminology. However, support in those initial years is crucial to prevent dropping out or loss of confidence.



The National Translation Mission under the Ministry of Education translates books into and from Indian languages.



Above: Bilingual textbooks in English and Telugu.

Availability of research-based study materials in different languages

This is a significant challenge. While school-level content in various languages has grown, the gap at the higher education level is stark. Most advanced textbooks, reference books and research articles are in English. Some issues and steps regarding this:

- **Textbook translation.** There have been initiatives to translate popular textbooks (for example, the NCERT has translated some undergraduate science books into Hindi; some states have translated medical glossaries). However, translations need to be of high quality and updated with new editions, which is a continuous process. The NEP 2020's vision includes setting up a national translation mission to translate global knowledge into Indian languages. An existing National Translation Mission under the Ministry of Education translates books into and from Indian languages; scaling it up for higher education is on the agenda.
- **Original writing in Indian languages.** One long-term solution is to encourage scholars to write new material in Indian languages. Some professors have started doing this – for example, writing a calculus textbook in Bengali or publishing popular science books in Malayalam. Academic promotion systems could incentivize the creation of learning materials in local languages as a service to the nation.

- **Digital resources.** The internet can democratize access. For instance, Wikipedia in Indian languages, science explainers on YouTube in Hindi/Tamil, and open courseware with subtitles can supplement official materials. The Indian Government's Study Webs of Active-learning for Young Aspiring Minds (SWAYAM) platform for massive open online courses (MOOCs) offers courses in regional languages or dubbed versions, at least. The Bharatvani portal by the CILL is trying to provide knowledge resources in multiple languages.
- **Journals and research.** Beyond textbooks, research literature is nearly all in English. A few journals accept papers in Hindi or other languages (especially in the social sciences and humanities), but they are not mainstream. Imagining a vibrant research culture in Indian languages is part of the vision, where perhaps a physics researcher can publish in Tamil and still be taken seriously. This represents a significant cultural shift, and it is likely a long way off in technical fields; however, in the social sciences, we are seeing a growing acceptance of vernacular scholarship.

Currently, the scarcity of higher education material in Indian languages means that even if undergraduate courses are offered in those languages, postgraduate students still have to rely on English sources. This is why the push for translation and new content creation is critical. It is as much an opportunity as a challenge – a new generation of academics could fill these gaps if policy and funding support it.

Adult education, women and mother tongues

The NEP 2020 replaces the term “adult literacy” with the broader concept of “adult education”, which is treated not as a stand-alone term but includes five components: foundational literacy and numeracy; critical life skills; vocational skills development; basic education; and continuing education. (NEP 2020, Section 21.3). It frames adult education within the vision of lifelong learning, stating: “The policy seeks to promote lifelong learning and provide opportunities for individuals to learn at any stage of life.” (Section 21.1). The section on adult education does not name specific mediums of instruction, but the wider policy commits to “making widely available reading material in Indian languages” through strengthened libraries and digital access (p. 52), implying that adult programmes work in learners’ own languages.

This orientation is reflected in implementation: the Padhna Likhna Abhiyan targeted 5.5 million adult illiterates in 2020–21 using flexible, community-based models and state-developed primers (MHRD, 2020; PIB, 2021). The New India Literacy Programme (2022–27) further mandates that “learners will be encouraged to access the content in local languages... through DIKSHA” (NEP, 2022), operationalizing the NEP 2020’s call for context-relevant, livelihood-linked adult education. It states that “adults in many cases will require different teaching-learning methods and materials than those designed for children. It will be implemented in convergence with Department of Higher Education and other relevant Ministries/Departments at Central and State/UT level” (p. 19).

The 2011 Census recorded an adult literacy rate of 59.3 per cent for females compared to 78.8

per cent for males, underscoring the persistent gendered inequities in access to adult learning (MoSPI, 2011, p. 67). More recent data from the Periodic Labour Force Survey 2023–24 show only modest improvement, with female literacy (7+ years) at 74.6 per cent, still significantly lower than the 87.2 per cent recorded for males (MoSPI, 2024). Research consistently shows that women, particularly from rural, tribal and minority-language communities, remain disproportionately excluded from adult education (UNESCO, 2022). Since a large proportion of non-literate women rely on home and community languages, the absence of learning materials in these languages restricts both comprehension and participation. The NEP 2020 emphasizes that adult education must be delivered “in a manner that is relevant to local contexts, languages and needs” (Section 21.2), making culturally rooted, mother-tongue materials indispensable for genuine inclusion. Without linguistically accessible and contextually meaningful resources, adult education programmes risk reinforcing, rather than reducing, gender gaps in lifelong learning.

As per the recommendations of the NEP 2020, a National Centre for Literacy (NCL) was established in 2021 at the NCERT, with a proposal to support the creation of state-level counterparts, State Centres for Literacy (SCLs) at SCERTs. The NCL is dedicated to “education for all”, with a special emphasis on adult education, aiming to build synergy with and expand upon the NCERT’s existing expertise in developing high-quality curricula for literacy, numeracy, critical life skills, basic education, vocational skills and more. The NCL aims to contribute towards achieving the national goal of 100 per cent literacy by 2030.

Below: A wall dedicated to the community of the Saora tribe. Government Upper Primary School, Gajapati.



Women, particularly from rural, tribal and minority-language communities, remain largely excluded from adult education.

School and higher education linkages for effective implementation of multilingual education programmes at all levels

To sustain MTB-MLE through all levels, there must be synergy between schools and higher educational institutions, particularly those specializing in languages, education and culture.

- **Research and evidence.** Universities (especially departments of linguistics, education, sociology) can research the outcomes of multilingual education and feed this into policy. For instance, a university research centre might conduct longitudinal studies on children who received mother-tongue-based instruction versus those who did not, and how each set fared in secondary/higher education. Such evidence can strengthen or refine MTB-MLE approaches. The research agenda could include documenting success stories, developing assessment tools for multilingual proficiency, studying language acquisition patterns among Indian students, and more.
- **Curriculum development and teacher training.** Colleges of teacher education should align with what schools need. District Institutes of Education and Training (DIETs) and Bachelor of Education (B.Ed.) courses could collaborate with university language departments to ensure new teachers are trained in multilingual pedagogy. If a state decides to introduce, say, MLE in ten tribal languages, it could tie up with a university's tribal language centre to design teacher training certificate courses in those languages. Some universities have folklore study centres or tribal study centres; these can contribute by providing content (folk-tales, songs) that can be used in school materials, as well as by training community resource persons who then work with schools.
- **Translation and language technology.** Linguistics and computer science

departments in universities can work together on translation tools, dictionaries and learning apps that would directly benefit schools, as part of student projects or research. For example, a university might develop a mobile app that helps teachers translate key terms between languages, or software to translate children's stories. The BHASHINI project (Indian language AI) could involve universities, and its outputs (speech-to-text in local languages, for instance) could be used in classrooms for assessments or content creation.

- **Pipeline of multilingual educators.** If students experience good MTB-MLE through school and then see that their languages are valued in higher education, some of them may pursue careers in these fields – as language teachers, translators or researchers. Universities could offer integrated programmes or incentives for multilingual students to pursue careers as educators or academics. For example, someone fluent in an endangered language could get a fellowship to document and create a curriculum for that language at a university, then implement it in schools, thus forming a closed loop of sorts.
- **Policy dialogue.** Forums where school educators and higher education experts regularly interact are essential. For instance, state education departments could have annual seminars inviting university experts on language to discuss the progress of MTB-MLE in schools. University experts (in linguistics/education) could serve on textbook committees or advisory boards for school curricula (which many already do via NCERT or SCERT panels). This ensures that the latest linguistic research and local historical knowledge (from folklore studies, etc.) inform school education.

Linking school and higher education creates a continuum – children learn in their mother tongues, they grow up seeing that advanced learning and careers are possible in their languages (or, at least, that their languages are respected at college), and some even choose to advance the field. This ecosystem approach is what can make multilingual education sustainable across generations. It also aligns with the idea of lifelong learning in one's own language – a person should be able to acquire knowledge from the ABCs to a Ph.D. in an Indian language if she so chooses, or seamlessly switch to another tongue with support, truly democratizing education.

Facing page: Teaching vernacular language to school children. Kalinga Institute of Industrial Technology, Odisha.

If students experience good MTB-MLE through school and then see that their languages are valued in higher education, some of them may pursue careers in these fields. For example, someone fluent in an endangered language could get a fellowship to document and create a curriculum for that language at a university, then implement it in schools, thus forming a closed loop.



Summary

Mother-tongue-based instruction strengthens comprehension and reduces early learning gaps, especially for tribal, rural and minority-language children.

Using familiar languages and cultural knowledge across STEM and humanities disciplines deepens conceptual clarity and engagement.

Bi/multilingual textbooks, teacher-led scaffolding and multilingual assessments provide clearer access to meaning and reduce linguistic bias.

UNESCO and NCF frameworks position L1 as the base for FLN, with gradual language layering supported by NCERT/SCERT materials, teacher training and community engagement.

Case studies from India and global contexts show that culturally rooted, multilingual teaching improves participation, confidence and conceptual mastery.

Assessment in multiple languages and modes reduces linguistic bias and presents a more accurate picture of children's conceptual and linguistic development.

NEP 2020-driven reforms expand Indian-language instruction, bilingual STEM pathways and inclusive sign language approaches, building continuity from school to higher education.

Higher education in India is gradually shifting towards multilingual access, with the NEP 2020 enabling Indian-language instruction, bilingual STEM pathways and multilingual examinations to widen participation and conceptual understanding.

Expanding high quality academic materials, translation ecosystems, digital resources and faculty capacity in Indian languages is essential to sustaining multilingual learning from school to university.



A dance presented in traditional attire from multiple tribes. The patterns on the saris represent different tribes. Kalinga Institute of Social Sciences, Bhubaneswar.



CHAPTER

5

Education of tribal children

When the first day of school begins in an unfamiliar language, learning loss is not gradual, it is immediate, structural and cumulative.

Mother-tongue-based multilingual education translates difference not into deficiency and pedagogic obstacles but into cognitive strength.

Teaching mathematics and science through tribal children's mother tongues, grounding concepts in local ecologies, oral algorithms, indigenous measurement systems and community practices, deepens comprehension significantly, strengthens conceptual transfer, and allows for the creation of a culturally rooted, cognitively powerful pedagogy.



5

Education of tribal children

The National Education Policy (NEP) 2020 recognizes the importance of mother tongues/home languages in education and advocates multilingual education (MLE) as essential for equity and inclusion, particularly for tribal and minority children who enter school speaking languages different from the medium of instruction. It calls

for decolonizing pedagogy by training teachers in culturally responsive multilingual education and by developing teaching-learning materials (TLMs) in students' mother tongues. Language, here, is not merely pedagogical but an issue of epistemic justice, as dominant linguistic hierarchies continues to marginalize Indigenous epistemologies and the right to live a dignified life.

Widening retention gaps: what do enrolment figures tell us about the education of tribal children?

Below: Students, mostly belonging to various tribes, arrive at their school. The school wall is adorned by murals of different tribes. Kalinga Institute of Social Sciences, Bhubaneswar.

The Unified District Information System for Education Plus (UDISE+) 2023–24 presents a complex picture of access and participation among Scheduled Tribe (ST) students. While enrolment in the early grades is nearly universal, participation declines sharply in higher classes.

At the primary stage (Grades 1 to 5), the gross enrolment ratio (GER) for ST students stands at 98.3 per cent, reflecting the success of access-

oriented interventions. At the upper primary level (Grades 6 to 8), the GER remains high, at 95.2 per cent, yielding a combined elementary GER of about 97.1 per cent. However, the transition to secondary schooling (Grades 9 to 10) is marked by a sharp fall in GER to 76.9 per cent, and further to 48.7 per cent at the higher secondary level (Grades 11 to 12) – indicating that fewer than half of ST students remain enrolled beyond Grade 10 (Ministry of Education, 2024, p. 27).





Above: Primary class in Government Upper Primary School, Gajapati.

The UDISE+ data from the earlier year confirms this downward drift: the higher secondary GER declined from 53.4 per cent in 2022–23 to 48.7 per cent in 2023–24, underscoring persistent retention challenges through the post-elementary cycle (Ministry of Education, 2023, p. 91; 2024, p. 97). Government reviews link these losses to home-school language mismatch, distance to secondary schools, limited residential/hostel capacity, gendered safety concerns and migration-related interruptions (Ministry of Education, 2024).

State-wise data from Open Government Data (OGD) India, 2025, and UDISE+ 2023–24 of GER for ST students show sharp variations across primary, secondary and higher secondary levels. States like Andhra Pradesh and Kerala maintain consistently high participation, while Odisha records moderate decline. By contrast, Jharkhand and Madhya Pradesh reveal steep declines beyond elementary schooling, with

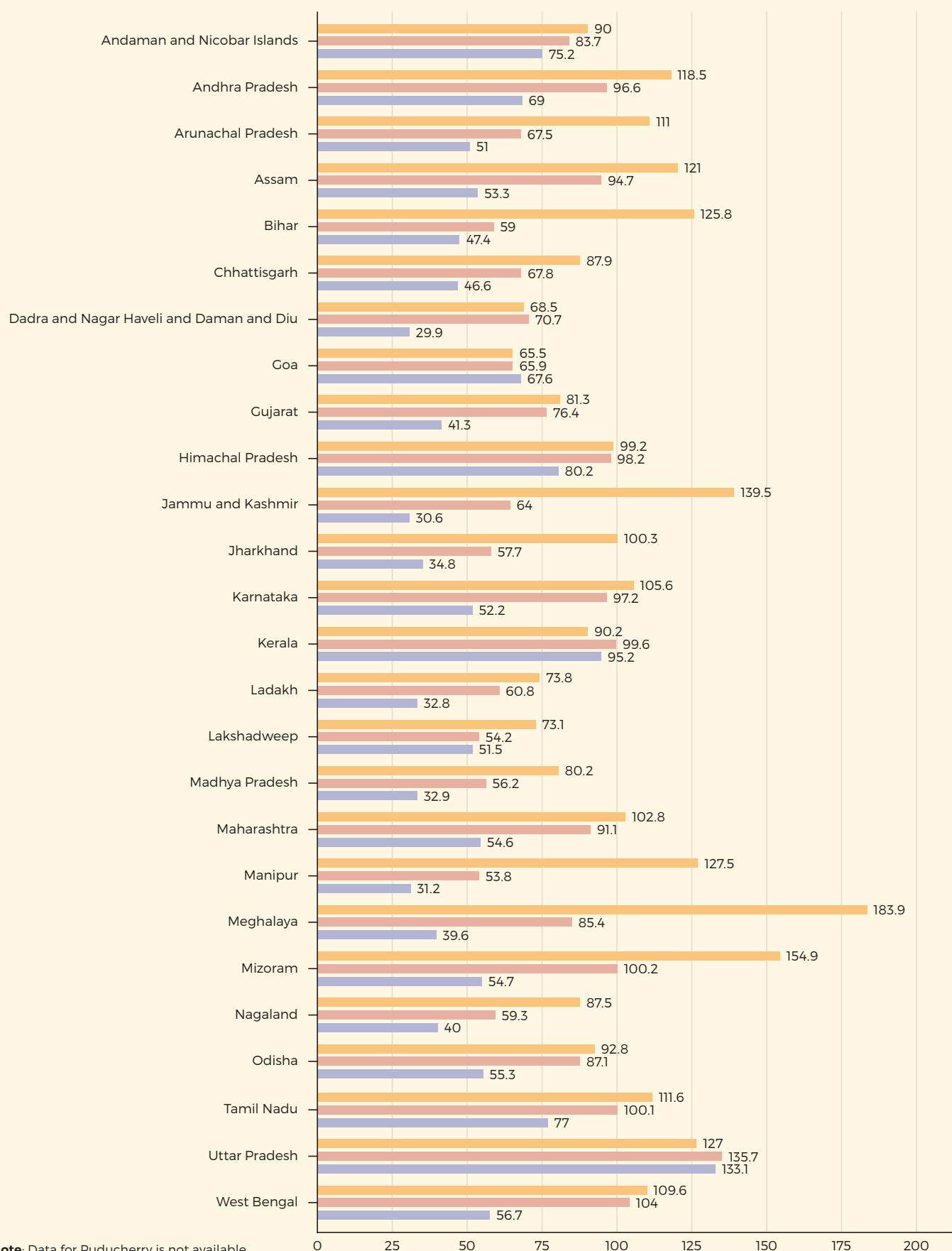
GERs dropping below 35 per cent in both states at the higher secondary level. North-eastern states such as Meghalaya and Mizoram show very high GERs at the primary stage but substantial declines at higher levels.

Overall, the data highlight near-universal access at the primary stage across most states but reveal major challenges in retention and transition to secondary and higher secondary levels, especially in Jharkhand, Madhya Pradesh, Chhattisgarh, Ladakh and Nagaland, where fewer than half the tribal students continue to study beyond Grade 10.

The enrolment trajectory in India demonstrates that while access has been achieved at the foundational and preparatory levels, meaningful participation and completion of schooling remain critical concerns. Gender disparities and socio-economic constraints further compound the issue, as tribal girls are disproportionately affected by cultural norms and household responsibilities that limit their chances of continuing education beyond secondary school. Moreover, while the GER captures enrolment, it does not reflect the quality of learning outcomes or age-appropriate progression, both of which remain weak among ST learners. There is an urgent need for targeted policy interventions – such as improved secondary infrastructure in tribal districts, language-responsive pedagogy, gender-sensitive facilities, and economic support measures – to ensure not just enrolment but quality learning and the successful transition of ST students across all stages of schooling and higher education.

While the GER captures enrolment, it does not reflect the quality of learning outcomes or age-appropriate progression, both of which remain weak among ST learners. There is an urgent need for targeted policy interventions – such as improved secondary infrastructure in tribal districts, language-responsive pedagogy, gender-sensitive facilities, and economic support measures – to ensure not just enrolment but quality learning.

FIGURE 5.1

Gross enrolment ratio (GER) for scheduled tribes by educational stage in 2023-24 (in percentage)**Note:** Data for Puducherry is not available.**Source:** Ministry of Education, 2024, Table 6.3, p. 97.

Primary (Grades 1-5)

Secondary (Grades 9-10)

Higher Secondary (Grades 10-12)

Learning outcomes

The National Achievement Survey (NAS) 2021 indicates that tribal-majority districts continue to underperform across grades and subjects, with persistent gaps in both language and mathematics learning outcomes. While the NAS does not publish disaggregated data for ST students on its public dashboard, analysis of district performance patterns shows that states with higher tribal populations, such as Odisha and Jharkhand, consistently record lower mean scores than the national average across all domains. These disparities reflect systemic factors identified by the NAS itself, including mismatches between home and school languages, limited cultural responsiveness in curricula, and high rates of poverty and migration that disrupt learning continuity (NCERT, 2022, pp. 34–36). Recent large-scale assessments show that learning gaps for ST students persist across grades. The 2024 Performance Assessment, Review, and Analysis of Knowledge for Holistic Development (PARAKH) national survey revealed that ST learners in Grade 3 correctly answered 61 per cent questions in languages and 57 per cent in mathematics, both below the national averages (NCERT/PARAKH, 2024, p. 3).

By Grade 9, the gap widens: ST averages are 47 per cent (languages), 32 per cent (mathematics), 36 per cent (science) and 36 per cent (social sciences), all lower than the national averages for the same (NCERT/PARAKH, 2024).

Complementing this picture, the Annual Status of Education Report (ASER) 2024, covering 650,000 rural children across India, shows modest post-pandemic recovery but continuing weakness in foundational literacy and numeracy. Among government-school students, only 23.4 per cent of Grade 3 and 44.8 per cent of Grade 5 students could read a Grade 2 text. In arithmetic, 33.7 per cent of Grade 3 could subtract; 30.7 per cent of Grade 5 could divide; and 45.8 per cent of Grade 8 demonstrated basic arithmetic competence (ASER, 2025). Although ASER does not disaggregate by social group, several state-level partner studies and census-linked district analyses confirm that districts with high tribal concentrations, particularly in central and eastern India, register substantially lower learning levels than the national mean. These findings highlight how linguistic and socio-economic barriers continue to hinder foundational learning for ST children, making early multilingual, community-based learning interventions crucial for bridging the gap.

TABLE 5.1
Reading and numeracy for Grades 3, 5 and 8

Grade / subject	Outcome	% of children achieving (all India)
Grade 3 – reading	Able to read Grade 2 text	23.4 in 2024 (up from 16.3 in 2022)
Grade 5 – reading	Able to read Grade 2 text	44.8 in 2024 (up from 38.5 in 2022)
Grade 8 – reading	Able to read at baseline level	67.5 in 2024 (up from 66.2 in 2022)
Grade 3 – arithmetic	Able to subtract, at least	33.7 in 2024 (up from 25.9 in 2022)
Grade 5 – arithmetic	Able to divide, at least	30.7 in 2024 (up from 25.6 in 2022)
Grade 8 – arithmetic	Able to do basic arithmetic	45.8 in 2024 (up from 44.1 in 2018)

Source: ASER 2024

The ASER 2024 shows a modest recovery in foundational learning after the shock of the COVID-19 pandemic, with reading ability improving across grades: 23.4 per cent of Grade 3 children could read a Grade 2 text (up from 16.3 per cent in 2022); in Grade 5, the ability reached 44.8 per cent compared to 38.5 per cent in 2022. Arithmetic skills also improved, with 33.7 per cent of Grade 3 students able to subtract and 30.7 per cent of Grade 5 students able to perform

simple division. The figures reflect gains but remain at worryingly low levels. By Grade 8, only 67.5 per cent of students demonstrated grade-appropriate reading, and only 45.8 per cent could handle basic arithmetic, indicating that learning deficits persist into upper grades. While the recovery trajectory is positive, foundational literacy and numeracy (FLN) remain fragile, especially for disadvantaged groups like tribal children who perform below national averages.

Linguistic and structural barriers to tribal education in India

The challenges of tribal education in India can be critically organized into eight interlinked, analytical themes:

- 1 linguistic exclusion in classrooms
- 2 curriculum and cultural irrelevance
- 3 teacher deployment and preparation
- 4 access and secondary stage bottlenecks
- 5 poverty, distress migration and irregular attendance
- 6 gendered barriers and safety
- 7 foundational learning deficits and weak remediation
- 8 data and accountability gaps

1 Linguistic exclusion in classrooms

Most tribal children enter school speaking a tribal mother tongue, yet instruction begins in the state's dominant language – or, increasingly, in English – causing early comprehension breakdowns and long-term learning loss. Research in Odisha and central India (Panda, 2009; Skutnabb-Kangas, 1995; Pattanayak, 2003; Kumar, 2008) shows that when first languages are excluded from teaching and curricular tasks, participation declines and classrooms fall silent. While multilingual education pilots demonstrate clear gains, institutional uptake across states remains limited.

2 Curriculum and cultural irrelevance

Longitudinal studies by the National Multilingual Education Resource Consortium (NMRC) and the National Council of Educational Research and Training (NCERT) in Odisha and Andhra Pradesh show that textbooks, tasks and assessments in non-MLE schools rarely represent local ecologies, livelihoods or cultural knowledge. This disconnect weakens identity affirmation and reduces children's sense of schooling as meaningful (NCERT, 2011; Panda et al., 2011; Panda & Mohanty, 2013). Even where "bridge models" exist, transitions away from tribal languages are rapid and superficial, rendering them tokenistic and providing little continuity into upper primary or secondary grades.

3 Teacher deployment and preparation

Tribal-area schools are often staffed by non-tribal teachers on short postings, many of whom lack proficiency in local languages and have little preparation for multilingual pedagogy (Panda et al., 2011; NCERT, 2011). Opportunities for sustained professional development in mother-tongue-based multilingual education (MTB-MLE) remain limited to small-scale pilots (Manocha, 2022). This combination of unfamiliarity with local contexts and inadequate training constrains teachers' ability to support comprehension, promote participation, or build on children's linguistic strengths.

4 Access and secondary stage bottlenecks

Although UDISE+ 2023–24 reports near-universal enrolment at the primary stage, participation drops sharply at secondary and especially higher secondary levels. Attrition is greatest in regions where schools are located far from habitations, transportation is unreliable, and hostel capacity is insufficient to support residential access. These constraints affect girls and first-generation learners particularly, as they require consistent proximity and institutional support to remain in school as the academic, linguistic and social demands of schooling increase.

5 Poverty, distress migration and irregular attendance

Seasonal migration for construction, brick-kilns, jute and garment industries, sugarcane harvesting and forest-based labour disrupts school calendars and learning continuity. Many children migrate with families or move repeatedly between destination and source villages, often without transfer certificates. This leads to irregular attendance, grade repetition and eventually dropping out. For households facing chronic precarity, the opportunity costs of schooling further intensify, making sustained participation difficult without flexible provisions or portable learning support mechanisms.



Above: Teacher and student engage with each other using the Ho language. Kalinga Institute of Social Sciences, Bhubaneswar.

6 Gendered barriers and safety

Safety concerns restrict schooling for tribal girls significantly, particularly at upper primary and secondary levels, where schools are distant and travel routes are unsafe. Lack of reliable transport, inadequate hostels and limited supervision increase vulnerability to harassment and violence, leading to reduced mobility and early dropout rates. School-home language gaps are a further disadvantage for girls, who receive less academic support than boys and rarely access private tuitions. Although KGBVs and EMRSs offer residential options, overcrowding and the shortage of women wardens weaken their impact. Ensuring safe transport, supportive hostels and community safety networks is essential.

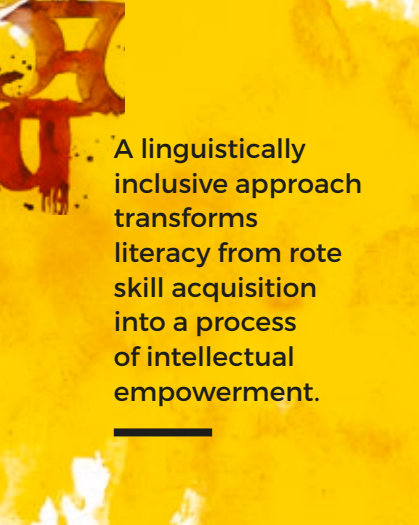
7 Foundational learning deficits and weak remediation

The ASER 2022 and 2024 show slow recovery in basic reading and arithmetic after the pandemic, especially in rural government schools. In districts with high ST populations,

these learning deficits intersect with language barriers, teacher shortages and access constraints, compounding academic failure in higher grades. While community-based catch-up initiatives show the potential for improvement, their coverage remains patchy and short-term. Because the ASER does not disaggregate by ST status, hidden disparities persist, underscoring the urgent need for systematic and well-funded remediation programmes.

8 Data and accountability gaps

Although the UDISE+, NAS and ASER generate valuable system-level indicators, routine state reporting rarely disaggregates data by tribal language group, migration status or hostel access, limiting the precision of planning and intervention measures. This lack of granularity obscures patterns of disadvantage and weakens the ability to design targeted responses. The NAS 2021 emphasizes the importance of interpreting score gaps in relation to demographic characteristics and school conditions at district levels, underscoring the need for stronger data systems and accountability mechanisms.



A linguistically inclusive approach transforms literacy from rote skill acquisition into a process of intellectual empowerment.



Above: Student carrying Odia textbook to class. Kalinga Institute of Social Sciences, Bhubaneswar.

Leveraging linguistic diversity for literacy

In India, over 40 per cent of children are taught in a language they do not understand, which hinders their emotional adjustment, self-confidence and learning outcomes. Despite the presence of multiple languages, clear hierarchies exist, often excluding children's home languages from formal classrooms.

Since the early 2000s, almost all government institutions and non-governmental partners have accepted the need to leverage linguistic diversity to optimize literacy and positive identity development in tribal areas. The most fundamental principle of this resolve is to first recognize children's mother tongues/home languages not as transitional tools but as cognitive and cultural assets central to learning at early stages. Tribal languages

like Kui, Saora, Santali and Gondi have rich oral traditions, semantic precision and ecological knowledge systems that can scaffold early reading and writing. When literacy instruction begins in these familiar languages, children develop phonological awareness, decoding skills and conceptual understanding more effectively, forming strong bridges to regional and national languages. Multilingual literacy practices – such as parallel storybooks, oral-to-print activities and translanguaging pedagogy – enable learners to navigate meaning across linguistic repertoires, enhancing comprehension and participation. A linguistically inclusive approach thus transforms literacy from rote skill acquisition into a process of intellectual empowerment, where every language becomes a pathway to knowledge, belonging and identity affirmation.

Multilingual education in tribal-area schools in Odisha

Government initiative in MLE in Odisha

The state government of Odisha experimented with MTB-MLE in Grades 1 to 4 in tribal areas under the Sarva Shiksha Abhiyan programme. Odisha's initiative began in 2006 and used a transition model in Grades 1 to 3, ensuring conceptual clarity before introducing L2 (regional language) and L3 (English) subjects in a phased manner (Mohanty et al., 2009). Region-specific primers and picture dictionaries embedding local flora, fauna and cultural motifs were developed. Odisha's Nua Arunima, a pre-primary guide that uses folk-tales to teach letters of the alphabet, was developed by UNICEF.

Odisha began with ten tribal languages – Kui, Saora, Juang, Munda, Bonda, Santali, Oram, Koya,

Kuvi and Desia – covering over 2,000 schools across tribal districts, and upscaled its MLE programme to eleven more tribal languages in subsequent years. The state developed primers (big book and small book), storybooks and teaching-learning materials in twenty-one tribal languages, incorporating oral traditions, local ecology and cultural practices. Instruction began in the mother tongue in Grades 1 to 3, with Odia and English introduced gradually as language subjects, aiming for a meaningful transition to higher concepts. Most MLE teachers were recruited from local communities and trained in multilingual pedagogy. Communities were engaged in preparing some of the academic content. Evaluations showed increased enrolment, reduced early dropout rates, and improved reading comprehension, with classrooms becoming more participatory and less alienating (NCERT, 2011; Panda and Mohanty, 2013).

9

Strengthening tribal education through mother-tongue-based multilingual education

Odisha's approach

Location: Odisha, implemented in seventeen tribal-dominated districts.

Duration: 2007–present.

Objective: To improve learning outcomes in education for tribal children by introducing MTB-MLE in the early grades, ensuring smooth transition from the first language (L1) to the state language (L2) and then to a national or international language (L3).

Focus area: Tribal education, foundational literacy, multilingual pedagogy, and community-based resource development.

Odisha has been a national pioneer in implementing mother-tongue-based multilingual education to address language barriers that impede learning among tribal children. Recognizing the vital role that classroom language plays in determining both quality and equity, the state launched its MLE programme in 2007–08 following the recommendation of the state's Tribal Advisory Council, chaired by Odisha's chief minister.

The programme's approach, anchored in the principle of “first language first”, ensures that children begin learning in their own mother tongue before gradually transitioning to Odia (L2) and other languages (L3). This design supports comprehension, retention and self-confidence among learners.

In line with Section 29(2) of the Right to Education (RTE) Act, 2009, which mandates instruction in the child's mother tongue “as far as practicable”, Odisha's model provides structured pathways for multilingual learning at the primary level.

PROCESS

1 Policy framework

- The MLE Policy and Implementation Guidelines for Odisha were approved by Government Notification No. 14118, dated 1 July 2014.
- The programme is implemented by the Department of School and Mass Education, with support from the Scheduled Tribe and Scheduled Caste Development Department.

1 Languages covered

Twenty-one tribal languages are covered by the scheme: Santali, Saura, Koya, Kui, Kuvii, Kishan, Oram, Munda, Juanga,

Bonda, Gadaba, Ho, Gondi, Paroja, Kharia, Didayi, Bhinjal, Bhuiyan, Bhumia, Bhatra and Bhunjia. The programme is implemented across seventeen tribal-dominated districts: Angul, Balasore, Bargarh, Dhenkanal, Ganjam, Gajapati, Kandhamal, Kalahandi, Koraput, Keonjhar, Malkangiri, Mayurbhanj, Nawarangpur, Nuapada, Rayagada, Sambalpur and Sundargarh.

2 School and teacher deployment

- 1,317 MLE schools are operational under the programme.
- These schools serve 89,942 ST children.
- The state has deployed 3,188 MLE teachers (shiksha sahayaks) and 143 language instructors/education volunteers who are fluent in local tribal languages.

3 Learning materials and curriculum

- Textbooks and supplementary reading material have been developed in all twenty-one tribal languages for primary grades.
- These include primers, workbooks, teacher guides and locally contextualized stories and poems.
- The materials are culturally rooted, helping connect classroom learning with tribal life and oral traditions.

4 Tribal Resource Centres

- Tribal Resource Centres (TRCs) have been established in every MLE district. The TRCs serve as repositories of Indigenous tribal knowledge, language, culture, literature, artefacts and oral histories.
- The centres store textbooks, reading material, training manuals, community photographs and traditional artefacts to help children connect formal learning with community heritage.

IMPACT

- 1 Odisha's MLE programme covers twenty-one tribal languages, the largest linguistic inclusion effort of any Indian state.
- 2 Over 89,000 tribal children are receiving education in their home languages through 1,317 schools.
- 3 3,188 teachers and 143 language instructors/education volunteers are engaged in classroom delivery.
- 4 TRCs have strengthened community-school linkages, ensuring that tribal culture and heritage are valued within formal education.
- 5 The programme has institutionalized mother-tongue-based education as a state-level practice and model for tribal inclusion.



Multilingual Education Plus (2006-2012): a path-breaking initiative in tribal education in Odisha

The Multilingual Education Plus (MLE Plus) programme piloted a culturally responsive multilingual education model between 2006 and 2012 in eight tribal-area schools – four each in Gajapati and Kandhamal districts of Odisha. The MLE Plus programme evolved into one of the most transformative multilingual education initiatives in India.

The programme began by appointing eight community mobilization workers (CMWs) and two language leaders from the Saora and Kui communities. It was led by two programme directors and six graduate researchers from

Jawaharlal Nehru University, local teachers (including MLE teachers), school headmasters and community language experts. Its major activities included village-specific sociolinguistic surveys, documentation of traditional knowledge systems, development of a community book titled *My Village, My Story* in collaboration with community members, establishment and management of village libraries, collective reading of evening newspapers at local tea stalls, and sustained work with teachers on multilingual pedagogy, teaching-learning materials and storytelling. The CMWs lived in the villages, bridging schools and communities, while the academic team conducted regular workshops in language, mathematics and environmental studies (EVS), leveraging children's, teachers' and community knowledge to translate and transform academic concepts.

Above: Students of Kalinga Institute of Social Sciences, Bhubaneswar.

10

Building multilingual classrooms

Telangana's approach

**Location:** Telangana.**Duration:** 2019–present.**Objective:** To strengthen MTB-MLE by ensuring that children begin learning in their home language, promoting linguistic inclusion, improving foundational learning, and celebrating India's linguistic diversity through the promotion of Indian languages.**Focus area:** Mother-tongue-based education, multilingual resource development, digital integration, teacher capacity building, tribal language inclusion.

Recognizing that children learn best in the language they understand, the Government of Telangana has systematically advanced the implementation of MTB-MLE and multilingual learning. The initiative integrates regional and tribal languages into the formal education system while also promoting English and Hindi as additional languages.

Telangana's model emphasizes:

- the development of textbooks and digital materials in multiple languages;
- capacity building for teachers in multilingual pedagogy; and
- community-based initiatives that celebrate linguistic diversity.

The state currently supports instruction in eight Indian languages: Telugu, Hindi, Urdu, Kannada, Marathi, Tamil, Bengali and English, and is among the few states to offer bilingual textbooks and digital learning resources.

PROCESS

1 Development of multilingual learning resources

- Textbooks have been developed for Grades 1 to 10 in eight languages, totalling 274 titles.
- These include 48 in Telugu, 43 in Hindi, 43 in Urdu, 28 in Kannada, 28 in Marathi, 28 in Tamil, 48 in English and 8 in Bengali.
- Textbooks are designed to reflect students' home languages and cultural contexts, using local idioms, settings and examples.
- Each textbook includes QR codes linked to the Digital Infrastructure for Knowledge Sharing (DIKSHA) and Telangana Skills, Academic and Training (T-SAT) platforms, providing free access to digital lessons in Telugu, Urdu and English.

2 Digital and media integration

- T-SAT, Telangana's satellite channel, broadcasts lessons in Telugu, Urdu and English.

- All content is mirrored on mobile apps and web platforms for flexible access.

- A multilingual dictionary has been created in nine languages (Telugu, Hindi, English, Urdu, Kannada, Marathi, Tamil, Gujarati and Bengali). It is available both digitally and in print.

3 Classroom practices and multilingual initiatives

- Bilingual textbooks (Telugu–English and Urdu–English) support a smooth transition from the home language to English-medium learning.
- A study on bilingual textbooks found measurable gains in vocabulary, comprehension, and reduced language anxiety.
- Under the Ek Bharat Shreshtha Bharat (EBSB) programme, Telangana partnered with Haryana, introducing daily Haryana Word segments in morning assemblies and hosting cultural exchange days.

4 Capacity building and teacher training

- Comprehensive teacher training is conducted in Telugu, Urdu and English.
- Training modules include multilingual teaching methods, culturally responsive pedagogy and the use of bilingual materials.
- In 2025, a three-tier cascade training model was launched for primary and high school teachers, supported by digital tracking and certification via the Telangana Education App.

5 Language promotion and community engagement

- Schools celebrate Language Days and Library Week, featuring reading corners with books in eight Indian languages.
- Bhasha Sanghams (language associations) were formed in schools to promote peer learning and multilingual exchanges.
- Students participate in daily multilingual activities, greetings and dialogues across Indian languages.

ACHIEVEMENTS AND IMPACT

- Bilingual textbooks have improved comprehension, increased vocabulary acquisition, and reduced learning anxiety among students transitioning to English-medium instruction.
- The integration of QR-coded digital textbooks has improved accessibility, allowing lessons to reach remote learners through DIKSHA and T-SAT.
- Language diversity in schools has expanded, with Bengali-medium instruction now available up to Grade 5.
- 1,212 Bhartiya Bhasha summer camps held in 2025 reached over 121,200 students, integrating literacy, arts, culture and foundational learning.
- Tribal language inclusion: the educator Thodasam Kailas developed storybooks in Lambadi, Gondi and Kolami, preserving tribal languages and fostering inclusivity.

The village as a site of knowledge and pedagogy

The tribal village functions as a living classroom where knowledge is produced, transmitted and practised through everyday activities like farming, weaving, house-building and storytelling. Each act of labour encodes mathematical reasoning, ecological understanding and linguistic creativity, making the village a dynamic epistemic space rather than a pre-schooling stage. In such contexts, elders, artisans and storytellers serve as teachers, and learning occurs through

participation, observation and narration rather than formal instruction. When schools engage with this knowledge ecology – as at a “maths and science *mela*” – they transform local practice into curriculum and local language into cognition. The tribal village, therefore, stands not at the periphery of education but at its foundation, embodying an Indigenous pedagogy rooted in relationships, rhythms and reciprocities with the natural world.

BOX 5.1

NMRC's Maths and Science Mela in Gajapati district of Odisha: a pedagogic encounter between school and community (2012-2015)

The Maths and Science Mela (fair) conducted by the NMRC was as a pioneering initiative in bridging formal education with community knowledge. Conceived by Panda (2012), the *mela* brought together teachers, students, community elders, artisans and traditional knowledge holders to co-create learning experiences that integrated Indigenous wisdom with school mathematics and science within a multilingual education framework.

CONTEXT AND RATIONALE

Tribal learners in Odisha's hilly regions often experience alienation in classrooms where teaching is delivered in Odia or English, languages unfamiliar to their home environment. The *mela* was designed as a dialogic and experiential pedagogic space where classroom science and mathematics could intersect with Saora and Kui communities' Indigenous systems of reasoning, measurement and observation.

DESIGN AND APPROACH

The *mela* was guided by activity-theoretical principles, emphasizing social interaction, community mediation and multilingual meaning-making.

KEY FEATURES

- **Collaborative learning spaces.** Elders, farmers, masons and weavers demonstrated Indigenous techniques such as step cultivation, bamboo house construction, and weaving patterns to illustrate local applications of mathematical and scientific principles.
- **Bilingual pedagogy.** Community knowledge was translated into classroom lessons using both Saora/Kui and Odia, allowing learners to conceptualize scientific ideas through familiar linguistic frames.
- **Contextual curriculum development.** Teachers documented oral algorithms, number systems and

problem-solving strategies from community members to design bilingual learning materials for classroom use.

EXAMPLES OF INTEGRATION

- **Mathematics.** The Saora's dual-base (12 and 20) number system was used to undertake arithmetic operations and place value through oral counting and local games.
- **Physics.** Concepts of force, balance, gravity, angle, etc. were discussed by parents in relation to step cultivation, house construction and agricultural tool-making, while teachers explained how these concepts and ideas are discussed in schools. These activities were adapted into lessons on gravity, equilibrium and friction.
- **Environmental science.** Community classification of soils, seeds and herbs became part of classroom discussions on biodiversity and ecological systems.

OUTCOMES AND INSIGHTS

The *mela* produced multidimensional outcomes in both learning and community engagement.

- **Cognitive gains.** Students demonstrated improved conceptual understanding by linking abstract ideas to familiar everyday practices.
- **Language empowerment.** Children's home languages became legitimate tools of reasoning, increasing participation and confidence.
- **Epistemic reciprocity.** Teachers gained insights into oral and intuitive reasoning, while communities saw their knowledge validated in formal education.
- **Resource creation.** Bilingual charts, posters and numeracy games developed during the fair were incorporated into regular classroom instruction.

Source: NMRC, JNU

Language-in-education policies to support MTB-MLE implementation



Above: Children look at the timetable. Model High School, Osmania University, Hyderabad.

Several Indian states have articulated their own language-in-education policies that adapt national guidelines to local sociolinguistic realities. States such as Odisha, Assam, Chhattisgarh and Andhra Pradesh explicitly support mother-tongue-based multilingual education for tribal and minority learners, developing primers and curricular resources in dozens of Indigenous languages. Some other states operationalize the Three-Language Formula differently, balancing regional language primacy with English as a strong second language. Yet other states emphasize bilingual or trilingual models that strengthen home-language literacy while gradually introducing additional languages for wider communication and mobility.

However, Odisha pioneered the development of a language-in-education policy for a state. It aligns formally with constitutional provisions (Articles 29, 30, 350A of the Indian Constitution) and the Right to Education Act, 2010, and provides several workable guidelines for implementing a multilingual education programme in Odisha. The state language policy prioritizes the use of children's mother tongues in the early years of schooling and as language subjects in higher classes.

1 FIRST-LANGUAGE-FIRST

Instruction for tribal children should begin in their mother tongue (L1) in the early primary grades (typically Grades 1 to 3), with the gradual introduction of Odia as L2 and English as L3.

2 TRANSITION PLAN FOR MEDIUM OF INSTRUCTION

The policy specifies a transition scheme where L1 is dominant in Grade 1 and the medium of instruction shifts gradually to Odia by Grade 5.

3 SELECTION OF SCHOOLS AND LANGUAGES

Schools in tribal-majority hamlets with children speaking a given tribal language are to be selected for MLE. Tribal languages are to be used as the medium of instruction in areas where over 90 per cent of students share that L1.

4 CURRICULUM AND INSTRUCTIONAL MATERIALS IN TRIBAL LANGUAGES

Textbooks, supplementary readers and workbooks are to be developed in tribal languages (using Odia script or appropriate scripts, e.g. Ol-Chiki for Santali) and aligned with the national and state curriculum frameworks (NCFs/SCFs).



Above: Morning prayers being conducted for students studying Sanskrit. Kalinga Institute of Social Sciences, Bhubaneswar.

5 TEACHER RECRUITMENT AND TRAINING

Teachers belonging to tribal-language communities are to be recruited and trained in MLE pedagogy, bilingual/multilingual instruction, and culturally responsive teaching.

6 CULTURAL CONTEXTUALIZATION

Curriculum and textbooks must incorporate local cultural knowledge, traditions and community languages to strengthen comprehension and the relevance of schooling for tribal children.

7 MONITORING, ASSESSMENT AND RESEARCH

The policy mandates continuing monitoring of MLE implementation, assessment of learning

outcomes of tribal students under MLE, and research to refine the programme.

8 SCALE-UP ACROSS TRIBAL DISTRICTS AND LANGUAGES

The policy calls for expansion of MLE from the initial ten tribal languages to twenty one tribal languages (there are twenty-one recognized tribal languages in Odisha) across the state's tribal-dominated districts.

In recent years, both Chhattisgarh and Jharkhand have been trying to develop state-level policies on language-in-education to support child-centric MLE programmes in tribal areas.

Mother-tongue-based multilingual education in Rajasthan

In Rajasthan, multilingual education has advanced through partnerships involving the School Education Department, the Language and Learning Foundation (LLF), UNICEF, Room to Read and local civil society groups. The Government of Rajasthan, with support from LLF, developed programmes in Wagdi, Bhili and Garasiya that have improved early literacy by integrating children's home languages into classroom instruction and creating locally rooted materials. UNICEF-supported initiatives have strengthened teacher training and multilingual resource development in tribal blocks, enabling teachers to use Hindi alongside community languages. Room to Read has contributed by developing bilingual storybooks and read-

aloud practices that support early-grade literacy in diverse linguistic settings. Organizations like Aajeevika Bureau and Vagad Shiksha Sahyog further support migrant children by addressing their language needs and ensuring continuity of learning across school transitions. Aajeevika Bureau works extensively with migrant families from southern Rajasthan, offering bridge schooling, language support and transition services that help children re-enter government schools after seasonal migration. Vagad Shiksha Sahyog complements this by running community learning centres in Wagdi- and Bhili-speaking areas, where tutors use children's home languages to strengthen foundational skills and maintain learning continuity during migration cycles.

Reading interventions using multilingual frameworks in Madhya Pradesh

A comprehensive reading programme by Room to Read emphasized simultaneous and integrated development of three core pillars: **orality** (phonological awareness and spoken fluency); **orthographic expertise** (mastery of writing conventions); and **exposure to diverse texts** (to build fluency and motivation). This non-linear, spiral curriculum is operationalized in Grades 1 and 2 through explicit, systematic

reading instruction in Hindi, which is reinforced by a parallel school library programme for Grades 1 to 5. The model's efficacy is driven by a holistic system that includes the provision of levelled materials, sustained professional development of teachers, continuous monitoring and community engagement, with the overarching objective of creating independent, lifelong readers.

11

Reading interventions

Room to Read in Madhya Pradesh

As an organization, Room to Read (RTR) sees itself in the league of thinking practitioners, focusing on what and how children learn in their literacy classes, and how literacy gains can be a sustained foundation for future learning. Since most of India is multilingual, RTR sees literacy in India evolve in diverse multilingual settings. Depending on the complexity of the situation, literacy instruction needs to be customized to suit children's learning needs.

Room to Read India conducted a sociolinguistic survey of Sirohi district (Rajasthan) and Barwani district (Madhya Pradesh) (Panda, 2018 and 2019) and implemented an intervention in some linguistically diverse regions of Madhya Pradesh. Here, children's mother tongues/home languages (e.g. Bhili, Wagdi) share lexical similarities with the school's instructional language, Hindi. Room to Read's approach leverages this connection strategically, transforming the classroom from a site of linguistic confusion to one of inclusive learning.

RTR's reading intervention programme was built on three core, interconnected strategies:

1 The oral strand

This foundational strategy acknowledges and actively incorporates the child's home language into classroom discourse. It validates the child's linguistic identity, creating a safe and confident learning environment from day one.

2 Systematized *akshara* instruction

A structured, phonics-based approach builds foundational literacy skills in the familiar script, ensuring children master

the alphabetic code (*aksharas*) effectively before transitioning to more complex reading. In a multilingually fluid situation, e.g. in Sirohi, where 'ch' is pronounced 's' (*chammach* is pronounced *sammas*), the phonetic variation of 'cha' as used in Gracia was also introduced along with the 'ch' sound in Hindi. That is, where there was more than one sound for a symbol, both sounds, even if from two different languages, were introduced by creating the right context, so that students could understand the otherwise effortless code-switch that they do in real-life interactions. If children need extra practise time with unfamiliar sounds, words or any other aspect of a language, then that practise time needs to be sensitively given. The idea is not to rush into a transfer model between home and school languages.

3 Multilingual learning materials

The use of purpose-designed multilingual storybooks, conversation charts and posters allows teachers to introduce concepts in the children's home languages and gradually scaffold instruction into Hindi. This bridges the gap between the known (home language) and the new (school language).

The scaffolding effect: how this works in practice

This triad of strategies creates a powerful scaffolding effect. Teachers use multilingual materials to explain stories and concepts first in the local language, ensuring comprehension. As children's understanding grows, instruction fluidly incorporates Hindi, building literacy skills in both languages simultaneously. This method ensures that the home language is not discarded but used as a vital resource to acquire the regional or official language, leading to stronger literacy outcomes in both.

Storytelling in the mother tongue: a cultural-linguistic scaffold to emergent literacy in tribal area schools

Storytelling operates as a powerful cognitive and linguistic scaffold for developing early literacy and fostering active participation in multilingual classrooms in India's tribal areas. Almost all multilingual programmes tried out in tribal-area schools (Andhra Pradesh, Odisha, Chhattisgarh, Jharkhand and Rajasthan) since the early 2000s emphasize storytelling as a culturally rooted pedagogic tool in multilingual education. When stories are drawn from children's home languages, they provide culturally familiar structures that facilitate phonological awareness, sequencing and inferential reasoning, all essential precursors to reading comprehension. Oral narratives strengthen working memory, attention and vocabulary development by engaging children in pattern recognition, prediction and dialogic exchange. As teachers transform local stories into print-rich activities, through picture reading, dramatization or shared writing, children begin to decode symbols anchored in meaning rather than abstraction. This process enhances word recognition, comprehension and narrative fluency, while affirming linguistic identity. Storytelling in the

mother tongue bridges oral and written modes of knowing, simultaneously nurturing cognitive flexibility, reading proficiency and participatory learning grounded in cultural belonging.

Building a digital commons for multilingual education in tribal areas

The Tata Trusts (2025) Multilingual Education Landscaping Exercise highlights some key gaps in the education of tribal children: an acute shortage of high-quality children's literature in tribal and minority languages, early transitions away from home languages, and inadequate teacher preparation to handle multilingual classrooms. These systemic challenges are compounded by fragmented ecosystems, where government, civil society and philanthropy often work in silos, limiting opportunities for scale and sustainability. Both the NCERT and Tata Trusts advocate building digital platforms for multilingual stories and readers.

Below: Students performing a puppet show in English and Telugu. Oxford Grammar School, Hyderabad.



12

Reviving local knowledge and language through community-school partnerships

Bastar Storytelling Festival

Bastar district, Chhattisgarh, is known for its linguistic and cultural diversity, with communities speaking Gondi, Halbi, Dorli, Dhurwa, Chhattisgarhi and other local languages. A predominantly tribal district, Bastar faces significant educational challenges, particularly in early grade learning due to language barriers between home and school. Recognizing this, the Neev Multilingual Education Programme introduced a Storytelling Festival as an innovative intervention. The Bastar Storytelling Festival has been organized annually since 2022, as part of the Neev Multilingual Education Programme led by the Chhattisgarh School Education Department in partnership with the Language and Learning Foundation (LLF).

OBJECTIVE

The primary objective of the Bastar Storytelling Festival is to bridge the gap between schools and communities by integrating local culture, oral traditions and home languages into early education. It aims to:

- 1 enhance children's listening, speaking, reading and writing skills;
- 2 instil pride in cultural traditions and mother tongues; and
- 3 foster stronger collaboration between schools, parents and community elders.

DESCRIPTION

The festival is celebrated during the Bagless Days of the academic calendar. On these days, children do not carry textbooks to school. Instead, classrooms come alive with folk-tales, riddles, myths and cultural narratives told by community storytellers, parents and elders. Children then illustrate these stories, write them in their home languages or dramatize them. Teachers document these stories as teaching-learning materials for continued use in classrooms.

Some innovations of the festival include the use of mother tongue narratives, engagement of intergenerational knowledge bearers, and integration with the NEP 2020's emphasis on multilingual and culture-linked education.

This process not only strengthens language proficiency but also provides children with a sense of cultural continuity and identity. It builds bridges between home and school, ensuring that learning is joyful and rooted in lived experiences.

ACHIEVEMENTS

- 1 Enhanced community ownership of schools and education.
- 2 Provided a culturally responsive pedagogy for tribal and multilingual areas.

KEY PEOPLE INVOLVED

- 1 **Teachers.** Coordinated storytelling sessions, documented material and integrated stories into lessons.
- 2 **Community elders and storytellers.** Shared oral traditions and mentored children.
- 3 **Students.** Actively participated in listening, retelling, writing and illustrating stories.
- 4 **Parents.** Became co-contributors to school-based learning.

IMPACT

- 1 1,100+ schools in Bastar district have hosted storytelling festivals.
- 2 10,000+ children have participated across Grades 1 to 8.
- 3 1,000+ storytellers and 5,000+ community members engaged.
- 4 700+ stories documented in written and illustrated form.

Source: LLF and Government of Chhattisgarh



Multilingual readers (primers) for foundational years

Above: Odia textbooks created in the children's mother tongue. Government Upper Primary School, Gajapati.

States like Odisha, Andhra Pradesh, West Bengal, Chhattisgarh and Jharkhand have developed primers in tribal languages. Odisha has developed primers in twenty-one tribal languages for Grades 1 to 2 (e.g. Kui, Desia, Koya, Kuvi, Saura, Munda, Santali, etc.), while Andhra Pradesh has developed primers in six tribal languages (Koya, Kuvi, Jatapu, Savara, Adivasi Oriya and Konda Dora) under MLE efforts between 2006 and 2014. West Bengal has developed two Santali primers (Santali-Bengali, Santali-Odia); Karnataka a Kodava primer; Chhattisgarh four primers; and Tamil Nadu a “language box” (Irula, Kurumba) and Tholkudi digital ethnography (Irulas, Todas, Narikuravars, Kanikkarars, Kurumbars).

Room to Read complements institutional MLE efforts by addressing the acute shortage of children's literature in minority languages. It has developed over 1,600 original titles in 44 local and minority languages across Asia and Africa, with more than 36 million copies distributed globally (Room to Read, 2023). In India, RTR partners with state governments and community organizations to publish books in tribal and regional languages such as Gondi, Kurukh, Santhali and Chhattisgarhi, thereby filling critical gaps in culturally relevant reading material. Its model emphasizes collaboration with local authors,

illustrators and storytellers, ensuring that books reflect Indigenous traditions, oral narratives and ecological knowledge familiar to children.

This approach has direct pedagogical impact: children are more likely to comprehend, engage and persist when texts mirror their life-worlds. By showcasing local festivals, farming practices and folklore, RTR publications promote both reading motivation and identity affirmation. Recently, RTR introduced the series *My Book, My Story*, which curates child-friendly narratives developed with community participation, encouraging children to see themselves as part of the story world. Beyond publishing, RTR organizes interactive story-reading sessions, read-aloud circles and library activities in schools and communities, further embedding reading as a joyful, social practice. Together, these initiatives strengthen foundational literacy and support multilingual education by rooting early reading in children's lived cultural and linguistic contexts.

The Language and Learning Foundation has developed multilingual and bilingual reading materials and classroom practices where instruction begins in the mother tongue (L1), then strategically introduces the language of instruction (L2). It adopts mixed-language use (L1 + L2) in early classes, using the child's language for new or complex ideas and the instructional language for familiar content. The LLF also complements these reading materials with community engagement initiatives like storytelling festivals and cultural museums in schools to surface local knowledge, narratives and languages in classrooms.

West Bengal has developed two Santali primers (Santali-Bengali, Santali-Odia); Karnataka a Kodava primer; Chhattisgarh four primers; and Tamil Nadu a “language box” (Irula, Kurumba) and Tholkudi digital ethnography.

Primers in Indian Languages

A collaboration between the NCERT and CIIL

Primers in Indian Languages is a collaborative initiative between the National Council of Educational Research and Training and the Central Institute of Indian Languages (CIIL) and represents a significant national step towards realizing the constitutional vision of multilingual and inclusive education articulated under Articles 29, 350A and 351 of the Indian Constitution and reaffirmed by the NEP 2020. Recognizing the centrality of language to foundational learning, the NCERT and CIIL jointly launched this project to strengthen foundational literacy and numeracy among children in diverse linguistic contexts, particularly tribal and minority language communities.

BACKGROUND AND RATIONALE

India's linguistic landscape, encompassing over 1,300 mother tongues (Office of the Registrar General and Census Commissioner, 2011), poses a challenge and offers an opportunity for equitable education. Research by the NCERT and CIIL has consistently shown that children learn best when instruction begins in their home language. However, in many tribal areas, the absence of learning materials in local languages has led to comprehension barriers, low reading achievement and early dropping out. The Primers in Indian Languages initiative emerged to address this gap by producing structured, pedagogically sound and culturally resonant reading materials in tribal and regional languages for the early grades (Grades 1 to 3), aligned with the NCF-FS (NCERT, 2022) and the National Initiative for Proficiency in Reading with Understanding and Numeracy (NIPUN) Bharat Mission objectives.

SCOPE AND LINGUISTIC COVERAGE

The project has developed 121 primers in 117 Indian languages, including 17 Scheduled, 81 tribal and 23 other regional and minority languages, thus reflecting India's multilingual diversity within the framework of Ek Bharat Shreshtha Bharat. Among the tribal languages represented are Halbi, Hmar, Gondi, Kui, Santhali, Jatapu (Kuvi), Juang, Ho, Bhili and Bodo, each primer developed in close collaboration with community experts, local teachers and linguists. The NCERT-CIIL collaboration ensured that each primer was contextually relevant, embedding local narratives, environmental features and familiar objects, to make early literacy instruction meaningful and relatable.

DESIGN PRINCIPLES AND DEVELOPMENT PROCESS

The CIIL's decades of expertise in language documentation, orthography development and pedagogical grammar provided the linguistic foundation for primer design, while the NCERT contributed its curricular and pedagogical framework based on learning outcomes for the foundational stage. The development followed a "mother tongue first" principle, ensuring that primers:

- promote early literacy through familiar oral patterns, songs and stories;
- reflect local cultural and ecological contexts to enhance engagement;
- use visual elements drawn from children's surroundings to support concept formation; and
- enable gradual transition from L1 to regional languages (L2) through bilingual bridging.

The process involved SCERTs, tribal language resource centres and community-based organizations, ensuring validation and adaptation to local speech varieties. The Kui and Desia primers, for instance, were co-developed with Odisha's SCERT and the NMRC, integrating tribal folklore and local art.

IMPLEMENTATION AND PEDAGOGICAL USE

The primers are now integrated into early-grade teaching in selected tribal districts in Odisha, Chhattisgarh, Jharkhand, Assam, Madhya Pradesh, Rajasthan and West Bengal. They form the basis of teaching-learning materials distributed under the Samagra Shiksha Abhiyan, where teachers receive orientation on using these materials for storytelling, shared reading and vocabulary development. These resources also complement state-level MLE programmes and are aligned with NIPUN Bharat's FLN indicators on oral language proficiency, decoding and comprehension.

IMPACT AND FUTURE DIRECTIONS

Early implementation reports from the NCERT's Department of Elementary Education and the CIIL's Language Data Centre highlight enhanced learner engagement, improved attendance and observable gains in reading readiness and oral expression among tribal children. States like Odisha and Chhattisgarh have begun integrating the primers into state textbooks and bridging modules, demonstrating the potential for scaling. The CIIL continues to expand digital versions of these primers through the Bharatvani portal,ⁱ making multilingual content publicly accessible.

ⁱ <https://bharatvani.in/>

Kalinga Institute of Social Sciences model of multilingual education

Below: Introducing students to Odia alphabets, Kalinga Institute of Social Sciences, Bhubaneswar.

Unlike most other educational interventions, the Kalinga Institute of Social Sciences (KISS) model presents a strong and scalable model of multilingual education for tribal children at all levels of education including higher education. It is rooted in a philosophy of linguistic inclusion, cultural respect and academic progression. While not without its challenges, it demonstrates

that when tribal languages are leveraged as mediums of learning rather than barriers at all levels of education, including higher education, tribal students' access to meaningful education improves significantly. Further attention to scalability, resource sustainability and monitoring will be key to leveraging this model for broader tribal education reform in India.



Multilingual education

Kalinga Institute of Social Sciences, Bhubaneswar, Odisha

Established in 1993 and granted deemed-university status in 2017, the Kalinga Institute of Social Sciences in Bhubaneswar, Odisha, is a fully residential and free educational institution for tribal children. It currently serves tens of thousands of tribal students (reporting enrolment of around 40,000 students in 2025) from over sixty tribal communities, covering education from primary grade through university-level studies.

RATIONALE FOR MOTHER-TONGUE-BASED AND MULTILINGUAL APPROACH

KISS recognized early on that tribal children often face significant language barriers in mainstream schooling, where instruction is typically in Odia or English, leading to dropping out, weak comprehension and low retention. To address this, KISS instituted an MTB-MLE programme in 2013, which begins schooling in students' home languages and gradually transitions to a common medium.

KEY PROGRAMME FEATURES AND IMPLEMENTATION

- **Initial learning in home language.** Instruction for early grades is delivered using Indigenous tribal languages and dialects, recognizing the children's linguistic repertoires.
- **Transitioning to a common language.** After building literacy and foundational skills in the home language, students transition to Odia and English, thereby preserving language roots while enabling broader communicative competence.
- **Curriculum and teaching-learning materials with cultural context.** The school incorporates tribal cultural narratives, local art forms and multilingual teaching materials to scaffold learning in both home and regional languages. For example, KISS runs a Centre for Indigenous Languages and Narratives and a School of Comparative Tribal Languages and Literatures to document, preserve and use tribal languages and literatures.
- **Teacher facilitation and residential model.** With its residential model, KISS offers immersive, language-rich environments with tribal peers, Indigenous teachers and culturally responsive pedagogy to reduce alienation and improve engagement.

IMPACT

- **Language and learning outcomes.** The programme improved attendance, retention and learner participation in multilingual classes at KISS, especially among students from tribal backgrounds.

- **Recognition and awards.** The MTB-MLE programme at KISS earned international recognition with the UNESCO King Sejong Literacy Prize in 2022 for its literacy innovations in Indigenous languages.
- **Parent and community attitudes.** A survey of 180 parents found a highly favourable attitude toward KISS's innovative practices, with 94.4 per cent of respondents scoring above the midpoint on an attitude scale. Female parents showed higher favourability (mean = 117.63) compared to male parents (mean = 111.93), $p < 0.01$.

CHALLENGES AND CONSIDERATIONS

- **Scale and resource intensiveness.** While promising, KISS's multilingual model demands substantial investment in teacher training, resource development in multiple tribal languages, and sustaining residential infrastructure. The question remains of how to scale the model broadly across tribal regions at state or national levels.
- **Maintaining language relevance.** It is to be noted that multilingual labs or Indigenous language initiatives at residential schools may become symbolic rather than deeply integrated into everyday practice.
- **Transition balance.** Ensuring that the transition from mother-tongue-based instruction to regional/English medium instruction does not undermine home-language continuity or cultural identity remains a delicate balance.

IMPLICATIONS FOR POLICY AND PRACTICE

- This case study underscores the importance of **starting foundational literacy in the child's home language**, as per multilingual education research.
- Institutional models such as KISS illustrate how a **residential, culturally grounded environment** can reduce dropout rates and alienation of tribal learners.
- The integration of **Indigenous languages and literatures into higher education and language documentation** suggests a sustainable pathway for preserving linguistic diversity while delivering academic progression.
- For widespread adoption, policy must allocate dedicated resources for developing **teaching materials in multiple tribal languages**, training multilingual teachers, and evaluating multilingual pedagogy as part of foundational learning missions (such as NIPUN Bharat).

Multilingual education programmes in tribal areas of the North-eastern states

Assam

Beyond long-standing Bodo-medium streams (now extending to higher secondary with Bodo-medium textbooks), Assam has begun introducing Indigenous languages as mediums

of instruction at the foundational stage. In 2025, Mising was adopted as the medium of instruction in 200 lower primary schools across eleven districts; Deori and Tiwa have also been announced as mediums of instruction for primary schools.

CASE STUDY

15

Tribal-language-medium schools

Assam's approach



Location: Assam



Objective: To introduce tribal languages as mediums of instruction for Indigenous students, with focus on foundational learning (Bodo, Garo, Mising) and middle and higher education (Bodo and Garo).



Focus Area: Development of mother tongue based multilingual education for tribal children

The people of Assam inhabit a multi-ethnic and multilingual society, with up to 207 mother tongues spoken in the state (Office of the Registrar General & Census Commissioner, India, 2011). To provide inclusive education to the speakers of these various languages, the Government of Assam offers fifteen languages at the foundational stage: Assamese, Bodo, Bengali, English, Hindi, Garo, Manipuri, Hmar, Karbi, Rabha, Tiwa, Deori, Dimasa, Santali and, most recently, in 2025, Mising. Textbooks are prepared by the SCERT with support from Sahitya Sabhas (literary groups) of each community, and freely distributed for elementary levels.

BODO

Bodo-medium education started in 1963 from Grades 1 to 4 and gradually began to be upgraded from Grades 5 to 10 in 1968. From 2023-2024, a Bodo curriculum has been implemented for the arts stream in Grade 11. For other streams, students have to transition to English to

continue their education. The script used for Bodo in Assam is Devanagari.

GARO

Garo-medium education in Assam began along with the introduction of Bodo. Currently, Garo is offered as a medium of instruction till Grade 8. Students must transition to English for further education. The script used for Garo in Assam is Roman.

MISING

As part of new policy measures, the Government of Assam has announced the revival of Mising-medium schools. The plan is to re-establish schools where Mising, the language of a significant tribal community in Assam, will once again be used as the medium of instruction in the early grades. This development is intended to ensure that children begin their schooling in their mother tongue.

Mising has been established as the ninth instructional medium at the primary level in Assam's state-run schools. Mising will be a medium of instruction in 200 schools up to Grade 5, and students will learn Assamese and English simultaneously. Furthermore, students in Grades 6 to 10 will be able to study Mising as an elective language.

IMPACT

- 1 Students develop fluency and pride in their mother tongues.
- 2 Greater recognition of tribal languages at the state level.

Arunachal Pradesh

The Ministry of Development of North Eastern Region records thirty to fifty tribal languages spoken, of which the government has introduced early readers in Nyishi, Wancho, Galo, Tagin, Kaman, Taraon, Idu and Tangsa. This marks a systemic move towards using community languages in the early grades, even as English remains the official language.

Meghalaya

In Meghalaya, policy is shifting from merely “allowing” Khasi and Garo in early schooling to requiring them. The State Curriculum Framework (SCF) for the Foundational Stage (2024) anchors mother-tongue use, and a 2025 proposal seeks to make Khasi or Garo compulsory up to Grade 4 across the state.

Nagaland

In Nagaland, alongside flexible state curriculum guidance, Nagaland University has launched a project to develop written grammars for eighteen Naga languages to feed directly into school textbooks, laying essential groundwork for sustained MLE.

Mizoram

Mizo is the medium of instruction up to Grade 8, with a mandated shift to English from Grade 9, an inflection point the state itself associates with increased dropout rates.

Tripura

The state reports a growing number of schools offering Kokborok as the medium of instruction from Grade 1, and Kokborok textbooks now extend up to Grade 12.

Sikkim

Sikkim has directed schools to offer eleven local languages (Bhutia, Lepcha, Limbu, etc.) as second-language options, and the Central Board of Secondary Education (CBSE) now includes Lepcha at the secondary level.

Manipur

Manipur offers eighteen tribal languages as additional language subjects in Grades 9 and 10. The NCERT and the CIIL have developed primers in Anal, Kabui (Rongmei), Maring and Bishnupriya Manipuri.

Below: Sign-language-based multilingual education initiatives in tribal areas by Happy Hands School for the Deaf, Odisha.

Multilingual education initiatives in tribal areas through sign language



Sign-language-based multilingual education initiatives in tribal areas mark an important intersection of linguistic diversity, inclusive education and equity for marginalized learners. Many tribal children with hearing impairments face a double exclusion – first, from mainstream education due to language barriers; and second, from special education systems that rarely acknowledge the multilingual realities of tribal communities. The initiative seeks to bridge this gap by integrating Indian Sign Language (ISL) within the multilingual education frameworks already active in tribal districts, such as those in Odisha, Chhattisgarh and Jharkhand. By combining visual-gestural communication with tribal and regional languages, the programme enables children to access foundational literacy and cognitive development in modalities suited to their lived experience. In doing so, it redefines both “multilingualism” and “accessibility”, emphasizing that sign language is not a compensatory tool but a full linguistic and cultural resource.



Children learn best in their home language.



NEP 2020

Gendered barriers and safety

Language and gender intersect powerfully in Indian schooling, shaping access, participation and learning pathways. Research shows that girls, particularly from rural, Adivasi and minority-language communities, face cumulative disadvantages when the school's medium of instruction is different from their home language, increasing cognitive load and reducing participation (Mohanty, 2019; Panda, 2012b; NCERT, 2023). Girls' linguistic repertoires are often undervalued within classrooms structured around dominant languages such as Hindi or English, leading to what Bourdieu (1991) terms linguistic capital deprivation. The NEP 2020 emphasizes that "children learn best in their home language" (Section 4.12), a provision particularly critical for girls whose educational trajectories are more vulnerable to dropping out early when comprehension barriers align with gendered household labour, mobility restrictions and socio-economic pressures (ASER, 2022). Language-inclusive pedagogy improves girls' engagement, literacy, and confidence, especially in upper primary and secondary grades.

In higher education, language continues to structure gendered access to opportunity. English-medium requirements for admission, coursework and workplace readiness disproportionately disadvantage first-generation women students from non-English backgrounds, reinforcing structural inequalities (Annamalai, 2004; Kumar, 2019). The NEP 2020 calls for "greater use of Indian languages in higher education" (Section 22.15) and underscores multilingual competence as a pathway to equity, yet implementation remains uneven across universities. Studies report that women students rely heavily on peer networks, bilingual materials and translanguaging practices to navigate English-dominant academic spaces (Rao 2020). At the same time, institutions such as central universities, state open universities, and women's colleges have begun offering bilingual textbooks, regional-language lectures and academic writing support, enabling women to participate more fully in disciplinary knowledge-making.

Challenges of early-exit mother-tongue-based multilingual education

Even though several states have started implementing mother-tongue-based education in tribal-area schools, tribal children continue to experience early-exit models where their mother tongues are ignored or phased out after Grade 2 (Panda, 2021; 2012a). Early-exit MTB-MLE programmes, where children's mother tongues are used only briefly before a rapid shift to the dominant school language, produce several well-documented cognitive and conceptual learning losses (Cummins, 2021):

Interrupted conceptual consolidation. When children exit their mother-tongue-based medium too early, they are forced to learn new concepts and a new language simultaneously. This overload prevents deep conceptual understanding and weakens the formation of stable cognitive schemas.

Weak transfer of higher-order skills. Foundational skills such as classification, inference, comparison and reasoning develop most effectively in the child's strongest language. Early exit prevents these skills from maturing fully, leading to weak transfer to the second language (Panda and Mohanty, 2009).

Shallow literacy and limited academic

vocabulary. Children do not get sufficient time to develop fluent reading and a rich academic lexicon in their first language. Because the shift to the dominant language happens before literacy is secure, both languages remain underdeveloped, a phenomenon often called "semilingualism."

Cognitive load and reduced working-memory efficiency. Early transition creates a high cognitive load: children must decode unfamiliar words, process meaning and grasp new disciplinary ideas simultaneously. This slows comprehension and reduces working-memory availability for reasoning and problem-solving.

Limited metalinguistic awareness. Strong metalinguistic skills emerge when children deeply understand the structure, logic and sound system of their first language. Early exit cuts short this development, weakening cross-language transfer and long-term bilingual proficiency.

Long-term academic underperformance. Studies from India and globally show that children in early-exit programmes often lag behind in mathematics, science and reading comprehension because conceptual foundations built in the first language were never strengthened before shifting to another language.



Summary

When tribal children encounter an unfamiliar medium of instruction on their first day of school, exclusion begins immediately. Early comprehension breakdowns compound across grades, ultimately appearing as weakened participation, poor learning outcomes and premature dropout rates.

Access is not the problem, retention and learning are. UDISE+ 2023–24 shows near-universal primary enrolment for Scheduled Tribes, but participation collapses at secondary and higher secondary levels, while NAS, PARAKH and ASER data consistently reveal deep learning deficits in tribal-majority districts.

The barriers tribal children face are not isolated, they are systemic and interconnected. Exclusion of mother tongues, culturally irrelevant curricula, unprepared teachers, secondary-stage bottlenecks, migration, gendered vulnerabilities, poor remediation and weak data systems form a chain that shapes tribal learners' entire educational journey.

When learning begins in children's home languages and draws on community knowledge, outcomes improve dramatically. Evidence from Odisha, Andhra Pradesh, NMRC's maths and science fairs, Room to Read, and LLF shows stronger comprehension, richer participation, and smoother transitions to regional and national languages.

Foundational literacy is strongest when it builds on orality, systematic script learning, and abundant, story-rich environments. Large-scale primer development (NCERT-CIIL), tribal-language storybooks, bilingual materials and mother-tongue storytelling festivals demonstrate how culturally grounded literacy accelerates both reading and conceptual learning.

States are gradually moving from symbolic inclusion to systemic multilingual policy. North-eastern states expanding tribal-language mediums, Odisha's formal MLE policy, emerging frameworks in Jharkhand and Chhattisgarh, and institutional models like KISS show how mother-tongue-based education can be extended across schooling and even into higher education.

Inclusive tribal education must centre gender, disability and equitable access across learning stages. Gender-transformative story programmes, sign-language-linked MLE initiatives, community-led pedagogies and residential multilingual models demonstrate that when language, culture and care align, tribal learners not only stay in school, they succeed.

29-10-2025

Wednesday

Ascending Order

(Smallest number to Biggest number)

ఆరోహణ క్రమం

చిన్న సంఖ్యల నుండి పెద్ద సంఖ్యల వరకు

eg:- 3





CHAPTER

6

Teacher education for mother-tongue- based multilingual education

India's teacher-education system cannot claim to advance equity when it continues to train teachers in languages children do not think in, turning the classroom into a site of quiet exclusion rather than learning.

Unless teacher education is radically reimagined around mother-tongue-based multilingual pedagogy, India will keep mistaking silence for deficit and structural injustice for a child's failure.

6

Teacher education for mother-tongue-based multilingual education

Teacher education is the cornerstone of implementing mother-tongue-based multilingual education (MTB-MLE) in India. Teachers are not only facilitators of subject knowledge but mediators of linguistic equity and cultural recognition. The UNESCO (2025) global guidance on multilingual education, *Languages Matter*, marks a pivotal moment in rethinking how teacher education must respond to linguistic diversity. It calls for systemic teacher recruitment, preparation and professional development strategies that prioritize fluency in learners' home languages and competence in multilingual pedagogy. It urges education systems to move beyond transitional or token models of MTB-MLE and to embed additive multilingualism as a pedagogical norm – one that strengthens comprehension, participation and identity formation. In this vision, teacher education becomes the foundation for social justice, ensuring that no learner is disadvantaged because their first language is not the dominant one in school.

In the Indian context, this global guidance resonates deeply with the goals of the National Education Policy (NEP) 2020 and the National Curriculum Frameworks for Foundational Stage (NCF-FS 2022) and for School Education (NCF-SE 2023), which together call for mother-tongue-based multilingual education in the foundational years. Yet, as UNESCO notes, effective implementation requires teachers who are not only linguistically capable but also pedagogically confident to design learning experiences that build bridges between home and school languages. Teacher education programmes must, therefore, cultivate multilingual awareness, critical reflection and community engagement. They should integrate multilingual and translingual practices, culturally responsive teaching and decolonial approaches that validate learners' linguistic and epistemic diversity. In this sense, multilingual teacher education is not merely a technical reform but a transformative project – linking cognitive development with equity, sustainability and the right to education in one's own language.

Below: Teacher training on MTB-MLE integration. Rajasthan.

Effective implementation of mother-tongue-based multilingual education requires teachers who are not only linguistically capable but also pedagogically confident to design learning experiences that build bridges between home and school languages.





“

... wherever possible, the medium of instruction until at least Grade 5, but preferably till Grade 8 and beyond, will be the home language/mother-tongue/local language.

”

NEP, 2020

Issues and challenges in teacher preparation

Teacher preparation for MTB-MLE faces challenges globally. UNESCO's *Global Education Monitoring Report 2025* draws on the OECD Programme for International Student Assessment (PISA) 2022 to show that students who do not speak the school language at home are 41 per cent less likely to achieve minimum reading proficiency than peers who do, with achievement gaps widening to 60 per cent in some systems (UNESCO, 2025).

In India, state-level evidence reveals the same pattern. Rajasthan's language mapping of Grade 1 learners found that over 72 per cent had limited proficiency in Hindi, the school medium, and only 42 per cent of teachers reported fluency in their students' home languages (Panda, 2019b). In Chhattisgarh, a UNICEF-supported study identified twenty-four home languages in Grade 1; three-quarters of the children and nearly half the teachers lacked adequate proficiency

in the medium of instruction, creating barriers to comprehension (LLF, 2025). These mismatches often lead to low engagement, weak retention and reduced learning outcomes.

Prevailing attitudes compound these gaps. Longitudinal studies by the National Multilingual Education Resource Consortium (Panda et.al. 2011) found that multilingual practices are already widespread in Indian classrooms. These practices need legitimization through a process of destigmatization and pedagogic scaffolding (Panda, 2025) by the state. The MultiLiLa (2016–20) project found a similar context in urban poor neighbourhoods of Delhi and Andhra Pradesh. It finds that teachers and learners are consistently using a mix of languages... not to be discouraged. It emphasized the need for training that helps teachers employ such practices “systematically and purposefully”.

Policy dimensions of teacher education

Above: Teachers are trained to use performance-based methods to make learning more engaging. Oxford Grammar School, Hyderabad.

The NEP 2020 anchors multilingual education in teacher preparation, directing that “wherever possible, the medium of instruction until at least Grade 5, but preferably till Grade 8 and beyond, will be the home language/mother-tongue/local language” (MoE, 2020, Section 4.12). It also mandates reforms in teacher appointment

and pre-service curricula, and continuous professional development to prepare teachers for multilingual classrooms. The historical precursors of these ideas are the Fundamental Right to Education 2009, a position paper on Indian languages (2006) and Odisha's language-in-education policy (2014).

Since around 2016, the NMRC's field-based work has helped destigmatize mixed-language pedagogy in tribal and border-area classrooms, emphasizing that learners' natural multilingual practices are legitimate instructional resources.

The UNESCO Bangkok (2023) report, *Addressing Barriers to Language-in-Education: Global Report on Inclusion and Multilingual Education*, highlights that recruiting, training and deploying teachers proficient in learners' home languages is vital for effective multilingual education. Released on International Mother Language Day 2023 as part of the International Decade of Indigenous Languages (2022–2032), the report underscores inclusive, language-responsive teacher preparation.

The NCF-SE 2023 incorporates multilingual pedagogy across all stages of schooling, requiring teachers to be trained in using local-language resources and assessments (NCERT, 2023). The National Professional Standards for Teachers (NPST) 2023 make multilingual pedagogy a professional expectation, while the Students' and Teachers' Holistic Advancement through Quality Education (SARATHAQ) framework (2021) instructs states to align training at District Institutes of Education and Training (DIETs) and State Councils of Educational Research and Training (SCERTs) with these goals (NPST, 2023; Ministry of Education SARATHAQ, 2021).

At the institutional level, the National Council for Teacher Education (NCTE), the National

Council of Educational Research and Training (NCERT), SCERTs and DIETs are increasingly tasked with mainstreaming MTB-MLE into teacher-education curricula and professional development. NITI Aayog has also emphasized strengthening DIETs and state teacher-education institutions to deliver multilingual training. The NMRC was instrumental in instituting the idea of MTB-MLE with a strong component of teacher education in multilingual pedagogy (Mohanty and Panda, 2010). Since around 2016, the NMRC's field-based work has helped destigmatize mixed-language pedagogy in tribal and border-area classrooms, emphasizing that learners' natural multilingual practices are legitimate instructional resources. Over the following years, the consortium's engagement with schools and teacher-education partners drew increasingly on emerging frameworks such as multilinguality, translanguaging and, more recently, transknowledging, reflecting broader developments in multilingual education research (Panda, 2025).

Since 2018, the Language and Learning Foundation (LLF) has organized several short training programmes on multilingual pedagogy. Common to the approaches of UNESCO and UNICEF, the NCERT, NCTE, NMRC and LLF is that this new multilingual pedagogy respects students' mother tongues/home languages, preserves their linguistic rights and treats their full linguistic repertoires as resources, where language-switching enables transknowledging, a dynamic process of translating and exchanging local and academic knowledge to create inclusive and dialogic classrooms.

Global guidance on recruitment, professional development and multilingual pedagogy

UNESCO's *Languages Matter* (2025) underscores that teachers are the cornerstone of effective and equitable MTB-MLE. Recognizing that language is both a medium of learning and a marker of identity, the global guidance calls for comprehensive strategies to recruit, prepare and empower teachers who can bridge linguistic diversity with pedagogical innovation. It emphasizes that quality MLE depends not only on teacher availability but also on their capacity to use learners' home languages as cognitive and

cultural resources. Building teacher competence in multilingual pedagogy – through pre-service and in-service training, mentorship and professional communities – ensures that linguistic inclusion is embedded in everyday classroom practice. By aligning teacher education, recruitment and pedagogical design with sociolinguistic realities, education systems can transform classrooms into spaces of belonging, comprehension and creativity, where language becomes a vehicle for holistic learning and social justice.



BOX 6.1

Languages Matter: Global Guidance on Multilingual Education (UNESCO, 2025)

Above: Teaching numbers in an engaging way. Government Upper Primary School, Gajapati.



TEACHER RECRUITMENT, PROFESSIONAL DEVELOPMENT AND PEDAGOGY FOR MLE

- Prioritize the recruitment of teachers fluent in learners' home languages and in national or regional languages of instruction
- Deploy educators strategically so that multilingual classrooms have staff capable of culturally and linguistically responsive instruction, especially in regions with minoritized or Indigenous languages
- Recognize and credential multilingual teacher competence as a professional standard. For example, language proficiency in both L1 (the mother tongue) and in the language(s) of wider schooling



STRENGTHENING PRE- AND IN-SERVICE TRAINING TO ENSURE QUALITY MLE INSTRUCTION

- Provide specialist pre-service training modules on multilingual pedagogy, including how to leverage learners' home languages and transition smoothly to additional languages
- Offer continuous professional development, via online platforms, peer communities, mentoring and coaching, that supports teachers in implementing mother-tongue-based and multilingual instruction effectively
- Ensure teacher-education programmes include awareness of sociolinguistic diversity, inclusive pedagogy, and how to design instruction in learners' languages for stronger cognitive and socio-emotional outcomes



ENHANCING PEDAGOGICAL APPROACHES

- Adopt pedagogies that use learners' home language(s) as the medium of instruction for initial years, while gradually introducing and integrating additional languages, particularly for multilingual learners
- Embed translinguaging practices, language-responsive teaching, culturally relevant learning materials and learner-centred strategies that affirm linguistic identity and support comprehension
- Align curricula, assessment and materials with multilingual instruction – making sure that teaching methods reflect the actual language practices of the community and the classroom, not just monolingual norms



Above: A teacher training session on early literacy. Rajasthan.

Core competencies for teacher educators

To strengthen teacher preparation for multilingual and minority-language contexts, teacher educators must develop a clearly defined set of core competencies that integrate linguistic, pedagogical and ethical dimensions. These competencies reflect the principles of the NEP 2020 and UNESCO's global guidance on multilingual education (UNESCO, 2025), which position linguistic diversity as a resource for inclusive and equitable learning.

Language awareness and sensitivity

Teacher educators must cultivate critical language awareness, recognizing that all language varieties and repertoires hold equal cognitive and cultural value. They should be equipped to analyse the sociolinguistic dimensions of any language and the impact of linguistic hierarchies that often marginalize non-dominant languages. Building on the NEP 2020, teacher education must include field-based learning where educators engage with children's actual linguistic environments. Such immersion enables teachers to move beyond token inclusivity and adopt transformative pedagogical practices that validate students' home languages as means of learning.

Multi/translingual pedagogy

A core competency lies in preparing educators to design and implement multilingual and

translingual pedagogies. Teacher-education curricula should integrate systematic training in multilingual resource development, using local storybooks, bilingual materials, visual aids and Indigenous narratives to support instruction across subjects. Concepts such as translanguaging, code-switching and transknowledging should be formally embedded into training programmes, enabling teachers to draw meaningfully on learners' full linguistic repertoires. Teacher educators should also facilitate critical discussions on sequencing L1 (home language), L2 (regional language) and L3 (English) across different curricular areas and ensure that multilingual classroom ecologies remain fluid and inclusive.

Assessment and reflective practice

Multilingual assessment should capture learners' conceptual understanding across their full linguistic repertoires. Teacher educators must train teachers to design formative, multi/translingual assessments, including oral, written and visual modes, that privilege comprehension over linguistic conformity. Reflective practice is essential for analysing how language mediates learning and for adapting pedagogy to scaffold understanding across home and school languages in multilingual contexts.

**MODULE
6.1**
Teacher-education module for multilingual education
NISHTHA 3.0 (FLN)
Module 7: Multilingual Education in Primary Grades

The NEP 2020 and the National Initiative for Proficiency in Reading with Understanding and Numeracy (NIPUN) Bharat Mission emphasize the use of children's mother tongues/home languages/local languages in early education to strengthen foundational literacy and numeracy. The National Initiative for School Heads' and Teachers' Holistic Advancement (NISHTHA) 3.0, implemented through the NCERT's Digital Infrastructure for Knowledge Sharing (DIKSHA) platform, translates this vision into practice. Among its twelve teacher-training modules, Module 7 (Multilingual Education in Primary Grades) focuses exclusively on enabling teachers to integrate multilingual pedagogy into classroom practice. Teachers complete videos, readings, practice tasks and an assessment, earning certification on achieving a score of 70 per cent or higher.

This nationwide programme has reached teachers across thirty-three states and union territories, aligning teacher capacity development with NIPUN Bharat's foundational learning goals.

Course design and learning objectives

Module 7 develops teachers' ability to:

- Understand and describe India's multilingual reality.
- Analyse children's linguistic contexts and behaviour.
- Recognize the role of familiar languages in comprehension.
- Explain MLE principles.
- Apply inclusive classroom strategies.

The course covers the home-school language gap, policy frameworks, multilingual early learning in Balvatikas, and examples of language-inclusive pedagogy.

Pedagogical core and key strategies

Teachers learn five central strategies for multilingual classroom practice:

1 Initial L1 immersion Begin Grade 1 using children's home languages to ensure comfort and comprehension.	2 Balanced L1-L2 progression Gradually introduce textbooks and the second language (L2), maintaining L1 for explanation and discussion.	3 Accepting translanguaging classroom practices Encourage mixed L1-L2 use to support meaning-making and identity formation.	4 Embedding culture and context Link texts and discussions to children's cultural experiences, recognizing knowledge encoded in home languages.	5 Leveraging L1 for literacy Use familiar L1 words in the school script to teach letters, phonics and early writing.
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Implementation and impact pathway

Module 7 is accessible to primary teachers nationwide via DIKSHA in multiple languages. Teachers completing the course gain conceptual understanding and tools to analyse classroom language ecologies, plan L1-led oral tasks, gradually introduce L2, and

integrate local songs, texts for comprehension. The course exemplifies how digital teacher education can advance multilingual, culturally responsive foundational learning, bridging national policy with classroom transformation.

Source: NCERT, 2021

Linking language mapping to language-in-education policy

Linking language mapping to teacher education is essential for translating linguistic realities into meaningful pedagogic practice. By identifying the languages children use at home and in their communities, language mapping offers a precise empirical base for designing teacher preparation programmes that reflect classroom diversity. It enables states to tailor pre-service and in-service training, particularly in multilingual pedagogy, local language proficiency and culturally grounded instruction, and align it with NEP 2020, NCF-FS 2022 and NCF-SE 2023. When integrated into teacher education, mapping ensures

that MTB-MLE is not implemented through assumptions or administrative convenience but through actual linguistic evidence. States such as Odisha, Jharkhand, Madhya Pradesh, Rajasthan and Chhattisgarh demonstrate how mapping can guide the recruitment of local teachers, the development of tribal-language primers, and the creation of contextually relevant instructional strategies. Organizations like the NMRC, LLF and Room to Read used sociolinguistic surveys as a foundation for preparing teachers who can deliver equitable, multilingual and comprehensible education in linguistically diverse classrooms.

TABLE 6.1
Linking language mapping to teacher training in Rajasthan, Chhattisgarh and Jharkhand

State	Background	Intervention	Results
Rajasthan	A 2025 survey of 243,532 Grade 1 learners in 20,298 schools revealed that over 72 per cent had limited proficiency in Hindi, while only 42 per cent of their teachers reported fluency in the students' home languages	Findings informed DIET training modules on bilingual pedagogy and guided the creation of multilingual materials	Early implementation showed stronger teacher preparedness and classroom engagement (LLF Annual Report, 2024)
Chhattisgarh	A UNICEF-supported survey identified 24 home languages in Grade 1, with 75 per cent of students and nearly half their teachers reporting inadequate proficiency in the school medium	The state developed a teachers' handbook and delivered in-service training on multilingual pedagogy	Classrooms applying these approaches reported higher participation and improved Hindi vocabulary acquisition
Jharkhand	The Palash programme (2024–29) was launched to integrate MTB-MLE in 1,000 schools across 7 districts	Palash combines multilingual textbooks, classroom libraries and continuous professional development for teachers	Embedding materials with training ensures teachers use them effectively, improving FLN outcomes

Source: UNICEF and LLF

TABLE 6.2
Material preparation and teacher development: reading programmes in Madhya Pradesh and Rajasthan

State	Background	Intervention	Results
Madhya Pradesh and Rajasthan	Room to Read, with academic support from the NMRC, carried out sociolinguistic surveys to integrate multilingual contexts in reading programmes in Barwani district of Madhya Pradesh and Sirohi district of Rajasthan	Using the linguistic fluidity and multilinguality of classrooms for phonetic awareness and development of critical reading skills	Embedding multilingual materials with training ensures teachers use them effectively for reading

Source: Room to Read and NMRC, JNU

Facing page: Primary class students learning the alphabet in English and Hindi. Sardar Patel Vidyalaya, Delhi.

Palash: multilingual education for foundational learning in Jharkhand

The Government of Jharkhand in collaboration with the Language and Learning Foundation

Location: 1,000 government primary schools across seven districts of Jharkhand: Latehar, Simdega, Lohardaga, Khunti, West Singhbhum, Sahibganj and Dumka.

Duration: July 2024-present

Objective: To strengthen foundational literacy and numeracy outcomes of tribal children by systematically integrating their home languages, Santhali, Mundari, Ho, Kurukh and Kharia, into early grade teaching and learning, addressing the linguistic gap between home and school.

Focus areas: Foundational learning, multilingual education, teacher capacity building

The Government of Jharkhand launched the Palash Multilingual Education Programme in collaboration with the LLF, introducing five tribal languages – Santhali, Mundari, Ho, Kurukh and Kharia – into foundational classrooms. The programme seeks to create linguistically inclusive learning environments by embedding children's familiar languages in classroom processes. This approach aims to improve comprehension, early literacy and numeracy performance, classroom

participation, and overall confidence among young learners.

A language mapping survey conducted in the state in early 2024 across 8,244 primary schools in seven districts revealed a significant linguistic mismatch: only 23 per cent of Grade 1 children understand Hindi, the state's official language, while the majority speak home languages such as Santhali, Mundari, Ho, Kurukh, Kharia and Nagpuri. The survey found that in nearly 80 per cent of schools, children face moderate to severe learning challenges due to this language gap, while many teachers, too, have limited proficiency in the children's home languages.

Through the development of culturally responsive teaching-learning materials, teacher-training programmes, on-site academic support and periodic internal and external assessments, the Palash programme attempts to address the challenges outlined by the survey. The process iterates on learnings from similar programmes implemented by the LLF in Bastar.

PARTNER ORGANIZATIONS

Government of Jharkhand (Jharkhand Council of Educational Research and Training, Samagra Shiksha), Language and Learning Foundation (LLF), Maitri Trust, UNICEF.

Source: LLF



Integrating community knowledge in pedagogy through multi/translanguaging and transknowledging

A distinctive strength of the NEP 2020 lies in its commitment to integrating community knowledge systems and local expertise into pedagogy, reaffirming that “India’s rich tradition of knowledge systems and learning will be rejuvenated and integrated with modern knowledge” (MoE, 2020, p. 8). Realizing this vision requires preparing teachers to work meaningfully with community knowledge holders – elders, cultural practitioners, artisans and families – so that curriculum development and classroom instruction reflect children’s lived experiences, linguistic worlds and cultural epistemologies. This orientation aligns with international evidence showing that education that honours Indigenous knowledge and builds bridges between home and school contexts leads to stronger learner engagement, comprehension and achievement (UNESCO, 2016; 2017), particularly in communities where children’s home languages differ sharply from the languages of formal schooling.

Teacher education, therefore, needs to prepare educators not only to value local knowledge but also to use language and culture as scaffolds for conceptual learning. The NMRC’s teacher-education programmes on MTB-MLE draw on ideas of “funds of knowledge” (Moll, 1993), transknowledging (Heugh, 2021) and multi/translingual pedagogy (Panda, 2025) for training teachers in how to help children’s gradual movement from oral communication to conceptually mediated literacy. Through such training, teachers learn to design lesson plans that intentionally mobilize children’s full linguistic repertoires for meaning-making, allowing them to express ideas across languages and deepen understanding by shuttling between linguistic resources rather than compartmentalizing them. In this integrated vision, multilingualism becomes both a cognitive asset and a pedagogical strategy, enabling learners to connect community knowledge with school learning and participate more equitably in classroom discourse.

Below: Teachers converse with students in Ho and Kuwi. Kalinga Institute of Social Sciences, Bhubaneswar.



Teacher education for integrating community knowledge and language

MLE Plus programme

Instead of limiting MLE interventions to language transition and textbook development, the MLE Plus programme at Jawaharlal Nehru University – led by Minati Panda and Ajit K. Mohanty, with funding support from the Van Leer Foundation, the Netherlands – sought to integrate community knowledge systems, oral epistemologies and Indigenous pedagogies into teacher education to strengthen learning outcomes for marginalized linguistic communities, specifically the Saora and Kondh groups in Odisha's Rayagada and Kandhamal districts. The programme reimagined teacher education as a process of dialogic engagement across four interconnected forms of participation:

COMMUNITY-BASED CURRICULUM CO-CREATION

Teachers and community members collaboratively generated curricular content through village surveys and ethnographic studies focused on local understandings of science, ecology and mathematics. Storytellers, farmers, carpenters and mothers were invited into classrooms as knowledge bearers, bridging oral traditions and formal education. Participatory workshops produced valuable resources: twenty bilingual (Saora-Odia and Kui-Odia) early readers and two bilingual community-authored books, *My Village*, *My Story* (one each for Saora and Kondh community). These resources integrated local taxonomies, counting systems, agricultural practices and oral histories into school learning, reaffirming the legitimacy of community epistemologies. Participation and collaboration with people and their knowledge systems was key to teacher reform at this stage.

TEACHER EDUCATION AND CRITICAL PEDAGOGY

Teachers underwent four structured training cycles (five days each) that moved progressively from interrogating one's own concepts and ideas about tribal communities and their language and knowledge systems to critical epistemic reflection. Training modules included:

- Understanding students' sociolinguistic contexts and attitudes.
- Revisiting beliefs about learners' cognitive and intellectual capacities.
- Critically analysing epistemic hierarchies that marginalize oral knowledge.
- Designing bilingual learning activities using local experiences to teach abstract concepts. For example, using Saora counting to explore mathematical equality or employing oral algorithms to explain division and ratio.

These sessions helped teachers view multilingualism as a cognitive resource, enabling students to move from everyday to scientific reasoning through culturally grounded examples.

COMMUNITY WORKERS AND THE VILLAGE AS A SITE OF KNOWLEDGE

Eight community workers (four Saora and four Kondh) were trained as cultural-aspirational mediators to bridge schools and villages. Their work involved documentation, bi/multilingual literacy facilitation, running a village library and co-teaching. They created language and art corners using local materials and help, documented children's stories and the community's ecological experiences, and maintained a discursive environment through evening discussions in small gatherings, thus re-inscribing the village as a legitimate site of critical pedagogy. This collaboration blurred the school-community divide, reinforcing the view that knowledge is distributed and dialogic rather than hierarchically transmitted.

MATHS AND SCIENCE MELAS: PEDAGOGIC ENCOUNTERS BETWEEN SCHOOL AND COMMUNITY

Through maths and science *melas* (fairs), elders, artisans, singers, teachers and students co-examined both Indigenous concepts and ideas embedded in local number systems, measurements, architecture, step cultivation, house building, local art and games, etc. and corresponding classroom/academic concepts. For instance, the Saora's two base (12 and 20) number system, heuristics for oral counting addition and subtraction, and knowledge of force and gravity in house construction were transformed into bilingual classroom activities. These *melas* served as bidirectional assessments: teachers learned from community problem-solving repertoires, while communities witnessed their knowledge recognized in formal schooling.

PEDAGOGICAL OUTCOMES AND TEACHERS' REFLECTIONS

A significant transformation was observed in teachers' attitudes and knowledge. Teachers appreciated the emergent nature of bi/multilingual literacies combining Saora, Odia and some English and Telugu. They began to view themselves as co-learners, shifting from authority figures to facilitators of collective meaning-making. The experience reframed the classroom into a space where oral, ecological and performative knowledge coexisted with formal disciplines.

The Integrated Teacher Education Programme



The Integrated Teacher Education Programme (ITEP), introduced by the National Education Policy (NEP) 2020 and implemented by the National Council for Teacher Education (NCTE), represents a structural transformation in teacher preparation in India. By integrating disciplinary studies, professional education and practical school-based experience into a single four-year dual-degree programme, ITEP aims to prepare a new generation of teachers equipped to address linguistic and sociocultural diversity in classrooms. ITEP mandates coursework on multilingual education, early literacy, inclusive education and language-across-the-curriculum, reaffirming NEP's principle that children learn best in their mother tongue during the foundational years. Multilingual pedagogy is treated as a core competency rather than a specialised elective. The pre-service curriculum under ITEP includes focused training in multilingual language practices, including code-switching, translanguaging and transliterating. The programme further underscores the need for teachers to integrate local knowledge systems, oral traditions and cultural narratives into instructional design.

Above: Teachers use innovative and interactive teaching methods. Kalinga Industrial Institute of Technology International School, Bhubaneswar.

Teacher-education programme with focus on teaching English in a multilingual classroom

Teacher-education programmes in India are gradually moving beyond the monolingual mindset to equip teachers for multilingual English classrooms. As emphasized in British Council reports (Heugh et al., 2019; Lightfoot et al., 2022), effective English teaching depends on teachers' capacity to use learners' home languages as cognitive and cultural scaffolds. However, most pre-service and in-service courses still privilege English-only delivery, overlooking the linguistic repertoires that children bring to school. Embedding planned translanguaging strategies, bilingual resources and metalinguistic awareness into teacher-education curricula can help future and practising teachers connect English learning with children's familiar languages, thus enhancing comprehension, participation and learner confidence.

At the same time, multilingual Indian classrooms present enduring and systemic challenges. Mohanty, Panda and Pal (2010) describe a

persistent "double divide" between home, school and English, which deepens inequality in low-cost English-medium schools in Delhi. Teachers often work with large, linguistically diverse groups, with limited pedagogical preparation and with assessment systems that reward English monolingualism. Anderson and Lightfoot's (2019) research indicates that while some multilingual practices exist, they are rarely strategic or institutionally sustained. In response, the British Council's English Language Training (ELT) frameworks advocate for English teaching that builds on mother tongue/home language strengths rather than replacing them. State initiatives such as Odisha's DIET-based multilingual training and Andhra Pradesh's Supporting Andhra's Learning Transformation (SALT) reforms, which position teachers' professional development at the heart of foundational learning, demonstrate emerging models for integrating MTB-MLE into English teacher education.

MODULE
6.2
Teaching English using a multilingual framework

The British Council has a resource book titled *Using Multilingual Approaches: Moving from Theory to Practice* (Heugh et al., 2019) for teachers who teach English as a subject and for teachers who use English as the medium of instruction in classrooms with students in multilingual societies. This book serves as a module or guide for teachers on using a multilingual framework for English language teaching and for teaching mathematics in multilingual English-medium classrooms. The British Council offer related online courses and professional development

resources under the umbrella of “English in a Multilingual World”.

Language mapping and planning

Teachers begin by creating a rapid language profile of each class, noting every learner’s home language and literacy level. Two sets of objectives are set: English learning goals (vocabulary, syntax, fluency) and cross-linguistic goals (translation, bilingual word banks, mediation). Lesson plans identify clear phases for L1, L2 and English-only practice, supporting “planned translanguaging” as proposed in the British Council guide.

Core multilingual routines

Teachers use six recurring multilingual routines to make English learning meaningful and equitable:

1
Preview-view-review

Introduce key ideas in home languages, teach in English, then revisit meaning through L1.

2
Bilingual word walls

Post frequent words with equivalents in home languages.

3
Contrastive noticing

Compare structures (e.g. SOV vs. SVO order) to enhance metalinguistic awareness.

4
Buddy mediation

Peer dialogue in home languages before English production.

5
Multilingual texts

Use teacher-glossed English texts with local-language margins, gradually reducing scaffolds.

6
Exit tickets

Learners summarize understanding in any familiar language.



Source: British Council, 2019

Lesson arc example

A sample lesson on sequencing connectors follows this structure: activate prior knowledge (L1 narration), input English text, peer sense-making in L1, bilingual sentence production, and reflective assessment.

Assessment and outcomes

Each learner maintains a dual-language portfolio that combines English tasks with home-language reflections, enabling progress tracking through English assessments and L1 diagnostic discussions. This approach results in 20–30% vocabulary gains, 70% coherent three-sentence retells, and improved learner engagement.

Harnessing India's digital and media ecosystem for multilingual teacher education

India's expanding ecosystem of digital and media resources offers powerful opportunities for training teachers in multilingual pedagogy. Platforms such as DIKSHA, Swayam Prabha, NISHTHA and state learning portals can host modular courses, interactive videos and model classrooms that demonstrate multilingual scaffolding and culturally grounded instruction. StoryWeaver's vast open-access library of bilingual and tribal-language storybooks enables teachers to see how local languages can be meaningfully

integrated into literacy instruction, while digital mathematics resources – including visual maths apps, multilingual concept videos and interactive problem-solving tools – show how home languages can support conceptual understanding. Regional language television, community radio and mobile-based applications further extend access to pedagogic materials, pronunciation guides and community language primers, making high-quality, context-sensitive training scalable even in remote and tribal regions.

Designing teacher-education modules for multilingual pedagogy

By embedding the broader semiotic strategies of multilingual and translingual practices across different stages, teacher education can produce educators who view linguistic diversity not as an obstacle but as the most powerful resource for both foundational and higher learning. Vygotsky's (1978) social mediation, Cummins's (2000) pedagogic translanguaging, Canagarajah's (2011; 2012) translingual practice Panda's (2025) multi-/translingual pedagogy, and India's sociolinguistic theories position teachers as cultural mediators who empower learners to participate, learn and think critically in India's multilingual classrooms.

and English are introduced as supplementary resources while retaining the mother tongue as the primary cognitive anchor. This approach prevents language and knowledge displacement and equips teachers to design learning trajectories, using both translanguaging and code-switching, that sustain children's everyday concepts alongside their acquisition of additional linguistic codes.



Foundational and preparatory stage (Grades 1 to 5)

PEDAGOGICAL EMPHASIS ON TRANSLANGUAGING AND CODE-SWITCHING

Teachers are trained to use the child's mother tongue or home language as the primary medium of instruction in the foundational stage, ensuring that early learning builds on familiar linguistic resources. Instead of suppressing natural translanguaging, teachers are encouraged to allow children to move fluidly across languages to make meaning. This approach recognizes linguistic diversity as an asset, enabling comprehension and participation for all learners.

At the preparatory stage, teachers are trained to foreground the additive model of multilingualism where state/regional languages

LEARNING THROUGH ACTIVITIES

Teachers are trained to make pedagogic use of local resources like folk-tales, local rhymes, proverbs, games and songs in classrooms to help children bring their cultural knowledge and move from empirically oriented everyday understanding to abstract concepts. These resources provide semiotic support to vocabulary development, storytelling, imagination, and literacy and numeracy readiness.

MULTILINGUAL ASSESSMENT

Multilingual assessment in foundational and preparatory stages should allow children to demonstrate understanding in their mother tongues/home languages, ensuring that evaluation reflects cognitive ability rather than proficiency in a dominant school language. Such assessments could include oral narratives, picture-based tasks, games and multimodal responses for assessing children's linguistic and academic competence. In the preparatory phase, multilingual assessment includes bilingual reading tasks, oral retellings and simple writing

exercises that allow learners to demonstrate literacy and numeracy development across their mother tongues/home languages and other newly introduced languages. Such assessments ensure that the gradual introduction of regional languages or English does not compromise comprehension or early conceptual learning.



Middle stage (Grades 6 to 8)

PEDAGOGICAL EMPHASIS

At the middle stage, teacher preparation must stress discipline-specific multilingual strategies, where mathematics, science and social studies concepts are introduced in L2 with scaffolding in the mother tongue/home language to ensure comprehension. Teachers mediate abstract ideas through familiar linguistic frames so that every child participates and classroom dialogue prevents conceptual gaps when transitioning to regional- or English-medium instruction.

MULTI/TRANSLINGUAL PRACTICE AND TRANSKNOWLEDGING

Teachers are trained to view code-switching and translanguaging not as interference but as resources for transknowledging and developing higher-order reasoning in academic subjects. By normalizing these language practices, teacher education reframes multilingualism as a cognitive and participatory strategy that strengthens critical thinking and enables equitable participation for learners from non-dominant linguistic backgrounds. For example, in a mathematics class, a teacher might introduce the concept of ratio and proportion

by first explaining it in the students' home language (e.g. Odia or Santali) using everyday examples like sharing rice or dividing land, and then guide learners to express the same concept formally in English. This translingual approach bridges intuitive understanding with academic terminology, ensuring both conceptual clarity and disciplinary literacy.

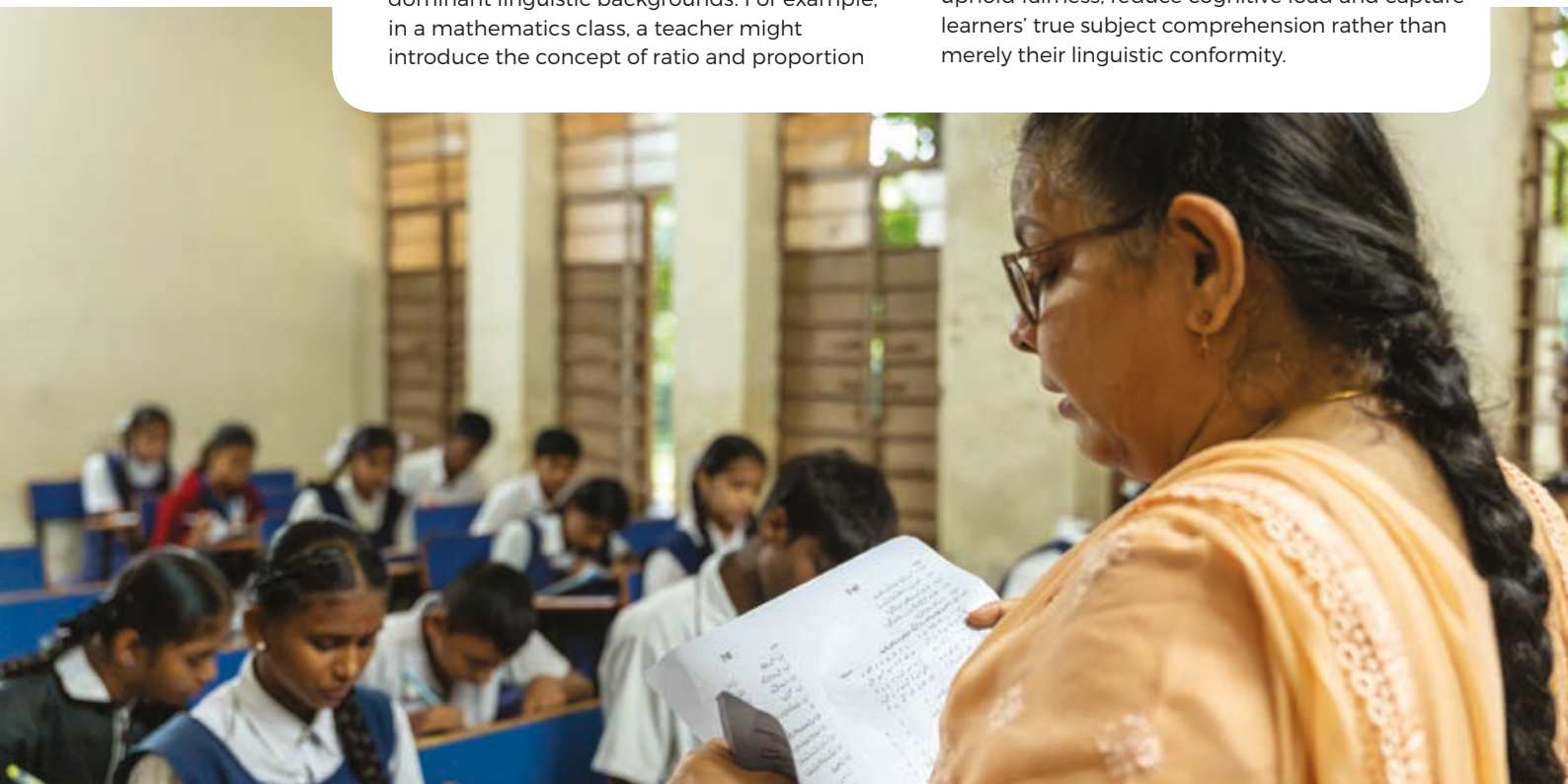
DESIGNING CULTURAL AND SEMIOTIC INPUTS

Training modules equip teachers to design and use graphic organizers, labelled diagrams, multilingual digital tools, storytelling and puppetry that support multimodal resources. These cultural and semiotic resources, presented across multiple scripts and languages, often promote cross-linguistic transfer while strengthening learners' cognitive flexibility, problem-solving and abstract reasoning, and supporting teacher pedagogy in linguistically diverse classrooms.

STUDENT ASSESSMENT IN L2 WITH SENSITIVITY TO MULTILINGUAL SCAFFOLDS FOR TEACHERS AND EDUCATORS

Teachers may be trained to design monolingual assessments with scaffolds that acknowledge learners' multilingual repertoires, such as glossaries, contextual prompts or allowance for brief explanatory responses in the home language at the middle school level. While the target language of assessment is maintained, students are not penalized for strategically drawing on other linguistic resources. Multilingual assessments uphold fairness, reduce cognitive load and capture learners' true subject comprehension rather than merely their linguistic conformity.

Below: Urdu being taught to middle school students. Model High School, Osmania University, Hyderabad





Secondary stage (Grades 9 to 12)

PEDAGOGICAL EMPHASIS ON MULTILINGUAL SCAFFOLDS

Teacher preparation at this stage prioritizes academic literacy and disciplinary mastery, where multilingual scaffolding is used in oral dialogues to mediate complex concepts in science, mathematics and the humanities. Teachers are trained to design lessons where such concepts are introduced through home or regional languages and gradually consolidated in the target school language. This scaffolding enables students to transition into advanced subject learning while retaining their cognitive anchor in home and regional languages.

TEACHERS' ROLE IN LANGUAGE PRACTICE

In secondary school, multi/translingual practices by teachers enable students to move fluidly between languages – for instance, discussing a physics concept in Odia while writing its formal explanation in English – treating linguistic flexibility as a resource rather than a deficit. Teachers can also facilitate cross-linguistic analysis in literature, such as comparing Telugu idioms with their English or tribal language equivalents, to enhance comprehension and critical reflection. Similarly, while teaching linear equations in two variables, a teacher might begin by contextualizing the problem in the home language – for example, describing the cost of two local market items (such as jackfruit and rice) in Kui, Santali or Saora. Students then discuss and solve the problem orally in their familiar

language, after which the teacher scaffolds the solution into a formal algebraic representation and explanation in Odia or English, thereby linking lived experiences with abstract symbolic reasoning.

SEMIOTIC SCAFFOLDS FOR SECONDARY SCHOOL EDUCATION

In secondary school classrooms, semiotic scaffolds use multiple sign systems, such as symbols, visuals, graphs and digital media, to enhance comprehension of abstract concepts. For example, when teaching the quadratic function in mathematics, the teacher can present the algebraic equation alongside a graph plotted on digital software, with annotations in both English and the students' home language. This multimodal representation allows learners to connect symbolic notation, visual patterns and linguistic explanations, reinforcing disciplinary understanding across languages.

LANGUAGE-SENSITIVE CONTINUOUS AND FORMATIVE ASSESSMENT

At the secondary stage, assessment should evaluate both disciplinary mastery and academic literacy by allowing students to demonstrate reasoning through multilingual scaffolds, such as oral explanations in the home language paired with written responses in the school language. This dual approach ensures that subject knowledge is assessed independently of linguistic dominance, promoting fairness while preparing learners for higher academic demands in the target language.

Below: Teacher helps students learn a new language. Gujarati, Tamil, Bengali and Urdu are offered to students in Sardar Patel Vidyalaya, Delhi.





Above: Teacher training forum, Jharkhand.

Training methodologies

Effective teacher training uses active, community-linked methods. Workshops and seminars can introduce theory, but field-based training is critical: trainee teachers should visit multilingual schools, observe classes and practise teaching local language lessons under supervision.

CLASSROOM DISCOURSE ANALYSIS

Teachers examine real or simulated classroom transcripts where students shift across languages. Through guided analysis, they identify how translanguaging and code-switching support or hinder access to subject content, thereby cultivating awareness of language as a resource for equity and cognition.

MULTILINGUAL LESSON DESIGN LABS

Participants collaboratively design lessons in mathematics, science or social studies that use inputs in the mother tongue, regional language and English. Each design is analysed for its inclusivity, scaffolding techniques and potential to sustain disciplinary mastery.

ROLE-PLAY AND MICROLEACHING

Teachers engage in role-play scenarios where they must teach a concept using semiotic scaffolds and multilingual strategies. Peer and mentor feedback highlight how different linguistic resources mediate comprehension and learner confidence.

GENDER-EQUITY MEASURES

Expand recruitment of female tribal-language teachers, with targeted scholarships and childcare support during training. Teacher modules must include gender-sensitivity training – for example, addressing how girls' language skills might differ and how to engage them.

COMMUNITY LANGUAGE MAPPING

Workshops involve teachers in documenting the linguistic repertoires of local communities and schools. This exercise sharpens awareness of sociolinguistic realities and trains teachers to adapt pedagogy to diverse learner profiles.

REFLECTIVE DIALOGUES AND CRITICAL JOURNALING

Participants maintain journals to critically reflect on their assumptions about language hierarchies, while dialogue circles encourage shared insights. These reflective practices nurture critical consciousness about multilingualism as a pedagogical and ethical imperative.

ASSESSMENT REDESIGN CLINICS

Teachers redesign monolingual assessments into multilingual or semiotically rich formats. By comparing outcomes, they recognize how assessment can either reproduce inequities or foster equitable evaluation of cognition.

Network of teacher-education institutions and training support through mentoring

Training requires sustained support. Teacher educators should establish mentoring systems, whereby experienced multilingual teachers (or local elders) can mentor young teachers in implementing curricula. Networks of teachers (often online or through DIET clusters) allow the sharing of lesson plans and materials. State governments can sponsor in-service workshops where teachers present innovations in areas

with high tribal concentrations. For example, Odisha organizes periodic training programmes specifically to improve teachers' skills in a multilingual environment. Such continuous professional development helps teachers refine their practice with feedback. Official support – e.g. stipends for learning tribal languages or recognition awards for MLE innovations – also encourages teacher engagement.

Implementation and institutional strengthening through partnerships

National and state agencies must collaborate with NGOs, universities and local bodies. For example, the NCTE and NCERT can partner with the CIIL, the English and Foreign Languages University (EFLU) and the British Council to develop certification programmes in tribal languages. SCERTs should collaborate with tribal research institutes, such as Odisha's Scheduled Castes and Scheduled Tribes Research and Training Institute (SCSTRTI), and cultural councils to create localized content. The British Council, the NMRC and NGOs like the LLF, Room to Read, Patang, Nirantar and Multilingual Education Networks can co-host teachers' workshops and publish resources.

Odisha's school board and its departments for tribal development and tribal welfare initiated

an MLE programme, with support from UNICEF and the NMRC. The programme trains teachers in multilingual skills and distributes bilingual dictionaries. In Chhattisgarh, the SCERT, with support from the LLF and UNICEF, worked with local literature bodies like Sahitya Sabhas and folk artists to prepare textbooks in Gondi, Sadri and Halbi. In Rajasthan, Room to Read, with support from Tata Trusts, instituted mother-tongue-based reading activities.

The governments of Andhra Pradesh and Odisha have initiated several training programmes for teachers in multilingual pedagogy through various collaborations to amplify the reach and relevance of mother-tongue-based multilingual teacher-education programmes.

Facing page: Teacher uses storytelling and puppetry in class. Sardar Patel Vidyalaya, Delhi.

Below: Teacher accompanies students on the way to class. Kalinga Institute of Social Sciences, Bhubaneswar.





Summary

Teacher education is central to the success of MTB-MLE, with teachers positioned as linguistic and cultural mediators who ensure that children begin learning in familiar languages. UNESCO's *Languages Matter* and India's NEP 2020 jointly affirm that mother-tongue-based pedagogy strengthens comprehension, participation and identity formation.

Major gaps persist between teachers' language skills and learners' home languages, especially in tribal and multilingual regions. Evidence from Rajasthan, Chhattisgarh and Jharkhand shows that many teachers lack fluency in learners' languages, leading to silent classrooms, weak foundational learning and long-term inequities.

National policy frameworks now mandate multilingual teacher preparation, including the NCF-FS 2022, NCF-SE 2023, NPST 2023 and NISHTHA training modules. However, DIETs, SCERTs, the NCERT and NCTE still require systemic strengthening to embed multilingual pedagogy, culturally responsive teaching and inclusive assessment across pre- and in-service programmes.

International guidance calls for reforms in recruitment, deployment and professional development, prioritizing teachers fluent in learners' home languages and ensuring structured training in translanguaging, bilingual resource development and sociolinguistic awareness. Community-integrated models such as Odisha's MLE Plus demonstrate how working with elders and local knowledge systems transforms pedagogy.

Stage-specific multilingual pedagogy is essential across schooling, with L1 immersion in foundational years, bilingual scaffolds and transknowledging strategies in middle grades, and semiotic and disciplinary multilingual supports in secondary education. These approaches ensure cognitive continuity and prevent learning loss during language transitions.

Sustained institutional partnerships, mentoring networks and digital ecosystems are critical for scale and quality. Collaborations between the NCERT, NCTE, SCERTs, TRIs, universities, NGOs (LLF and Room to Read), the British Council and DIKSHA-linked platforms enables continuous teacher learning, high-quality multilingual materials and locally grounded professional support.



Students interact with their teacher during a Tamil Language class, Sardar Patel Vidyalaya, Delhi.



CHAPTER

7

Use of technology in mother-tongue-based multilingual education

What does a “digital revolution” mean for India’s learners if the platforms meant to empower them cannot even speak the languages of their homes, their forests, their stories or their identities?

Can technology claim to democratize education when a Santali, Gondi, Irula or Mundari child must first translate herself into the language of the machine before she can even begin to learn?

If digital public goods promise inclusion, then why do so many tribal and minority languages still remain unheard, unseen and unencoded –waiting for a system that recognizes them as more than data gaps?



7

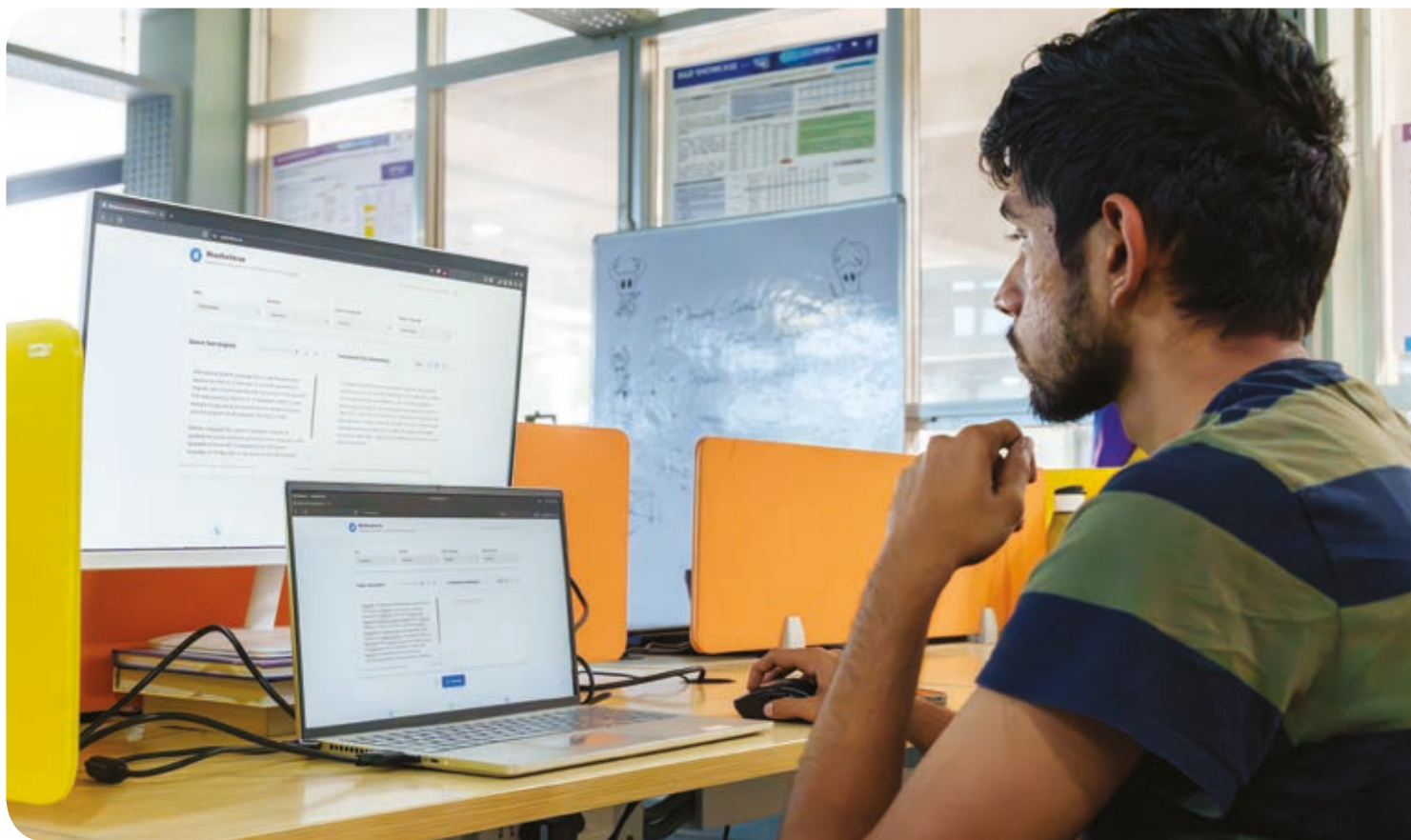
Use of technology in mother-tongue-based multilingual education

The integration of technology into language-inclusive and holistic education represents a transformative step towards realizing the vision of equitable, multilingual learning in India. By bridging the gap between linguistic diversity and digital innovation, technology enables the creation, adaptation and dissemination of educational content in multiple Indian languages – thus ensuring that all learners, regardless of linguistic background, can access quality learning materials. Platforms such as the Digital Infrastructure for Knowledge Sharing (DIKSHA), the BHASHa INterface for India (BHASHINI) and Pratham Books' StoryWeaver exemplify how digital ecosystems can support mother-tongue-based multilingual education (MTB-MLE) by offering localized resources, interactive tools and teacher training in regional and tribal languages. When combined with inclusive pedagogies, such technologies not only enhance literacy and learning outcomes but also enhance learners' linguistic and cultural identities. As India advances toward a technology-enabled education system aligned with the National Education Policy (NEP) 2020 and UNESCO's principles of linguistic inclusion, the challenge lies in ensuring equitable access, community participation and contextual relevance, so that technology becomes a genuine enabler of holistic, language-responsive learning.

Digital platforms create open-access repositories of multilingual children's literature, offering theoretical knowledge across India's numerous languages. StoryWeaver's achievement of hosting over 64,000 stories in over 370 languages (Pratham books, 2023) suggests a radical departure from monolingual frameworks, enabling communities to create culturally relevant content. Similarly, AI-driven initiatives like Microsoft's Project ELLORA (Enabling Low Resource Languages) and Google's Indic speech programmes promise to bridge the digital divide by making technology accessible in vernacular languages (Bali et.al., 2019). These initiatives exemplify the goals of MTB-MLE by expanding access to learning materials in diverse languages and promoting inclusive, culturally responsive education. These efforts illustrate how technology can advance multilingual literacy while fostering linguistic diversity in education systems. UNESCO's Roadmap on Multilingualism in the Digital Era offers a global framework for inclusive digital access, promoting open datasets, interoperable tools, and language technologies for low-resourced and Indigenous languages while emphasising stronger policies, research ecosystems, and partnerships. India's advances in speech recognition, machine translation, and AI-enabled tools demonstrate how national innovation can operationalise the roadmap's recommendations and ensure meaningful digital inclusion for all language communities.



Right: Computer classes. Sardar Patel Vidyalaya, Delhi.



Above: BhashaVerse has developed translation models for 36 Indian languages. International Institute of Information Technology, Hyderabad.

Technology in advancing multilingual education in India

India's multilingual digital public infrastructure for school education includes: DIKSHA (a national platform for content and teacher development); PM eVIDYA direct-to-home (DTH) TV channels and radio for low-connectivity delivery; ePathshala [e-texts from the National Council of Educational Research and Training (NCERT)]; Bharatvani (multilingual dictionaries and glossaries); and BHASHINI [translation / automated speech recognition (ASR) / text-to-speech (TTS) / application programming interface (API)] (NEP, 2020). Together, these systems localize, produce and deliver educational material in various mother tongues and offer teacher support across states and union territories (UTs), aligning with UNESCO's guidance to use technology as an enabler of equitable multilingual learning (UNESCO, 2022b). Digital repositories and broadcast channels expand the availability of materials in Indian languages via mobile phones, TV and radio, and via offline modes. Their effectiveness depends on curriculum alignment, teacher mediation, age-appropriate design, and assessment integration; access alone does not guarantee learning.

Online training and professional communities on DIKSHA and the National Initiative for School Heads' and Teachers' Holistic Advancement (NISHTHA) are building capacity for mother-tongue-based instruction at scale. NISHTHA reached approximately 1.7 million teachers and heads in 33 states and UTs for face-to-face interactions; its online rollout targets 4.2 million elementary teachers in multiple Indian languages. DIKSHA's usage indicators³ show 19,698 courses, 182.3 million enrolments, 145.7 million completions, and content across 36 Indian languages – evidence that digital professional development has moved beyond pilots to system scale.

UNESCO's theme for International Mother Language Day 2022, "Using technology for multilingual learning: challenges and opportunities", underscores that technology, when integrated with pedagogy and teacher development, can widen participation and reduce exclusion based on language.

³ UNESCO case brief on DIKSHA: https://www.unesco.org/en/dtc-financing-toolkit/diksha?utm_source



Above: Student watch interactive digital content in Indian languages. Sardar Patel Vidyalaya, Delhi.

Media and communication landscape

The media and communication landscape in India is evolving rapidly to include multiple languages, creating new opportunities for mother-tongue- and tribal-language-based education. Traditional media, like radio, and newer digital media platforms are increasingly broadcasting or hosting content in Indigenous languages, thereby supporting multilingual learning and cultural preservation.

Community radio and local media

Community radio is one of the most accessible and cost-effective tools for delivering education in local languages. It fosters real-time connections with communities by offering culturally resonant programming that includes songs, folklore, health messages and educational content. For example, **Radio Kotagiri** in Tamil Nadu broadcasts in Irula, Toda, Kota and Kurumba – languages rarely represented in mainstream education or media. **Radio Mattoli** in Kerala broadcasts in Indigenous tribal dialects, enhancing the visibility of tribal languages and cultural identity in Wayanad district. **Bultoo Radio** in Chhattisgarh employs Bluetooth and “narrowcasting” to circulate content in tribal languages, especially in remote Adivasi areas. **Radio Madhuban**, though not exclusively educational, has been studied as an ally in tribal women’s empowerment, broadcasting content that supports language maintenance and awareness. In addition, **community radio stations** in Koraput district of Odisha and **Mandakini ki Awaaz** in Uttarakhand also broadcast in local tribal languages, further

expanding the reach of multilingual education through community media.

National policies, such as the Supporting Community Radio Movement (SCRM) 2021–26, have enhanced the reach and viability of these stations. Additionally, platforms like **PM eVIDYA** and the state broadcaster **Akashvani’s regional services** use radio to deliver lessons in multiple Indian languages, expanding the potential of this medium for inclusive education. These initiatives show how community radio (and radio-like systems) can support multilingual education: by broadcasting in Indigenous languages, by involving local speakers in content production, and by reinforcing language identity within local communities.

Vernacular digital content

Demand for regional-language content is high; for many users, local-language content is considered more reliable than English content. The rapid growth of rural internet use and vernacular-first platforms indicates that educational content must be produced and distributed in learners’ languages in order to be trusted and used (TRAI, 2025).

Television and multimedia

Educational TV and multimedia are adapting to linguistic diversity. Kerala produced digital video lessons for Grade 1 in Adiya, Paniya, Oorali and Kattunaikka, with Malayalam subtitles – a replicable model for remote delivery through TVs or projectors.

A community media network and narrowcasting system in the forests of Aracode, Tamil Nadu

Kwol Radio

 **Location:** Aracode region of the Nilgiris hills in Tamil Nadu.

 **Duration:** 2020 to present.

 **Objective:** Kwol Radio is a community-owned wireless mesh (COW-mesh) network, a tech infrastructure that enables narrowcasting initiatives in the Aracode region of the Nilgiris hills in Tamil Nadu, serving several forest-fringe villages, including Garikyur and Mallikoppaiyur. Launched in 2020 as a hyper-local Irula-language media network, it uses simple broadcasting tools such as routers, Raspberry Pi and ThinkCentres to share regional news and features on health, nutrition, forest rights, biodiversity and human-wildlife interactions, besides traditional cultural expressions and ecological knowledge, and community updates. Operating in an ecologically remote area with limited access to mainstream media, Kwol Radio has become a vital platform for cultural expression and grassroots communication. On 11 March 2025, the station celebrated its fifth year of continuous community broadcasting.

Kwol Radio functions through a narrowcasting model, where programmes are recorded locally and broadcast to a small, geographically specific audience within the Aracode forest region. Its infrastructure enables access to media content through mobile phones on a local mesh network. The entire programming and production process is supported by a local youth group and community foundation called Porivare. Since 2018, the Porivare group has initiated efforts to support traditional music amongst the youth and elders, and its media network has also bolstered language revitalization initiatives in the community.

The community media network is named after a flute called the “kwol”, one of the instruments used in the traditional Irula musical quartet, “arakol”. Youth volunteers take the lead in recording stories, conducting interviews, documenting songs, and creating informational content. Elders contribute cultural knowledge through folk narratives and traditional histories, ensuring intergenerational learning and building the overall capacities of the community, ensuring that traditional expressions find local platforms and space to thrive.

These efforts are supported by two NGOs, Keystone Foundation (Nilgiris) and Janastu (Tumkur), who provide

technical training, equipment support and guidance on audio production, helping community members develop ownership of the broadcasting process. The operations are sustained by a small team that has been trained to maintain the infrastructure and has also begun work on a community archive model for in-situ access to sensitive and essential community data in the Irula language

ACHIEVEMENTS

Kwol Radio has emerged as a significant cultural and social resource.

1 Intergenerational cultural transmission

The platform encourages young people to learn and document folk stories, songs and music from their elders, strengthening the continuity of oral traditions and cultural resilience.

2 Youth leadership

Young people gain confidence, communication skills and a sense of purpose by taking part in infrastructure maintenance and content creation, uploading and airing.

3 Community participation and social cohesion

Residents feel encouraged to contribute to programmes after hearing familiar voices on air, expanding the diversity of content in local Irula dialects.

4 Indigenous data sovereignty

The community-owned management of media content has helped to envision a decentralized model for community archives, embedded in principles of data sovereignty.

5 Local knowledge sharing loops

The system relays timely information relevant to daily life, enhancing local coordination and awareness in an ecologically sensitive area fraught with climate risks, wildlife conflict and other social issues.

Kwol Radio demonstrates that low-cost, locally managed narrowcasting systems can effectively serve remote forest communities, especially where mainstream media or internet access is limited. Investing in community media enhances cultural vitality, fosters youth engagement, and strengthens local communication infrastructure. Scaling similar models could support culturally responsive education, community-centric conservation and knowledge preservation, participatory development planning, and enhanced access to information, especially in fragile ecosystems. Kwol Radio shows that when communities control media tools, communication becomes more meaningful, culturally grounded and socially empowering.

Language technology innovations: emerging tech solutions

India is witnessing rapid growth in language technologies that widen access across linguistic communities, including tribal groups, addressing both the digital divide and language equity.

Natural language processing and artificial intelligence initiatives

Launched in 2022 under the National Language Translation Mission (NLTM), BHASHINI (MEIT, 2023) is India's flagship natural language processing (NLP) and artificial intelligence (AI) platform for language technologies. It offers machine translation, automatic speech recognition, and text-to-speech services across 22 Indian languages, creating interoperable tools for education, governance and digital access. In the classroom, BHASHINI allows for voice-enabled pedagogy and real-time translation, supporting multilingual learning environments. By providing scalable APIs and multilingual interfaces, it advances India's vision of inclusive digital ecosystems, where linguistic diversity is treated as an asset rather than a barrier (MEIT, 2023).

⁴ Corpora are large, structured collections of spoken or written language data – recordings, transcripts, and texts – that train computers to recognize, translate or generate human language. Datasets such as BhashaAnuvaad and IndicVoices use corpora to improve ASR and TTS systems for Indian languages.

Open models and data sets

Research groups such as AI4Bharat are releasing open corpora⁴ and models. Platforms such as BHASHINI's BhashaAnuvaad and IndicVoices, supported by AI4Bharat and its collaborations with institutions like the Indian Institute of Technology (IIT) Madras, Microsoft, Google and IBM, advance the vision of MTB-MLE by developing open, AI-driven language tools that enable translation, speech recognition and content creation across Indian languages. BhashaAnuvaad, a speech translation dataset comprising approximately 44,400 hours across 13–14 Indian languages, focuses on spontaneous speech, code-switching and dialectal variation. IndicVoices offers extensive speech resources for training automatic speech recognition and text-to-speech software. Such initiatives expand access to learning materials in diverse languages and promote inclusive, culturally responsive education. They strengthen the role of local languages in digital and classroom spaces, supporting equitable participation in knowledge creation and furthering the goals of MTB-MLE.



Language technologies are being adapted to support learners with disabilities and those in low-literacy environments.



Above: Assistive learning for learners with disabilities. Kalinga Industrial Institute of Technology International School, Bhubaneswar

Facing page: Extensive digital work helps preserve and revitalize tribal languages. International Institute of Information Technology, Hyderabad

Script support and Unicode

Technical support for India's diverse writing systems is central to ensuring digital inclusion, linguistic equity and mother-tongue-based multilingual education. Over the past decade, updates from the Unicode Consortium and open-source font projects have expanded script coverage, with over sixty-five Indian scripts now supported in varying capacities. Recent inclusions such as the Warang Citi script (Ho language), Ol Chiki script (Santali language) and Meitei Mayek script (Manipuri language) enable native-script content creation, digital publishing and classroom use. Parallel advances in input tools, keyboards and rendering engines, developed through collaborations between IIT Madras, Indic computing groups and community initiatives, have opened pathways for Indigenous users to participate more fully in education, governance and online communication.

Assistive and inclusive technologies

Language technologies are being adapted to support learners with disabilities and those in low-literacy environments.

Innovative tools include:

- **Text-to-speech** and **speech-to-text (STT)** solutions in Indian languages, enabling visually impaired students or those with reading difficulties to access content in their own language.
- **Print-to-audio converters** such as the **Akshar** project (IIT Bombay), which turns textbooks and learning materials into audio content using optical character recognition (OCR) and TTS.
- **Talking pens** that read printed tribal language texts aloud when touched to the page – a powerful tool in early literacy for pre-literate or low-literacy learners. Talking pens have been effectively used in India to support tribal literacy, notably by the Jan Shikshan Sansthan (JSS) in Malappuram, Kerala, under its Vidya Programme. These pens read printed text aloud in tribal languages like Paniya, allowing learners to simultaneously hear and recognize alphabets and words. The programme's use of local language materials and technology earned it the UNESCO Confucius Prize for Literacy in 2016. The Ministry of Tribal Affairs supports bilingual primers to integrate with such tools, while the International Institute of Information Technology (IIIT) Hyderabad's AI developments for tribal languages like Santali and Mundari offer future possibilities for enhancing audio-based learning in tribal education.

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Automatic speech recognition and text-to-speech for early literacy

Google's Read Along app

**Location:** Rural Uttar Pradesh**Duration:** 2019 to present**Objective:** Children read aloud into the app; the system listens, provides instant feedback, and plays back natural-sounding voices. Offline functionality makes it practical for low-resource contexts.

Google's Read Along app was launched in India as Bolo in 2019, through a pilot project in 200 villages of Uttar Pradesh, supported by the Annual Status of Education Report (ASER) Centre. It uses offline-capable automatic speech recognition and text-to-speech technology to support early reading in multiple Indian languages. The Uttar Pradesh pilot found that 64 per cent of children improved reading proficiency within three months. A 2020 multi-state study

by Sattva, with about 6,000 learners, showed that 40 per cent to 42 per cent of Hindi and English learners progressed by at least one fluency level, with significant gains compared to control groups. Now rebranded as Read Along, the app offers over 1,000 stories in 11 languages, integrated into state foundational literacy initiatives in Uttar Pradesh (Mission Prerna), Gujarat, Telangana and Tamil Nadu. By 2024, it helped over 1 million beginner readers improve oral reading fluency by 40 per cent, demonstrating strong potential as a supplemental literacy tool in multilingual, low-resource contexts.

IMPACT

The app has been piloted in primary grades across rural and low-connectivity settings, showing potential to improve pronunciation, fluency and confidence where trained multilingual teachers are scarce.

Source: https://www.sattva.co.in/wp-content/uploads/2020/09/Sattva_Google_Read-Along-IA-Report.pdf interactive-neural-machine-translation-inmt/?utm_source

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Neural machine translation for Gondi textbooks

Microsoft Research INMT project

**Location:** Developed in Chhattisgarh**Duration:** 2020 to present**Objective:** Interactive Neural Machine Translation (INMT) combines neural translation engines with human-in-the-loop validation, enabling rapid production of localized textbooks while building community translator capacity.

The Microsoft Research INMT project for Gondi textbook translation has been active since 2020, with significant public updates and impact reports emerging between 2023 and 2025. The initiative forms part of Microsoft's broader Project ELLORA

(Enabling Low Resource Languages), launched to enable AI solutions for low-resource Indian languages, including Gondi. Impact-wise, it has successfully created the first comprehensive set of school textbooks in Gondi, enabling classroom instruction in this language for the first time and building local translation capacity. The authenticity of this data is well-supported by official Microsoft Research publications, news releases and academic presentations detailing the INMT approach and its socio-educational outcomes in India.

IMPACT

The project delivered textbooks in a previously unsupported tribal language, making classroom instruction in Gondi feasible for the first time.

Source: https://www.microsoft.com/en-us/research/project/interactive-neural-machine-translation-inmt/?utm_source

Such inclusive technologies are crucial for ensuring universal access, particularly in multilingual classrooms where learners may

encounter difficulties with reading or navigating content in unfamiliar scripts or languages.

Digital solutions by and for the community to preserve and promote tribal languages

Community-, civil-society- and state-led efforts are increasingly using digital tools to revitalize tribal languages by expanding educational content, documentation and online presence.

Community-driven innovation. In parts of Jharkhand and Odisha, young community members are leading projects in the Warang Citi script for the Ho language and in the digital preservation of Mundari – building smartphone keyboards⁵/apps, YouTube channels,⁶ and community dictionaries so children and adults can read, record and share stories in their mother tongues, bringing long-under-represented languages into the digital sphere.

Language apps and dictionaries. Language apps and trilingual dictionaries developed by the community provide early literacy support by giving children and adults access to vocabulary and text in their own mother tongues plus regional and global languages. These resources create a scaffold for future translation, voice-recognition and other digital language tools. For instance, a Mundari-English-Hindi dictionary published by the Central Institute of Indian Languages (CIIL) under the Ministry of Education

is listed on the Bharatavani portal.

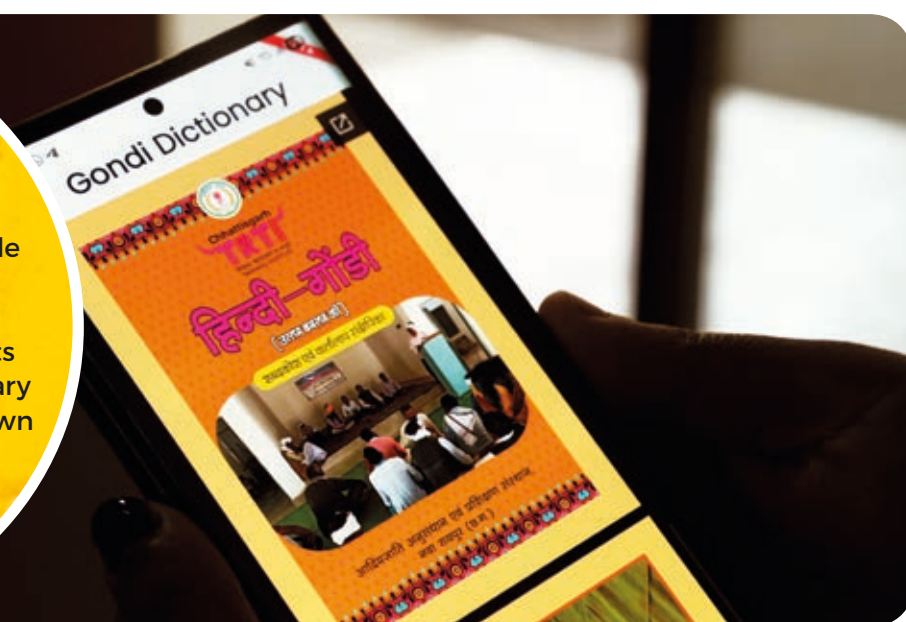
State-led content. States like Kerala, Odisha, Jharkhand and Andhra Pradesh have deployed bilingual e-books, QR-code-linked audio lessons, video content and community radio to operationalize MTB-MLE in tribal areas, leveraging platforms like DIKSHA and state-specific portals. The latter include the Kerala Infrastructure and Technology for Education (KITE)-Samagra; the Odisha School Education Programme Authority (OSEPA) e-Pathshala; DigiSATH in Jharkhand; and the State Council of Educational Research and Training (SCERT)-DIKSHA in Andhra Pradesh. QR-coded “energized textbooks” make multilingual audio-visual learning resources directly accessible through textbooks, ensuring inclusion and continuity of learning in students’ home languages (NCERT, 2021; Ministry of Education, 2022; OSEPA, 2022; Samagra Shiksha PAB Minutes, 2022-23).

Below: A digital dictionary created to preserve and support Indigenous languages. International Institute of Information Technology, Hyderabad.

⁵ https://keyman.com/keyboards/warang_citi?utm

⁶ <https://www.youtube.com/channel/UCf6qyBG8u-nIVIXWSL9F9BA>

Language apps and trilingual dictionaries provide early literacy support by giving children and adults access to vocabulary and text in their own mother tongues.





Above: A wall that highlights the endangered tribes of Odisha. Kalinga Institute of Social Sciences, Bhubaneswar.

Open repositories of online platforms.

StoryWeaver brings together writers, translators and educators to create open, child-friendly books in many Indian languages – including smaller tribal ones like Santali, Gondi and Kui – so children can learn to read in their own mother tongues. This online platform by Pratham Books makes all its stories freely available to read, translate and adapt (Pratham Books, 2023).

Archival and digital preservation of tribal languages.

Communities and researchers are using technology to record and archive oral traditions, folk songs and stories that embody tribal linguistic and cultural heritage. These initiatives, supported by institutions like the Indira Gandhi National Centre for the Arts (IGNCA), the CIIL and the Ministry of Tribal Affairs, create digital repositories and multimedia archives to ensure that endangered languages and their oral art forms are not lost to future generations.

BOX 7.1

The Indira Gandhi National Centre for the Arts

The Indira Gandhi National Centre for the Arts has been engaged in the archival and digital preservation of tribal languages since at least 2015, with ongoing projects through the 2020s. The IGNCA's digital archives include extensive multimedia collections of oral traditions, folk songs and tribal stories that preserve linguistic and cultural heritage.

These archives are open for researchers and local communities to support language revitalization and documentation. The IGNCA's digital repository is an authentic and authoritative source, supported by the Ministry of Culture, ensuring the preservation of endangered tribal languages for future generations.

As part of UNESCO's International Decade of Indigenous Languages (IDIL) 2022–2032, national and community initiatives are documenting and archiving folklore, songs and oral histories in endangered languages. The Scheme for Protection and Preservation of Endangered Languages (SPPEL), run by the CIIL, has listed 117 endangered languages for documentation, producing primers, bilingual/multilingual dictionaries, grammatical sketches and pictorial glossaries. In collaboration with the NCERT, the CIIL has already produced 121 primers in these 117 Indian languages. The CIIL's Bhasha Sanchika

portal now hosts hundreds of audio, video and text samples drawn from community collaborations, enabling the integration of these corpus materials into curricula and teacher training. Through partnerships with grassroots groups, the CIIL crowdsources local idioms, stories and pronunciation guides, weaving them into digital learning platforms to support both language revitalization and curricular enrichment – so learners see their heritage reflected in school content and feel valued in the education system.

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An archive of Tamil Nadu's tribal languages and cultures

Tholkudi Digital Repository



Location: Tamil Nadu



Duration: Launched in 2025



Objective: The Tholkudi Digital Repository is a state-led initiative of the Tamil Nadu Adi Dravidar and Tribal Welfare Department, aimed at safeguarding the linguistic and cultural heritage of the state's tribal and Indigenous communities. Tamil Nadu has identified at least thirty-one tribal languages spoken across the state and has committed to documenting and creating resources for all of them.

Launched on 9 August 2025, on the International Day of the World's Indigenous Peoples, Tholkudi aligns with UNESCO's International Decade of Indigenous Languages (2022–2032) and responds to global concerns highlighted in UNESCO's *Atlas of the World's Languages in Danger* (2010). It embodies the vision of the Los Pinos Declaration (UNESCO and Government of Mexico, 2020), which emphasizes Indigenous leadership and the principle "Nothing for us without us".



ACHIEVEMENTS

Tholkudi provides Tamil Nadu's first systematic infrastructure for preserving tribal languages, traditions and worldviews. The archive safeguards oral literature, ritual practices, place-based ecological knowledge, folk songs, narratives, lexicons and visual culture. It supports intergenerational learning, encourages youth leadership, strengthens cultural identity and provides high-quality resources for curriculum developers, teacher educators, researchers and community educators. By positioning tribal languages within a formal preservation ecosystem, Tholkudi advances cultural revitalization and linguistic dignity.

IMPACT

Tholkudi demonstrates how state-led archives can successfully localize global frameworks for Indigenous rights and cultural preservation. Investing in ethical, community-led digital documentation strengthens multilingual education, cultural continuity and equitable development. The initiative offers a scalable model for India and beyond, showing how governments can partner with Indigenous communities to build responsible, participatory and future-facing archives that honour linguistic diversity as a pillar of cultural justice.

Tholkudi operates on an open-source archival platform built around clear standards for file formats, metadata, consent, naming, transcription and cultural protection. Documentation is carried out by teams that combine academic experts and community-led groups, with a strong emphasis on encouraging tribal women and youth to take active roles in collecting, narrating and curating materials.

The repository applies the CARE Principles for Indigenous Data Governance – Collective Benefit, Authority to Control, Responsibility, and Ethics – to allow participating tribal communities to retain meaningful authority over how recordings, knowledge and cultural materials are archived, accessed and reused. A four-tier access system (open, restricted, protected, embargoed) balances openness with cultural sensitivity.

Source: <https://tholkudi.in/about-us>

Building a digital commons for multilingual education in India

StoryWeaver

CONTEXT AND CHALLENGE

India's linguistic diversity – around 19,500¹⁰ languages and dialects – presents both opportunities and challenges for education. Nearly 40 per cent of children enter school facing a language different from their home language, leading to poor comprehension, low learning outcomes, and high dropout rates, especially among girls and marginalized communities.

The Tata Trusts Multilingual Education Landscaping Exercise (MLENA, 2025) identifies key gaps: a shortage of quality children's literature in tribal and minority languages, early transitions away from home languages, and teachers unprepared for multilingual classrooms. Fragmented efforts among government, civil society and philanthropy further limit scale and sustainability.

UNESCO and the NEP 2020 advocate mother-tongue-based multilingual education, emphasizing home-language instruction, additive bilingual models, and open, high-quality resources.

INTERVENTION: STORYWEAVER'S OPEN DIGITAL COMMONS

StoryWeaver, Pratham Books' open digital platform, bridges these gaps by creating and sharing multilingual children's books at scale. All content is openly licensed, enabling free reading, downloading, adaptation and translation by teachers, parents, governments and communities.

As India's first digital public good for early grade reading, StoryWeaver reduces duplication, supports localization and empowers educators and volunteers to contribute translations and adaptations. Government partnerships ensure curricular alignment, while integration with platforms like DIKSHA

extends reach nationwide. Today, StoryWeaver hosts over 64,000 books in nearly 370 languages, making it one of the world's largest repositories of multilingual children's literature.

IMPACT ON LEARNING

- **Community-driven multiscript storybooks.** In West Bengal, teachers and volunteers translated over 600 storybooks into Santali and Kora, using five scripts including Ol Chiki. These books, often the first in their mother tongue for many, fostered classroom engagement and cultural pride.
- **Digital libraries for Urdu literacy.** In Maharashtra, a partnership with the SCERT, the Regional Academic Authority (RAA) and UNICEF enabled a foundational Urdu digital library and statewide training. Urdu readership on StoryWeaver rose by 248 per cent, with over 70 district "reading ambassadors" mobilized. This demonstrates the importance of government ownership and multilateral support for scale and sustainability.
- **Forgotten Folklore Project.** In Meghalaya, StoryWeaver collaborates with Khasi, Garo and Jaintia communities to transform oral narratives into open children's books. Community authorship ensures cultural relevance, while open licensing enables wide dissemination and adaptation.

¹⁰ The Government of India's 2011 Census recorded 121 languages with over 10,000 speakers and 270 mother tongues overall, recognizing 22 scheduled languages in the Constitution. While the census used broader classifications and consolidated 19,569 raw mother tongue entries, a more recent survey by the Ministry of Home Affairs documented 576 languages and dialects with field videography, reflecting efforts to comprehensively map linguistic diversity beyond the scope of the census.



BOX 7.2

Appu: an LLM-powered adaptive companion

Appu is a GenAI-powered learning companion developed by Rocket Learning with support from Google, targeting children aged 3 to 6 years across India. It uses large language models (LLMs) along with speech recognition and adaptive algorithms to offer personalized storytelling, quizzes and

interactive tutoring aligned to each child's pace. By bridging children's mother tongues with school-readiness skills, Appu enables localized multilingual learning, particularly in settings where teacher support in local languages is scarce.

Use of technology in language-inclusive holistic education

Technology serves as a catalyst for language-inclusive holistic education in diverse contexts. The Kerala Infrastructure and Technology for Education programme integrates digital learning through its E³ (Enjoy, Enhance, Enrich) model, combining e-libraries, interactive language labs and televised lessons to strengthen bilingual proficiency and foster creativity among primary learners. By contrast, Papua New Guinea's

Rapidly Improving Standards in Elementary (RISE) project leverages low-cost, solar-powered digital tools to provide multilingual reading materials in local and trade languages, even in remote villages. When technology is embedded within community-driven, context-sensitive pedagogies, it enhances literacy and engagement and makes linguistic diversity a foundation for inclusive learning

23

Promoting language proficiency**Kerala Infrastructure and Technology for Education (KITE) project**

Objective: As part of Kerala's Public Education Rejuvenation Mission, the Kerala Infrastructure and Technology for Education project has transformed over 45,000 classrooms and established 11,275 high-tech labs in primary schools. Its Samagra e-resource portal complements these efforts by providing digital learning resources across grades

speaking, reading, writing, grammar and vocabulary, with recording and self-assessment.

- **e-Broadcast:** lessons via the KITE Vickers channel covering conversational and academic skills.

In 2024–25, the model expanded with **E³ Hindi (Prayogshaala)**, piloted in 163 schools.

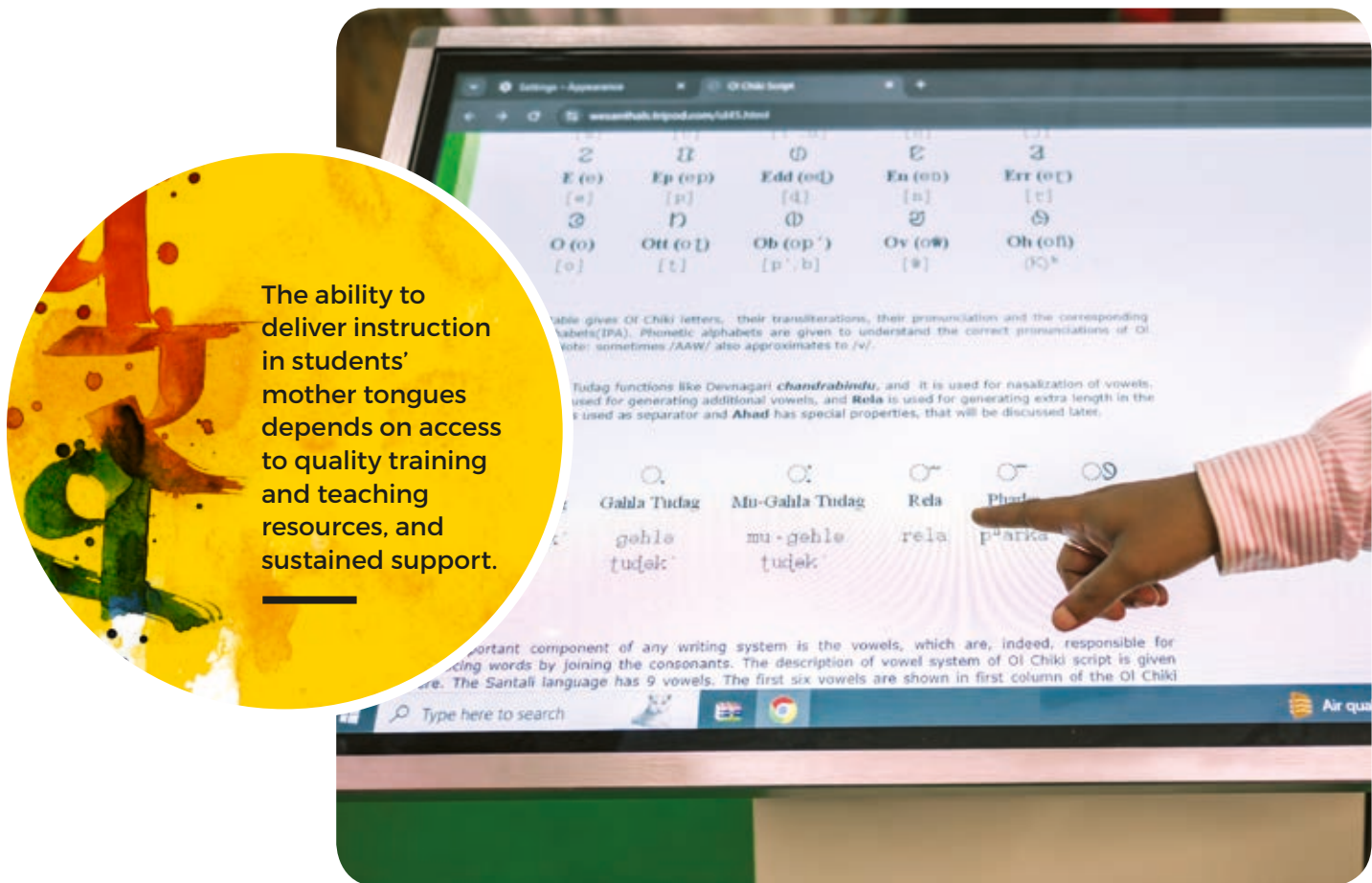
IMPACT

The programme has strengthened literacy, imagination and communication skills, supported bilingual proficiency, and laid a foundation for multilingual learning.

Source: <https://kite.kerala.gov.in/KITE/index.php/welcome/ict/23>

To strengthen structured language learning, the state launched an **E³ (Enjoy, Enhance, Enrich) English Language Lab Programme** in 2020 for Grades 1 to 8, with three components:

- **Samagra e-Library:** over 200 graded digital stories in text, audio, video and animated formats.
- **e-Language Lab:** interactive software for practising listening,



The ability to deliver instruction in students' mother tongues depends on access to quality training and teaching resources, and sustained support.

Technology for teachers and teacher-education programmes

Above: Learning Santhali through digital kiosks. Kalinga Institute of Social Sciences, Bhubaneswar.

Teachers are the cornerstone of multilingual education. In India's diverse linguistic landscape, the ability to deliver instruction in students' mother tongues depends not only on language proficiency but also on access to quality training and teaching resources, and sustained support. Digital technology is playing an increasingly vital role in scaling this support system through online training programmes, communities of practice and resource-sharing platforms, helping to bridge the gap between multilingual education policy and classroom practice.

Online training at scale

To strengthen teacher capacity in multilingual pedagogy, in alignment with the NEP 2020, India has expanded professional development through NISHTHA on the DIKSHA platform. Building on its earlier face-to-face phase, which reached over 1.7 million teachers and school leaders across 33 states and UTs, the programme also uses a digital format to extend training to millions more. The online rollout is designed to cover over 4 million

elementary school teachers, with modules offered in regional and tribal languages to ensure accessibility and impact in even the most remote areas.

- NISHTHA 2.0 provides 68 online modules in 10 languages for secondary educators, focusing on core subjects and inclusive practices.
- NISHTHA 3.0 addresses foundational learning, training over 2.5 million teachers in 11 languages on literacy, numeracy, multilingual education and information and communication technology (ICT) integration.

Training in the teachers' own languages ensures accessibility and uptake, while the digital format supports flexibility and reaches remote contexts. Complementing this, communities of practice on DIKSHA and NISHTHA allow teachers to exchange materials and strategies, and discuss classroom use of digital tools (voice typing, bilingual dictionaries, speech-to-text), thus supporting the preparation of multilingual learning resources.

Digital presence of Indian languages

Building a robust digital presence for Indian languages, including support for scripts, content availability, and tools and apps, is crucial for leveraging technology in multilingual education.

Unicode and input tools

All twenty-two scheduled languages of India now have Unicode support, ensuring consistent digital representation across platforms. Several tribal languages utilize distinct scripts—such as Ol Chiki for Santali, Warang Citi for Ho, Lepcha, and Meitei Mayek for Manipuri—which are

also supported in Unicode, enabling digital preservation and usage. Standardized keyboards including InScript, phonetic, and transliteration layouts are integrated into major operating systems, facilitating typing in languages like Gondi and Bodo. Many tribal languages remain primarily oral, with ongoing efforts to encode new scripts like Tangsa and Wancho, or to adapt existing technologies (such as IIT Madras/Indic tech) to broaden digital accessibility. These advancements reinforce linguistic inclusion and support the preservation and digitization of India's rich tribal language heritage.

Integration models: successful technology implementation approaches

The NEP 2020, with the National Educational Technology Forum (NETF) and National Digital Education Architecture (NDEAR), provides frameworks for interoperable, multilingual deployments. Some states pair early-grade instruction in mother tongues with devices preloaded with local-language content. Effective models align policy, community engagement and multimodal delivery.

- **One-nation platform, state customization.** DIKSHA enables states to upload, align and localize content in chosen languages; it proved vital for rapid multilingual remote learning under PM eVIDYA.
- **Hybrid broadcast + digital.** Educational TV/radio (PM eVIDYA's expansion from 12 to 200 DTH channels) plus on-demand DIKSHA access extends reach and continuity in low-connectivity settings.

- **Voice solutions for home learning.**

A "home learning playbook" from the Central Square Foundation (2024) highlights how voice-based tools such as interactive voice response (IVR) calls and WhatsApp storytelling (piloted in Maharashtra, for example) enable children without reliable internet to access home-learning material. These voice/text blends allow learners to listen, engage and respond using simple devices, and such models could be scaled via platforms like BHASHINI to serve diverse literacy and device contexts.

- **Government-NGO-community partnerships.**

Odisha's MTB-MLE shows how co-created, culturally relevant digital content, distributed via tablets or in print, translates policy into classroom practice and sustains usage through local ownership.

Right: Children reading picture books translated into the Indigenous language, Kora on StoryWeaver. West Bengal.



Digital divide and the limitations of technology

Despite technological advances, disparities in access persist, limiting the effectiveness of digital education, especially in rural, tribal and low-income communities and among marginalized groups.

The digital divide significantly limits the effectiveness of mother-tongue-based multilingual education in India, especially in rural and tribal areas where internet access, electricity and digital devices are scarce or unreliable. Sattva Consulting, and Oxfam India (2022) and Observer Research Foundation (2022) highlight that, despite technological advances, infrastructure gaps and uneven policy implementation hinder digital inclusion for MTB-MLE. These constraints affect content accessibility in local languages and restrict teacher training scalability, leaving many minority language learners underserved. The data underscores the need for integrated, context-sensitive solutions, combining technology with ground-level support to bridge this divide and fulfil MTB-MLE's promise in India's diverse linguistic landscape.

Language and content gaps. Many smaller languages still lack adequate digital interfaces or educational resources, discouraging adoption among minority-language speakers.

Digital literacy. Many teachers and students in tribal areas lack the skills to use platforms like DIKSHA effectively. As Oxfam (2022) notes, only 38 per cent of Indian households were digitally literate in 2022.

Language and dialect coverage. Public tests and real-world checks show that ASR and machine translation (MT) systems don't perform as well on spontaneous, mixed-language (code-switched) Indian speech as on clear, read speech because there isn't enough data available for many Indian languages and dialects.

Curricular alignment and teacher support. Where content in the mother tongue exists, its integration with lesson plans and assessments and continuous mentoring determine uptake. National professional development has scaled (e.g. NISHTHA 3.0 (NCERT, 2021) on DIKSHA (2017) targets 25 lakh teachers and school heads for training in FLN using multilingual modules); but school-level coaching remains uneven, limiting consistent classroom use.

Policy and administrative constraints. Fragmentation across agencies and platforms slows localization. NDEAR was introduced to enforce interoperability, open standards, and modular building blocks for multilingual deployments, signalling that this barrier is structural and well understood.

Usability and accessibility. Keyboard and font gaps, as well as single-language interfaces, persist for some communities. While DIKSHA meets Web Content Accessibility Guidelines (WCAG) 2.0 AA and supports over thirty content languages, its interface/localization depth varies by product and state rollout. Voice pipelines under BHASHINI expand ASR/TTS coverage, but accuracy is uneven across accents and code-switching, reinforcing the first barrier.

Facing page: Students in a computer lab. Sardar Patel Vidyalaya, Delhi.

Below: Students using digital kiosks at the annual science festival. Kalinga Institute of Social Sciences, Bhubaneswar.





Summary

India's multilingual education ecosystem demonstrates that when technology is purposefully aligned with linguistic inclusion, it can transform both access and pedagogy. Platforms like DIKSHA, BHASHINI, ePathshala and Bharatavani have created a national digital commons where learners, teachers and policymakers can collaborate to generate, localize and share resources in over thirty Indian languages.

This infrastructure, supported by NDEAR and guided by the NEP 2020 and UNESCO's global principles on equitable multilingual learning, enables scalable teacher development and inclusive learning pathways. At the same time, open repositories like StoryWeaver and state-level programmes like Kerala's KITE E³ demonstrate how decentralized models can nurture bilingual and trilingual proficiency through community engagement and creative pedagogy. These collective efforts signal a paradigmatic shift, positioning linguistic diversity not as a barrier but as the foundation of educational quality and cognitive equity.

Equally important is the growing role of AI, assistive technologies and digital archives in preserving and revitalizing India's linguistic heritage. The BHASHINI platform and research initiatives like AI4Bharat's BhasaAnuvaad extend the reach of translation, speech and text technologies to under-represented languages, while community media, radio networks and archives like Tholkudi and Sanchika ensure cultural continuity through localized content.

Technology's transformative potential depends on continuous teacher support, offline accessibility and community ownership.

When teachers are empowered to adapt materials, parents to participate, and learners to encounter their home languages in digital spaces, technology becomes a genuine enabler of holistic, language-responsive learning, strengthening not only literacy and numeracy but also belonging, identity and epistemic justice within India's multilingual democracy.





Students engage with their teacher in a multilanguage reading session. Sardar Patel Vidyalaya, Delhi.



CHAPTER

8

Way forward: teaching and learning in every tongue

Mother-tongue-based multilingual education is no longer an experiment or a peripheral reform, it is the only path to educational justice, epistemic dignity and meaningful learning for millions of tribal and minority-language children. Unless teacher education, policy design and technology transform together, multilingual ideals will remain rhetoric and classrooms will continue to fail those whose first language is not the language of power.



8

Way forward: teaching and learning in every tongue

India is at a decisive moment in its efforts to make schooling genuinely inclusive for learners from tribal, minority and other non-dominant language communities. The various chapters of this report have shown, in different ways, that mother-tongue-based multilingual education (MTB-MLE) is not a peripheral innovation but a central pathway for achieving the aims of the National Education Policy (NEP) 2020, the National Curriculum Frameworks (NCFs), and India's commitments under global frameworks such as UNESCO's Languages Matter guidance (2025) and the International Decade of Indigenous Languages (2022–2032).

Yet, the evidence from classrooms, teacher-education institutions and digital platforms is equally clear: without systemic changes in

teacher preparation, language-in-education policy and technology design, India will continue to reproduce silent classrooms for children whose first language is not the language of schooling.

The way forward, therefore, calls for a coherent, multilevel strategy that links:

- 1 policy vision with classroom realities;
- 2 teacher education with community knowledge systems; and
- 3 digital public infrastructure with local linguistic ecologies.

Such an approach can reposition linguistic diversity as a foundation of quality education, not a barrier to be managed.

Below: Assembly prayers offered in different languages. Kalinga Institute of Social Sciences, Bhubaneswar.



Recentre language-in-education reform

The analyses in this report show that language is not only a medium of instruction; it is a medium of participation, identity and epistemic justice. The NEP 2020 already affirms the use of mother tongues/home languages/local languages as the preferred medium of instruction in the foundational and preparatory years of schooling, and the NCFs extend this orientation across later stages of school. The way forward is to treat this not as an optional clause but as a central organizing principle of inclusive education.

This requires:

- Systematic language mapping at school and

cluster levels, especially in tribal and border areas, to inform medium-of-instruction decisions, teaching-learning-material (TLM) development and teacher deployment.

- Coherent language-in-education policies at the state level that explicitly operationalize MTB-MLE, rather than relying on ad hoc transitional models.
- Alignment between school education, teacher education and higher education so that learners can move through the system without being penalized for their linguistic identities and resources.

Transform teacher education into the backbone of mother-tongue-based multilingual education

Chapter 6 of this report, on teacher education, highlights that teachers are the decisive factor in whether multilingual classrooms become spaces of exclusion or spaces of learning. The way forward calls for reimagining teacher education (pre-service, in-service and continuous professional development) through a multilingual lens.

This includes:

- Making multilingual pedagogy, translanguaging and transknowledging core components of all teacher-education curricula, not specialized electives.
- Building stage-specific competencies: mother-tongue-based bilingual assessment in the foundational years; discipline-specific

multilingual scaffolds in middle school; and semiotic, multi-script and academic-literacy support at the secondary level.

- Equipping teacher educators themselves with critical language awareness, the skills to analyse classroom discourse, and the capacity to mentor teachers in multilingual lesson design, assessment and reflective practice.
- Recognizing multilingual teacher competence (in learners' home languages and languages of wider communication) as a professional standard in recruitment, deployment and career progression.

Without such a shift, policy commitments to MTB-MLE will remain disconnected from daily classroom practice.

Centre communities and Indigenous knowledge

Case studies show that MTB-MLE has the greatest impact when schools recognize villages and communities as sites of knowledge rather than just locations where schools are placed. The way forward must consolidate and scale models that:

- Involve elders, storytellers, artisans, cultural practitioners and parents in co-creating curricula, early-grade readers and learning

activities in mother tongues and local languages such as Saora, Kui, Kondi, Santali, Bhili, Khasi, Mizo, Toda and other languages.

- Train community workers and local resource persons as co-educators who can mediate between the school and the village, support bi/multilingual literacy, and document oral traditions and ecological knowledge.

- Use participatory spaces, such as maths and science fairs, village libraries and cultural events, to connect Indigenous epistemologies with formal disciplinary concepts.

Such approaches deepen learner engagement, strengthen intergenerational language transmission, and change teachers' perceptions of tribal and minority languages from "problems" to rich resources for learning.

Integrate a gender-responsive mother-tongue-based multilingual education framework across middle, secondary and alternative schooling

India should adopt a gender-responsive MTB-MLE framework that systematically addresses the learning and retention gaps faced by girls and young women – particularly from tribal, minority-language and rural communities – across schooling systems.

India should adopt a gender-responsive MTB-MLE framework that systematically addresses the learning and retention gaps faced by girls and young women – particularly from tribal, minority-language and rural communities – across Eklavya Model Residential Schools (EMRSs), Kasturba Gandhi Balika Vidyalayas (KGBVs), and private and alternative schooling systems. This framework should:

- Mandate mother-tongue-/home-language-based instruction and structured multilingual bridging in EMRSs, KGBVs and other residential schools to address gender-differentiated vulnerabilities arising from the home-school language gap, especially in literacy and numeracy.
- Recruit and deploy teachers who speak learners' home languages, prioritizing tribal and minority-language women educators, and provide continuous professional development on gender-sensitive multilingual pedagogy, classroom discourse and bilingual assessment.

- Develop graded bilingual and trilingual learning materials, including girl-friendly, culturally grounded texts that reflect Indigenous knowledge, local ecologies and women's narratives, enabling stronger comprehension and confidence among first-generation learners.
- Monitor gender-wise and language-wise learning outcomes through the Unified District Information System for Education Plus (UDISE+), the National Achievement Survey (NAS) and the Annual Status of Education Report (ASER) by incorporating indicators such as medium of instruction, teacher-learner language match, language transition points, and gender-specific learning loss during linguistic discontinuities.

A gender-responsive MTB-MLE approach will ensure that linguistic inclusion directly strengthens girls' access, participation, learning and progression throughout schooling, transforming India's linguistic diversity into an enabler of gender equity and educational justice.

Establish a national multilingual pedagogy and materials development framework

To strengthen classroom practice and ensure equitable learning for children from diverse linguistic backgrounds, India should adopt a unified framework for multilingual pedagogy and learning-materials development aligned with the NEP 2020 and the NCF 2023. This framework should:

- Standardize multilingual pedagogic practices, including translanguaging, oral-written bridging, multimodal semiotic strategies and bilingual assessment, across foundational, preparatory, middle and secondary stages,

ensuring consistency in teacher preparation, classroom discourse and academic language development.

- Create state- and cluster-level multilingual materials development cells that draw on State Councils of Educational Research and Training (SCERTs), District Institutes of Education and Training (DIETs), tribal research institutes (TRIs), universities, community knowledge holders and local artists to co-create high-quality, culturally rooted, graded bilingual and trilingual materials in tribal, minority and regional languages.

To strengthen classroom practice and ensure equitable learning, India should adopt a framework for multilingual pedagogy and learning-materials development aligned with the NEP 2020 and the NCF 2023.



Above: Exams can be written in either Telugu or Urdu. Model High School, Osmania University, Hyderabad.

- Ensure alignment between textbooks, supplementary readers, digital resources and assessment tools, so that multilingual materials are not add-ons but core components of subject learning, particularly in mathematics, science, social sciences and foundational literacy.
- Develop open, interoperable repositories of multilingual resources on the Digital Infrastructure for Knowledge Sharing (DIKSHA), Bharatavani and state edtech platforms, paired

with offline, low-tech formats (print, radio, audio books, talking pens) to ensure access in connectivity-poor regions.

A coherent national framework for multilingual pedagogy and materials production will enable teachers to teach more effectively, ensure that children's languages are systematically integrated into learning, and support India's transition towards inclusive, culturally grounded and academically rigorous multilingual education.

Harness technology as an enabler of linguistic inclusion, not as a substitute for teachers

India has made significant investments in digital public infrastructure for education – for example, DIKSHA, PM eVIDYA, ePathshala, Bharatavani, BHASHINI, etc. as well as partnerships with open platforms such as StoryWeaver, AI-enabled tools such as Read Along and Project ELLORA, and state-level initiatives like the Kerala Infrastructure and Technology for Education (KITE) and QR-coded “energized textbooks”. This report’s analysis of technology in MTB-MLE highlights both the promise and the risks of these developments.

The way forward is to anchor technology in pedagogical intent and linguistic justice, by:

- Ensuring that content in local languages is not an afterthought but a design requirement, developed in partnership with communities, TRIs, SCERTs and local non-governmental organizations (NGOs).
- Using digital platforms primarily to support teachers, through multilingual National

Initiative for School Heads’ and Teachers’ Holistic Advancement (NISHTHA) modules, communities of practice, and repositories of classroom-ready resources, rather than imagining technology as a replacement for human mediation.

- Investing in open, community-validated datasets and language technologies [automated speech recognition (ASR), text-to-speech (TTS), machine translation (MT)] that perform robustly on spontaneous, code-switched speech and underserved languages.
- Strengthening offline and low-tech pathways (radio, community audio, solar-powered devices, talking books and pens) that are essential for learners in connectivity-poor settings.

Technology must be treated as one strand in a broader ecosystem that includes teacher agency, community languages and accessible infrastructure.

By viewing digital and linguistic inclusion as mutually reinforcing goals, policy can avoid creating new hierarchies.



Above: Annual science project display by students of Kalinga Institute of Social Sciences, Bhubaneswar.

Address digital and linguistic divides together

Evidence presented in this report shows that digital access, digital literacy and language coverage are unequally distributed, with rural, tribal, low-income and girls' education bearing the brunt of these gaps. The way forward requires integrated strategies that address:

- Infrastructure gaps, electricity, connectivity, devices and maintenance as preconditions for meaningful digital learning in tribal and remote areas.
- Teacher and student digital literacy, including capabilities to navigate platforms like DIKSHA in

their own languages, create local content and critically evaluate digital materials.

- Language gaps in technology, by prioritizing the inclusion and accurate representation of non-scheduled and endangered languages in digital interfaces, keyboards, fonts and content repositories.

By viewing digital inclusion and linguistic inclusion as mutually reinforcing goals, policy can avoid creating new hierarchies in the name of innovation.

Build institutional partnerships and governance mechanisms

This report demonstrates that no single institution can deliver MTB-MLE at scale. Sustainable change requires coordinated action between the ministries of education, tribal affairs, culture, and technology, the National Council for Teacher Education (NCTE), National and State Councils of Educational Research and Training (NCERT and SCERTs), the Central Institute of Indian Languages (CIIL), DIETs, TRIs, universities, NGOs, community organizations and technology partners.

The way forward should include:

- Establishing cross-sectoral coordination platforms on MTB-MLE with clear mandates

for curriculum design, teacher education, technology deployment, research and community engagement.

- Formalizing state-level partnerships between SCERTs, TRIs, cultural institutions, universities, NGOs and edtech actors to co-develop multilingual training modules, textbooks, digital content and assessment tools.
- Strengthening institutional capacities, particularly of DIETs and SCERTs, for language mapping, teacher mentoring, materials development and multilingual data analysis.

Invest in research, monitoring and sustainable financing

An adequately financed approach will ensure that MTB-MLE is a long-term structural shift towards equity and social justice.

The way forward must be underpinned by rigorous evidence and predictable financing. Longitudinal evaluations of MTB-MLE programmes in India have already demonstrated reduction in educational wastage for girls, tribal students and other minority communities when mother tongues/home languages are used as the foundation for instruction. However, these studies remain unevenly distributed and often disconnected from policy design.

Moving ahead will require:

- Embedding monitoring and evaluation frameworks that track language-related indicators (language of instruction, teacher-learner language match, use of home languages, availability of materials, etc.) within existing systems such as the (UDISE+) and the NAS.

- Supporting research collaborations between universities, government agencies and community organizations on topics such as multilingual and translingual practices, digital tools for tribal languages, and gender-affirming experiences of MTB-MLE.
- Allocating dedicated budget lines for MTB-MLE in state and central schemes, including teacher recruitment and training, materials development, community participation and digital inclusion.

A sustained, evidence-informed and adequately financed approach will ensure that MTB-MLE is not a short-term project but a long-term structural shift towards equity and social justice.

Establish a national mission for mother-tongue-based multilingual education

To ensure coherence, scale and sustained impact, India should establish a time-bound national mission for MTB-MLE that integrates the implementation of the NEP 2020 and the NCFs, reforms in EMRS and KGBV systems, and investments in digital public infrastructure such as DIKSHA, BHASHINI and PM eVIDYA. The mission should function as a unified coordination and accountability platform spanning ministries, state governments, academic bodies and community institutions.

Its core functions should include:

- Mandating systematic school- and cluster-level language mapping to guide medium-of-instruction decisions, teacher deployment, TLM development and community engagement, particularly in linguistically diverse and tribal regions.
- Setting national standards for teacher multilingual competence and community participation, thus ensuring that all schools, public and private, provide early-grade instruction in learners' home languages and implement structured multilingual bridging throughout the elementary years.
- Embedding language-sensitive indicators, such as medium of instruction, teacher-learner language match, multilingual availability of learning materials and linguistically responsive assessment practices, within UDISE+, NAS and ASER for ongoing monitoring and public accountability.
- Creating ring-fenced financing lines at central and state levels to support multilingual teacher recruitment and training, community-developed materials, low-tech and offline access pathways, and research on language, learning and digital inclusion.

Below: Students learn in their mother tongue with English translations alongside. Model High School, Osmania University, Hyderabad.





Students take turns reading aloud from textbooks during class. Government Upper Primary School, Gajapati.



CHAPTER

9

Recommendations

The following recommendations translate this report's analysis into concrete, actionable steps for government, institutions and civil society partners. They are designed to be adaptable across states while remaining anchored in the National Education Policy 2020 and UNESCO's principles on multilingual education.



9

Recommendations

Recommendation 1

Adopt and operationalize clear state-level language-in-education policies grounded in MTB-MLE

States should develop or update language-in-education policies that explicitly commit to mother-tongue-/home-language-/local language-based multilingual education through the foundational and preparatory stages, at least, consistent with the National Education Policy 2020.

Policies should be based on school- and cluster-level language mapping and specify how languages will be used as mediums of instruction, as subjects and as co-curricular resources across stages, including pathways for languages like Santali, Condi, Kui, Saora, Ho, Mundari, Bhili, Khasi, Mizo, Toda and others.





Above: Urdu class in session. Sardar Patel Vidyalaya, Delhi.

Facing page: As a part of MTB-MLE, schoolchildren are taught to read in their mother tongue. Kalinga Institute of Social Sciences, Bhubaneswar.

Recommendation 2

Strengthen teacher recruitment, deployment and professional standards for multilingual competence

Recruitment rules should prioritize teachers proficient in learners' home languages and regional languages, especially in tribal and border areas, and recognize multilingual competence as an asset in selection and promotion.

The National Professional Standards for Teachers and state norms should explicitly include multilingual pedagogy and language-responsive assessment as core professional competencies.

Recommendation 3

Reform pre-service and in-service teacher education to embed multilingual pedagogy at all stages

The National Council for Teacher Education, National and State Councils of Educational Research and Training, universities and teacher education institutions should ensure that all pre-service programmes (including the Integrated Teacher Education Programme) and in-service courses (e.g. NISHTHA 2.0 and 3.0) integrate structured modules on multi/translanguaging, transknowledging, multilingual lesson design, and multi-/translingual assessment.

Teacher-education curricula should be stage-

differentiated: mother-tongue-led early literacy and numeracy; bilingual scaffolds and disciplinary language in middle school; and semiotic, multi-script and academic-literacy strategies in secondary education.

Teacher education programmes should also help teachers recognize gendered participation patterns, adapt classroom language use for learners with disabilities, and design inclusive, multilingual tasks that reduce linguistic and cognitive load.



Above: Men of the Saora community perform a traditional dance. Government Upper Primary School, Gajapati.

Recommendation 4

Institutionalize community participation and Indigenous knowledge in school practice

States and institutions should formalize mechanisms to engage elders, storytellers, artisans, artists, cultural practitioners and community workers in curriculum development, classroom activities and teacher-education modules.

Programmes should support schools to organize maths and science fairs/activities, village libraries, oral-history projects and community documentation in local languages, with structured reflection for teachers and learners.

Recommendation 5

Develop and distribute high-quality multilingual learning materials and assessments across grades

Standardize multilingual pedagogy across all stages and establish state- and cluster-level material development cells to co-create culturally rooted bilingual and trilingual learning resources. National and State Councils of Educational Research and Training, Tribal Research Institutes, the Central Institute of Indian Languages, the Bharatiya Bhasha Samiti and partners should collaborate to develop bilingual and multilingual primers, storybooks, subject readers, glossaries and digital open educational resources in tribal and minority

languages aligned with state curricula.

Textbooks and digital and supplementary materials should be aligned with assessment tools, and open-access multilingual repositories expanded to ensure equitable access in low-connectivity regions. Assessment bodies should pilot and scale multilingual, multimodal assessment models that allow learners to demonstrate understanding through oral, written and visual modes, across languages, especially in early and middle grades.

Recommendation 6

Integrate a gender-responsive MTB-MLE framework across middle, secondary and alternative schooling

India should institutionalize a gender-responsive MTB-MLE framework that ensures mother-tongue-/home-language-based instruction and structured multilingual bridging across all schools, including Eklavya Model Residential Schools, Kasturba Gandhi Balika Vidyalayas, private and alternative schooling systems, with a clear focus on girls and young women from tribal, minority-language and rural communities.

The framework should be informed by school- and cluster-level language mapping and specify how learners' mother tongues/home languages will be used as mediums of instruction, as subjects and as co-curricular resources, while prioritizing the recruitment and deployment of women educators fluent in minority and tribal languages.

Recommendation 7

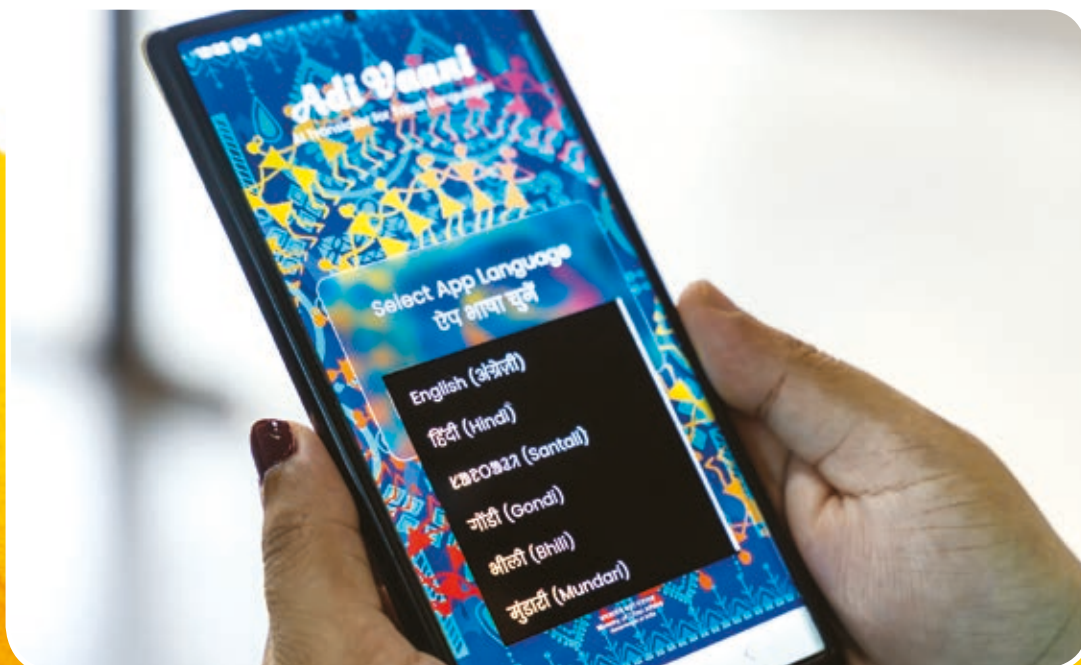
Leverage digital public infrastructure for inclusive, multilingual teacher support and learning resources

Below: Adi Vaani, a translation platform for tribal languages, has been developed as part of a STEM initiative. International Institute of Information Technology, Hyderabad.

DIKSHA, ePathshala, Bharatavani and state platforms (e.g. Kerala Infrastructure and Technology for Education, the Odisha School Education Programme Authority's e-Pathashala, DigiSATH, etc.) should prioritize curation and tagging of MTB-MLE resources, including tribal-language modules, classroom videos and

community-generated materials.

NISHTHA and other online programmes should continue to expand multilingual teacher-training modules, ensuring availability in tribal languages and building communities of practice focused on MTB-MLE.



Recommendation 8

Invest in inclusive language technologies and bridge the digital divide

Strengthen language-technology ecosystems for underserved Indian languages by expanding automated speech recognition, text-to-speech and machine translation coverage through BHASHINI, AI4Bharat and open datasets, and link these resources to educational tools such as reading platforms, talking books and assistive devices.

Prioritize digital infrastructure and low-tech/offline solutions in tribal, rural and low-income regions to ensure equitable access to multilingual learning, with focused support for girls and learners with disabilities.

Recommendation 9

Ensure sustainable, equitable and inclusive financing for MTB-MLE and language-responsive technologies

Establish dedicated MTB-MLE budget lines within central and state schemes, including Samagra Shiksha, to fund multilingual teacher recruitment and training, community participation, material development, digital inclusion, and research, with targeted allocations for girls, children with disabilities and non-dominant language communities.

Create strong national and state accountability

and financing mechanisms through the Unified District Information System for Education Plus, the National Achievement Survey, the Annual Status of Education Report and innovation incentives, to support partnerships, safeguard linguistic rights and ensure sustainable investment in multilingual materials, sign-bilingual models, Indian Sign Language/Braille resources, gender-responsive digital content and assistive technologies.

Recommendation 10

Establish a national mission for MTB-MLE and strengthen institutional coordination

Create a time-bound national mission to drive implementation of the National Education Policy 2020, harmonize school- and cluster-level language mapping, and ensure consistent mother-tongue-/home-language-based instruction and multilingual bridging across all schooling systems.

Establish robust multistakeholder coordination mechanisms at national and state levels, bringing together the ministries of education, tribal affairs, culture, and technology, and the National Council for Teacher Education,

National and State Councils of Educational Research and Training, the Central Institute of Indian Languages, Tribal Research Institutes, universities, non-governmental organizations, technology partners and community representatives to steer MTB-MLE implementation and language-technology integration, while embedding language-sensitive indicators into existing data systems such as the Unified District Information System for Education Plus and the National Achievement Survey.

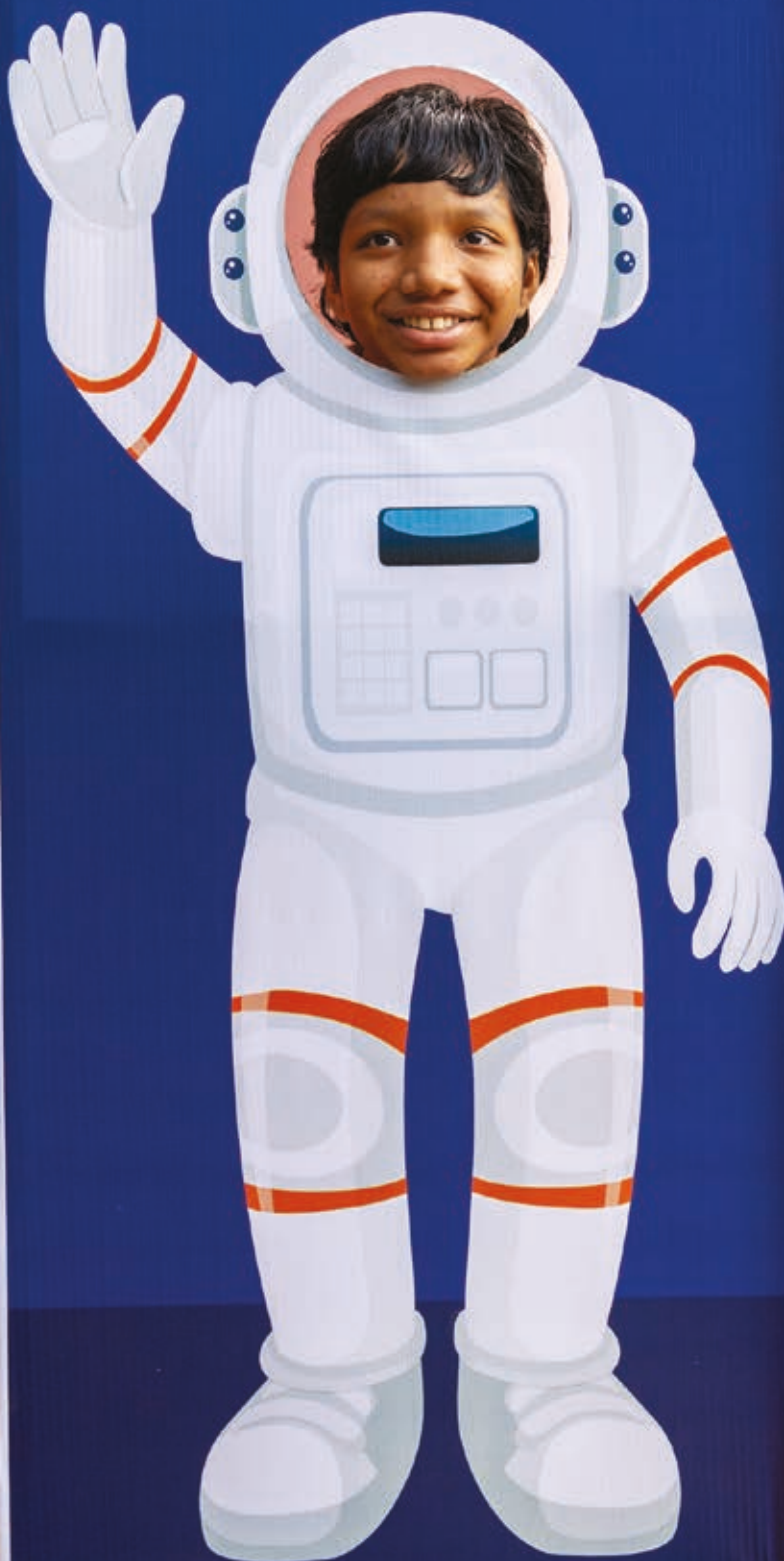
Facing page: Students engage with a project about the universe. Kalinga Institute of Social Sciences, Bhubaneswar.



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Student interacts with
a display at the annual
science festival, 2025.
Kalinga Institute of Social
Sciences, Bhubaneswar.

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UNICEF's Education Programme in India aims to ensure that, by 2027, children and adolescents have equitable access to and participate in quality learning and skilling in a safe and inclusive learning environment. UNICEF supports the Government of India and seventeen states in implementing the National Education Policy (NEP) and strengthening the Samagra Shiksha programme,

with a strong focus on improving early childhood education, foundational learning and secondary education, and enabling the school-to-work transition.

UNICEF works to strengthen policy and planning, ensuring that education reforms are evidence-based, child-centred and effectively implemented at all levels. It also promotes quality and equitable education by enhancing curriculum relevance, improving teacher education and strengthening learning assessment. Additionally, UNICEF supports efforts to make schools safer and climate resilient, and more engaging and conducive to learning for all children.



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Children display their drawings after taking part in a storytelling session at Government Primary School in Pandalnar village, Sukma district, Chhattisgarh, India.

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